Phase Quantisation from Optical Lattices to the Zeta Landscape

A Unified Framework Connecting Lattice Cryptography, Fresnel Optics, and Analytic Number Theory

Kim Young-Hu

Independent Researcher

April 19, 2026

Abstract

We identify a shared mathematical grammar—*phase quantisation on algebraic structures*—that connects three seemingly disparate domains: CRYSTALS-Kyber lattice-based post-quantum cryptography, free-space Fresnel diffraction, and the phase structure of the Riemann ξ-function on the critical line. The concrete nucleus is QROP-Net, a fully differentiable pipeline that encodes Kyber ciphertext as optical phase patterns and recovers them via deep-learning phase retrieval. Its $\sin^2$-based quantisation loss, which forces continuous phase predictions onto discrete cyclotomic lattice levels, is shown to be structurally isomorphic to the Bohr–Sommerfeld phase quantisation condition $\Delta\nu = m\pi$ that characterises "resonant primes" in the Resonant Prime Law framework. We further demonstrate that the LWE error threshold $\lfloor q/4 \rfloor = 1920$ in Kyber and the σ-lift deviation $\sigma^* - \frac{1}{2}$ from the critical line share the same role as hard error boundaries beyond which signal recovery fails. A multi-group (Polyadic) ODE-based zero detector, which self-calibrates without external ground truth, mirrors QROP-Net's blind test-time optimisation. Seven structural correspondences are formalised, and their mathematical, physical, and computational implications are discussed with explicit acknowledgement of the boundary between analogy and proof.

Seventy-four numerical results across ten verification axes—local curvature, global spectral statistics, topological invariants, Dirichlet L-function extension, negative controls, GL(2) extension, GL(3) extension, cross-rank monodromy, GL(4) extension, and GL(5) extension (GL(1) through GL(5) full coverage)—support the framework's internal consistency.

Note added in revision (2026-04-08). An attempted empirical realisation of the cyclotomic PQO programme on the critical line, using a discriminant network with $L = 4$ cosine PQO and the synthetic ideal phase $\Psi(n)$ of Equation (43) as supervisory target, returns a decisive null: across three seeds the resulting phase distribution fails to discriminate Riemann zeros from off-zero ordinates (χ^2 p-values 0.13/0.53/0.41; sectors $k = 2, 3$ exactly empty). The diagnosis (Remarks 3.2 and 4.2) is that the synthetic Ψ depends only on $\|X\|$ and contains no ξ-zero information whatsoever, so no term of the four-term loss ever injects a t-local zero label into training; Equation (31) (training potential) and Equation (32) (measurement) are operationally disjoint. The text below preserves the synthetic Ψ as the historical object and the cyclotomic PQO as the algebraic target; both are now flagged as awaiting an experiment with a supervised replacement $\Psi(t) := \arg \xi(\frac{1}{2} + it)$ (Equation (44)).

Keywords: Phase quantisation, cyclotomic polynomial ring, CRYSTALS-Kyber, fractional Fourier transform, Riemann ξ-function, Bohr–Sommerfeld condition, resonant primes, σ-lift, Berry phase, Dirichlet L-functions

Contents

1 Introduction **7**

I The Concrete System: QROP-Net **9**

2 QROP-Net: System and Mathematics **9**
- 2.1 Polynomial Ring and NTT . . . 9
- 2.2 Phase Modulation: From Integers to Optics . . . 10
- 2.3 Differentiable Optical Channel: FrFT . . . 10
- 2.4 Hybrid Loss Orchestration . . . 11
  - 2.4.1 Data Fidelity Loss . . . 11
  - 2.4.2 Periodic Soft Quantisation Loss . . . 11
  - 2.4.3 L^∞ Exponential Barrier Loss . . . 12
  - 2.4.4 Total Variation Loss . . . 12
- 2.5 Receiver Neural Network and Decoding . . . 12
- 2.6 Two-Stage Training and Physical-Layer Security . . . 12
  - 2.6.1 Stage 1: Amortised Pre-Training . . . 12
  - 2.6.2 Stage 2: Blind Test-Time Optimisation (TTO) . . . 13

3 The Phase Quantisation Principle **13**

II The Analytic Extension: Resonance Theory **15**

4 Phase Structure of the Completed Zeta Function **16**
- 4.1 The Riemann ξ-Function . . . 16
- 4.2 Amplitude–Phase Decomposition on the Critical Line . . . 16
- 4.3 The Ideal Phase Function $\Psi(n)$ . . . 17
- 4.4 Connection to the Riemann–Siegel ϑ Function . . . 20

5 Gate Structure: Phase Resonance as Prime Detection **20**
- 5.1 Gate A: Phase Quantisation (Alignment) . . . 20
- 5.2 Gate B: Phase Stationarity (Curvature) . . . 21
- 5.3 Combined Discriminant: Resonant Points . . . 22

6 The σ-Lift and Error Tolerance **22**
- 6.1 Deviation from the Critical Line . . . 22
- 6.2 Scattering Interpretation and Unitarity . . . 23
- 6.3 Berry Phase and Holonomy . . . 23

7 Multi-Group Zero Detection: The Polyadic Frame **23**
- 7.1 Target Functions and Evaluation Points . . . 24
- 7.2 Directional Channels . . . 24
- 7.3 Central Difference Log-Derivative . . . 25

7.4 Gauge ODE for Phase Correction . . . 25
7.5 Final Discriminant Function . . . 25
7.6 Numerical Verification and Convergence . . . 26
7.7 Cross-Validation: Three Independent Axes . . . 27
7.7.1 Axis A: Partial Euler Product Matching . . . 27
7.7.2 Axis B: Explicit Chebyshev $\psi(x)$ Recovery . . . 27
7.7.3 Axis C: Arithmetic Progression Bias Signal . . . 27

8 ■ The F_2-Discriminant as a Hardy-Z Sign-Change Detector 27
8.1 Hardy Z and the Analytic-Signal Identity . . . 28
8.2 Empirical Validation: 10/10 Zero Detection on $t \in [10, 50]$ . . . 28
8.3 Zero Difficulty Hierarchy and the Lehmer Phenomenon . . . 29

9 ■ Extension to High Zeros: $t \in [100, 200]$ 29

III Theoretical Foundations of the ξ-Bundle 30

10 Holonomy Characterization of Zeros 31

11 Curvature Concentration at Zeros 32

12 Gauss–Bonnet Quantization 42

13 Cross-Rank Universality 43

14 The Dual Criterion: Monodromy + Curvature 46

IV Failed Directions and Their Diagnoses 47

15 ■ Cyclotomic $\cos^2$ PQO training ($L = 2, 4$): Ψ-decoupling and the isolation cascade 47

16 ■ Phase-Gauge Gradient Descent (PGGD) 48

V ξ-Bundle Numerical Evidence 49

17 ■ Midpoint Curvature Nonlocality 49
17.1 Chern Number Interpretation: Numerical Verification . . . 51

18 Structural Correspondence Table 62
18.1 Mathematical Formalisation of the Correspondences . . . 62
18.2 ■ Dynamical and spectral correspondences . . . 63
18.2.1 Maslov correction as the metaplectic sheet shift . . . 63
18.2.2 Gauge ODE critical transition as Kuramoto synchronisation . . . 64
18.2.3 Lehmer phenomenon and GUE spacing statistics . . . 65

18.2.4 Weyl equidistribution and the PGGD angular walk 66

19 ■ The $|F_2|$ residual: a tentative mesoscopic observable **66**
19.1 Original observations (April 2026) 66
19.2 2026-04-08 partial reinterpretation 67
19.3 2026-04-10 follow-up: the "hard cluster" was a basis-bandwidth artifact . . . 68
19.4 What remains open 71

Summary of Numerical Evidence **72**

20 Discussion **80**
20.1 What the Correspondence Means 80
20.2 What the Correspondence Does Not Mean 80
20.3 Predictive Power and Future Directions 81

21 Related Work **87**

22 Conclusion **88**

A QROP-Net Full Equation Reference **92**

B Numerical Tables for Resonance Gates **93**

C Generalisation to Dirichlet L-Functions **93**
C.1 Numerical Verification of Bundle Properties 93
C.2 Blind Zero Prediction 94
C.3 Conductor Dependence: Topological vs. Quantitative Properties 95

D Beyond GL(1): GL(2) L-functions (Elliptic Curves and Ramanujan Δ) **96**
D.0.1 Mechanism of σ-uniqueness failure for GL(2) 97
D.0.2 t-Scaling Stability: GL(2) at High Imaginary Height (Result #49) . . 98
D.1 Extension to Maass Forms (Non-Holomorphic, Weight 0) 99

E Beyond GL(2): GL(3) Symmetric-Square L-functions **102**
E.1 Four-Property Verification: $\mathrm{sym}^2(11a1)$ and $\mathrm{sym}^2(37a1)$ 102
E.2 Blind Zero Prediction and the Zero-Finding Machine 103
E.3 σ-Uniqueness Failure: Dirichlet Series Convergence Origin 104
E.4 κ-Conductor Scaling and the κ_{near} Universal Constant 105
E.5 GL(3) ξ-Bundle $A(t_0)$ and Motivic-Centre σ-Uniqueness 106

F GL(4) Extension: Symmetric Cube L-Functions **107**
F.1 Four-Property Verification: $\mathrm{sym}^3\Delta$ and $\mathrm{sym}^3(11a1)$ 107
F.2 ξ-Bundle $A(t_0)$ Measurement (result #69) 108

G GL(5) Extension: Symmetric Fourth-Power L-Function **108**
G.1 Four-Property Verification: $\mathrm{sym}^4(11a1)$ 108
G.2 ξ-Bundle $A(t_0)$ Measurement (result #71) 109

H Negative Result: Gram Point Curvature **109**

I Notation Index **110**

1 Introduction

The polynomial ring $R_q = \mathbb{Z}_q[X]/(X^n + 1)$ with $n = 256$ and $q = 7681$ is the algebraic backbone of CRYSTALS-Kyber, the NIST-standardised post-quantum key encapsulation mechanism [1]. The defining relation $X^n + 1 = 0$—or equivalently $X^n \equiv -1$, the origin of our project codename $k^n = -1$—is a cyclotomic polynomial whose roots are the $2n$-th roots of unity:

$$\omega_j = e^{i\pi(2j+1)/n}, \quad j = 0, 1, \ldots, n-1. \tag{1}$$

These n phases are equally spaced on the unit circle, separated by $\Delta\theta = \pi/n$. This discrete, periodic phase structure is not merely an algebraic convenience for fast polynomial multiplication via the Number Theoretic Transform (NTT); it is, we argue, an instance of a more general *phase quantisation principle* that appears in at least two other domains:

1. **Fresnel optics**: The fractional Fourier transform (FrFT) maps a coherent wavefront $U_0 = e^{i\phi}$ through free-space diffraction, where the fractional order α encodes the propagation distance as a phase parameter.

2. **Analytic number theory**: The completed Riemann ξ-function $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ decomposes on the critical line $s = \frac{1}{2} + it$ into amplitude $A(t) = |\xi(\frac{1}{2}+it)|$ and phase $\Phi_\xi(t) = \arg \xi(\frac{1}{2}+it)$, with phase quantisation conditions intimately linked to the distribution of primes and the Riemann Hypothesis.

Thesis. The $\sin^2$-based quantisation loss of QROP-Net, the Bohr–Sommerfeld condition $\Delta\nu = m\pi$ on the critical line, and the cyclotomic root structure $X^n \equiv -1$ are three manifestations of a single mathematical grammar: *periodic phase penalty on an algebraic structure forces continuous signals onto discrete lattice levels, with a hard error boundary beyond which decoding fails.*

Contributions.

1. We present QROP-Net, a fully differentiable pipeline unifying Kyber lattice cryptography, Fresnel optics, and deep phase retrieval (Part I).

2. We abstract the phase quantisation mechanism into a general principle bridging discrete and continuous Fourier analysis (Section 3).

3. We apply this principle to the phase structure of $\xi(s)$, formulating a two-gate resonance criterion for prime detection (Part II, Sections 4–5).

4. We establish a structural correspondence between the LWE error threshold and the σ-lift from the critical line (Section 6).

5. We present a multi-group ODE zero detector that self-calibrates without ground truth, mirroring QROP-Net's blind test-time optimisation (Section 7).

6. We formalise seven structural correspondences and rigorously delimit the boundary between analogy and proof (Part III, Section 18).

7. **(Negative empirical result, 2026-04-08)** We attempt the cyclotomic PQO programme empirically with $L = 4$ on $t \in [100, 200]$, document a complete failure to discriminate zeros from non-zeros across three seeds, trace the cause to a Ψ-*decoupling defect* of Equation (43), and propose the supervised replacement Equation (44) as the necessary precondition for any future positive result (Part IV, Remarks 3.2 and 4.2).

8. **(Isolation experiment, 2026-04-09)** We isolate the target-injection, loss-weighting, and read-out channels and demonstrate that, with all three closed, the supervised chain $\arg \xi(\frac{1}{2} + it) \to \mathcal{L}_{\text{tgt}} \to \hat{\Psi}$ is sound at off-zero sampling points ($p = 1.18 \times 10^{-15}$, ratio $18.2\times$), localising the remaining defect to the network's output representation near zeros (Remark 4.3).

9. We establish the *rigorous theoretical foundations* of the ξ-bundle: the holonomy characterization of zeros (Theorem 10.1), curvature concentration (Theorem 11.1), Gauss–Bonnet quantization (Theorem 12.1), and cross-rank universality (Theorem 13.1). These are proved via the argument principle and Laurent expansion, showing that the bundle diagnostics are *theorems*, not conjectures.

10. We verify the fibre-bundle interpretation through 74 numerical results spanning ten independent axes: local curvature, global spectral statistics, topological invariants, Dirichlet L-function extension, negative controls, GL(2)/GL(3)/GL(4)/GL(5) extensions, and the κ-δ scaling law covering GL(1)–GL(5). A consolidated summary appears in Table 9. A Stirling-based analysis further identifies {monodromy, κ-concentration, blind prediction} as *degree-universal* diagnostics and σ-uniqueness as a degree-1-specific property, with the GL(2) failure explained by the exact identity $f_{\Gamma(s)} = f_{\text{zero}}$ (Appendix D.0.1). The GL(5) extension (results #70, #71) completes degree-1 through 5 coverage: GL(1)–GL(5) coverage: all three degree-universal properties (functional equation, κ-concentration, monodromy) hold; the ξ-bundle κ (σ-direction) and Hardy Z κ (t-direction) are identified as distinct observables (B-23 resolved); σ-uniqueness holds at GL(5) with $\sigma = 2.5$ uniqueness ratio $\approx 10000\times$ (result #70).

Reading Guide: Evidence Tiers

This work is a *consolidated research notebook*: it records the full trajectory of a four-month investigation including ideas that remain conjectural and empirical attempts that failed. We therefore classify every substantive claim into one of three tiers, signalled explicitly in each section heading and in the table of contents:

- **Tier 1 — Validated.** Claims backed by complete derivations and successful numerical experiments. Examples: the F_2-discriminant zero detector (Section 8), the supervised identification $\Psi(t) = \arg \xi(\frac{1}{2} + it)$ via the isolation experiment (Remark 4.3), the high-zero extension to $t \in [100, 200]$ (Section 9). These sections should be read as mature results.

- **Tier 2 — Ideas and partial results.** Analytic correspondences, theoretical constructions, or experiments with suggestive but incomplete evidence. Examples: Gate B

stationarity (Section 5), the σ-lift (Section 6), the seven structural correspondences (Section 18), the $|F_2|$ spatial observables with their 2026-04-08 reinterpretation (Section 19). These sections motivate and sketch; they should be read as research programme rather than settled theory.

- **Tier 3 — Failed attempts and their diagnoses.** Empirical programmes that did not succeed, preserved here with full diagnostic analysis. Examples: cyclotomic $\cos^2$ $L \in \{2, 4\}$ PQO training (Section 15), the phase-gauge gradient descent (PGGD) optimiser (Section 16). We retain these sections, compressed but intact, because the diagnostic chain is itself a contribution: it documents exactly which combinations of target design, loss weighting, measurement, and model expressivity are incompatible, saving future work from the same failure modes. A failed direction whose cause is understood is a partial positive result.

We will often refer to the tier of a given section or remark by the symbols ■/ ■/■. The reader may use the tier as a calibration for how much trust to place in a given statement. The structural division of the paper into Parts I–V respects this tiering: Part I and Part II collect Tier 1 and Tier 2 material, Part IV collects Tier 3 failure diagnoses, and Part V integrates all three into a unified picture.

Part I

The Concrete System: QROP-Net

2 QROP-Net: System and Mathematics

2.1 Polynomial Ring and NTT

Kyber operates over $R_q = \mathbb{Z}_q[X]/(X^n+1)$ where $q = 7681$ is chosen to satisfy $q \equiv 1 \pmod{2n}$, enabling the Number Theoretic Transform (NTT) for $O(n \log n)$ polynomial multiplication. The ring condition $X^n + 1 = 0$ factors as:

$$X^n + 1 = \prod_{j=0}^{n-1} (X - \omega_j), \quad \omega_j = e^{i\pi(2j+1)/n} \in \mathbb{C}. \tag{2}$$

Every element $a(X) \in R_q$ corresponds to a vector $\mathbf{a} = (a_0, \ldots, a_{n-1}) \in \mathbb{Z}_q^n$, and multiplication $a \cdot b \pmod{X^n + 1}$ becomes component-wise multiplication in the NTT domain.

Definition 2.1 (Kyber IND-CPA Encryption)**.** Given public key $(\mathbf{A}, \mathbf{t})$ where $\mathbf{A} \in R_q^{k \times k}$, $\mathbf{t} \in R_q^k$, and message $m \in \{0, 1\}^n$, the ciphertext $(\mathbf{u}, v)$ is:

$$\mathbf{u} = \mathbf{A}^\top \mathbf{r} + \mathbf{e}_1, \tag{3}$$

$$v = \mathbf{t}^\top \mathbf{r} + e_2 + \lceil q/2 \rfloor \cdot m, \tag{4}$$

where $\mathbf{r}, \mathbf{e}_1, e_2 \sim \beta_\eta$ are sampled from the centred binomial distribution:

$$\beta_\eta : \quad x = \sum_{i=1}^{\eta} (a_i - b_i), \quad a_i, b_i \sim \text{Bernoulli}(\tfrac{1}{2}), \quad x \in [-\eta, \eta]. \tag{5}$$

Definition 2.2 (Compression and Decompression)**.**

$$\text{Compress}_q(x, d) = \left\lceil \frac{2^d}{q} \cdot x \right\rfloor \bmod 2^d, \tag{6}$$

$$\text{Decompress}_q(x, d) = \left\lceil \frac{q}{2^d} \cdot x \right\rfloor, \tag{7}$$

where $\lceil \cdot \rfloor$ denotes ties-rounded-up: $\lfloor x + 0.5 \rfloor$. Parameters: $d_u = 11$ (2048 levels for $\mathbf{u}$), $d_v = 3$ (8 levels for v).

Theorem 2.1 (LWE Correctness Criterion)**.** *Decryption succeeds if and only if the total error $w = v - \mathbf{s}^\top \mathbf{u} \pmod q$ satisfies:*

$$\|w - \lceil q/2 \rfloor \, \hat{m}\|_\infty < \lceil q/4 \rfloor = 1920. \tag{8}$$

*A single coefficient exceeding this bound causes complete decryption failure (*DFR $= 1$*). The ring distance is:*

$$|e|_{\mathbb{Z}_q} = \min(e,\ q - e), \quad e \in \{0, 1, \ldots, q-1\}. \tag{9}$$

2.2 Phase Modulation: From Integers to Optics

Each compressed ciphertext coefficient $c_i \in \{0, \ldots, 2^{d_u} - 1\}$ is mapped to a phase angle and encoded as a unit-amplitude complex wavefront:

$$\phi_i = \frac{2\pi \cdot c_i}{2^{d_u}}, \qquad U_0(x, y) = e^{i\phi(x,y)}, \quad |U_0| \equiv 1. \tag{10}$$

The $2^{d_u} = 2048$ discrete ciphertext levels map to 2048 equally spaced phases on the unit circle, separated by $\Delta\phi = 2\pi/2048$. This is a direct physical realisation of the algebraic structure: the polynomial ring's discrete coefficients become discrete optical phases on the SLM.

A dimensional constraint links the cryptographic and optical domains:

$$N = \sqrt{k \cdot n + n} \times M = \sqrt{1024} \times 8 = 256, \tag{11}$$

where $M = 8$ is the macro-pixel expansion factor for noise resilience.

2.3 Differentiable Optical Channel: FrFT

When the wavefront U_0 propagates a distance d through free space, the diffraction integral is equivalent to a fractional Fourier transform of order p:

$$p = \frac{2}{\pi} \arctan\left(\frac{\lambda d}{N \Delta x^2}\right), \qquad \alpha = \frac{p\pi}{2}, \tag{12}$$

where $\lambda = 500\,\text{nm}$ is the wavelength and $\Delta x = 8\,\mu\text{m}$ is the pixel pitch. The propagation distance d is declared as a learnable parameter, enabling gradient-based auto-focusing:

$$d \xrightarrow{\partial} R = \frac{\lambda d}{N \Delta x^2} \;\rightarrow\; \alpha = \arctan(R) \;\rightarrow\; s_2 = \sqrt{1+R^2}. \tag{13}$$

The IP-DFrFT is implemented via chirp–FFT–chirp decomposition, reducing complexity from $O(N^4)$ to $O(N^2 \log N)$:

$$\boxed{U_d = \underbrace{(1 - i\cot\alpha)}_{A_\alpha^2} \cdot \mathcal{F}^{-1}\Big[\mathcal{F}\big[\underbrace{U_0 \cdot e^{i\pi(\cot\alpha - \csc\alpha)\, r^2}}_{\text{input chirp}}\big] \;\cdot\; \mathcal{F}\big[\underbrace{e^{i\pi\,\csc\alpha\, r'^2}}_{\text{kernel chirp}}\big]\Big] \cdot \Delta x^2} \tag{14}$$

where $r^2 = (m^2 + n^2)\Delta x^2$ on $N \times N$ and r'^2 on zero-padded $2N \times 2N$ to prevent circular aliasing.

The sensor captures intensity and adds physical noise:

$$I_{\text{clean}} = \frac{|U_d|^2}{1 + \cot^2\alpha}, \tag{15}$$

$$I_{\text{shot}} \sim \text{Poisson}(I_{\text{clean}} \cdot \text{FWC}), \tag{16}$$

$$I_{\text{meas}} = \text{ADC}_{\text{12-bit}}(I_{\text{shot}} + \mathcal{N}(0, \sigma_{\text{read}}^2)). \tag{17}$$

The straight-through estimator (STE) preserves gradient flow through the non-differentiable Poisson sampling and ADC rounding:

$$\text{STE}(x_{\text{noisy}}, x_{\text{clean}}) = x_{\text{clean}} + \text{stopgrad}(x_{\text{noisy}} - x_{\text{clean}}). \tag{18}$$

2.4 Hybrid Loss Orchestration

Four loss terms are combined under a curriculum scheduler.

2.4.1 Data Fidelity Loss

The predicted wavefront $\hat{U}_0$ is propagated through the FrFT to produce $\hat{I}$. Comparison in the amplitude domain ensures uniform gradient flow:

$$\mathcal{L}_{\text{fid}} = \frac{1}{N^2} \sum_{x,y} \left(\sqrt{\hat{I}(x,y) + \epsilon} - \sqrt{I_{\text{target}}(x,y) + \epsilon} \right)^2. \tag{19}$$

2.4.2 Periodic Soft Quantisation Loss

A $\sin^2$-based penalty forces continuous phases onto discrete Kyber lattice levels:

$$\boxed{\mathcal{L}_{\text{quant}} = \frac{1}{2}\left[\frac{1}{kn} \sum_{i=1}^{kn} \sin^2\!\Big(\frac{2^{d_u}}{2}\,\phi_i^{(u)}\Big) + \frac{1}{n} \sum_{j=1}^{n} \sin^2\!\Big(\frac{2^{d_v}}{2}\,\phi_j^{(v)}\Big) \right]} \tag{20}$$

The $\sin^2$ function creates a *valley–ridge landscape*: zero loss at exact lattice positions $\phi = 2\pi k/2^d$, maximum penalty at midpoints. This is the key equation that bridges to

analytic number theory (Section 3).

2.4.3 L^∞ Exponential Barrier Loss

An exponential barrier enforces the LWE threshold on the most dangerous outliers:

$$\mathcal{L}_{\max} = \frac{1}{|S_K|} \sum_{i \in S_K} \exp(\gamma \cdot \mathrm{ReLU}(|e_i|_{\mathbb{Z}_q} - \tau_{\text{safe}})), \tag{21}$$

where $\tau_{\text{safe}} = 0.8 \times 1920 = 1536$, $\gamma = 10$, and S_K is the set of top-5% error pixels.

2.4.4 Total Variation Loss

On the complex wavefront (not the phase angle, to avoid wrapping artefacts):

$$\mathcal{L}_{\text{TV}} = \frac{1}{N^2} \sum_{x,y} (|\hat{U}_0^{x+1,y} - \hat{U}_0^{x,y}|^2 + |\hat{U}_0^{x,y+1} - \hat{U}_0^{x,y}|^2). \tag{22}$$

2.5 Receiver Neural Network and Decoding

A LeWin U-Net (window-based multi-head self-attention + locally-enhanced feed-forward) maps the intensity measurement I_{meas} to a 2-channel $(\mathrm{Re}, \mathrm{Im})$ output, from which the phase is extracted via:

$$\hat{\phi} = \mathrm{atan2}(\mathrm{Im},\ \mathrm{Re}) \bmod 2\pi, \tag{23}$$

$$\hat{U}_0 = e^{i\hat{\phi}}, \quad |\hat{U}_0| \equiv 1. \tag{24}$$

Macro-pixel pooling uses circular statistics to avoid phase averaging catastrophe:

$$\bar{\phi} = \mathrm{atan2}\left(\frac{1}{M^2} \sum \sin \phi_{ij},\ \frac{1}{M^2} \sum \cos \phi_{ij}\right) \bmod 2\pi. \tag{25}$$

Final Kyber decapsulation:

$$\hat{\mathbf{u}}, \hat{v} = \mathrm{Decompress}_q(c_{\text{pred}}, d_u, d_v), \tag{26}$$

$$w = \hat{v} - \mathbf{s}^\top \hat{\mathbf{u}} \pmod q, \tag{27}$$

$$\hat{m}_j = \left\lceil \frac{2}{q} \cdot w_j \right\rfloor \bmod 2. \tag{28}$$

2.6 Two-Stage Training and Physical-Layer Security

2.6.1 Stage 1: Amortised Pre-Training

Synthetic ciphertexts are generated on-the-fly; AdamW jointly updates 119 U-Net parameters and the physical distance d.

2.6.2 Stage 2: Blind Test-Time Optimisation (TTO)

At deployment, only physics-based and mathematics-based constraints are available:

$$\mathcal{L}_{\text{blind}} = \mathcal{L}_{\text{fid}} + 0.5\,\mathcal{L}_{\text{quant}} + 0.05\,\mathcal{L}_{\text{TV}}. \tag{29}$$

The U-Net backbone is frozen at $\text{lr} = 10^{-5}$ while d receives $\text{lr} = 10^{-3}$, enabling rapid auto-focusing in only 15 gradient steps *without access to the ground-truth ciphertext.*

Theorem 2.2 (Avalanche Effect — Physical-Layer Security)**.** *An eavesdropper who possesses identical U-Net weights, PRNG seed, and Kyber secret key, but estimates* $d' = d + 0.1\,\text{mm}$*, experiences* DFR $= 1$*. The FrFT angle* α *is a nonlinear function of* d*; a 0.1 mm offset cascades through the chirp–FFT–chirp pipeline, causing* $\|e\|_\infty > 1920$ *for every ciphertext element.*

3 The Phase Quantisation Principle

We now abstract the $\sin^2$ quantisation mechanism (Equation (20)) into a general principle.

Definition 3.1 (Phase Quantisation Operator)**.** Let $\phi \in [0, 2\pi)$ be a continuous phase variable and $L \in \mathbb{N}$ a target number of discrete levels. The *phase quantisation penalty* is:

$$\mathcal{P}(\phi; L) = \sin^2\left(\frac{L}{2}\,\phi\right). \tag{30}$$

This function has the following properties:

1. $\mathcal{P}(\phi; L) = 0$ if and only if $\phi \in \{2\pi k/L : k = 0, 1, \ldots, L-1\}$ (the lattice positions).
2. $\mathcal{P}(\phi; L) = 1$ at the midpoints $\phi = 2\pi(k + \frac{1}{2})/L$.
3. $\nabla_\phi \mathcal{P} = \frac{L}{2}\sin(L\phi)$ provides a gradient that steers ϕ toward the nearest lattice point.
4. The landscape is $2\pi/L$-periodic, creating L equally spaced attraction basins on the unit circle.

In QROP-Net, the lattice levels are $L = 2^{d_u} = 2048$ (for $\mathbf{u}$-coefficients) and $L = 2^{d_v} = 8$ (for v-coefficients). The $\sin^2$ loss drives each continuous phase prediction toward the nearest discrete Kyber level, realising a *soft quantisation* that is fully differentiable.

Remark 3.1 (Cyclotomic Connection)**.** The L zeros of $\mathcal{P}(\phi; L)$ are precisely the L-th roots of unity $e^{2\pi i k/L}$. When $L = 2n = 512$ (the order of the cyclotomic polynomial $X^n + 1$), these are the roots $\omega_0, \ldots, \omega_{n-1}$ of Equation (1). The $\sin^2$ loss is thus the *analytic restatement* of the algebraic condition $X^n + 1 = 0$ in the phase domain.

Remark 3.2 (Cosine PQO and Higher-Resolution Cyclotomic Lattices)**.** The PQO admits a structurally identical *cosine* variant

$$\mathcal{P}_{\cos}(\phi; L) = \cos^2\left(\frac{L}{2}\,\phi\right), \tag{31}$$

whose zero set is $\phi \in \{(2k+1)\pi/L : k = 0, 1, \ldots, L-1\}$. These are precisely the *primitive* $2L$*-th roots of unity*, i.e. the L roots ω_j of $X^L + 1 = 0$ from Equation (1). Hence $\mathcal{P}_{\cos}(\,\cdot\,; L)$ is the L-th cyclotomic PQO, parallel to Remark 3.1 but realising the algebraic constraint $X^L + 1 = 0$ *directly* on the phase domain rather than the shifted unity-root constraint $X^L - 1 = 0$ associated with $\sin^2$.

Trivial collapse at $L = 2$. At $L = 2$ the attractor set degenerates to $\{i, -i\}$. Any signal trained with a non-trivially weighted $\mathcal{P}_{\cos}(\phi; 2)$ term collapses globally onto one of these two points: in particular, in a critical-line discriminant network whose loss includes such a term, the empirical phase $\arg Z(t)$ is driven to $\pi/2$ *for all t in the trained band*, irrespective of whether t lies near a Riemann zero. The $L = 2$ cosine PQO therefore carries no zero-localising information; it merely confirms that the attractor mechanism functions as designed.

Higher-resolution generalisation (attempted). For $L \geq 4$ the lattice acquires sufficient cardinality, in principle, to act as a sector classifier on S^1. Each candidate ordinate t is assigned a discrete label

$$k(t) \;=\; \arg\min_{0 \leq k < L} \left| \arg Z(t) - (2k{+}1)\pi/L \right| \pmod{2\pi}, \tag{32}$$

naming the cyclotomic sector $\omega_{k(t)}$ closest to the trained phase, with the hope that Riemann zeros would occupy a non-trivial sub-lattice and thereby provide an arithmetic stratification finer than the $L = 2$ case admits.

Empirical refutation at $L = 4$ (2026-04-08). A direct test of this programme was performed using a critical-line discriminant network with $L = 4$, hidden width 32, three seeds $\{42, 123, 777\}$, 200 training epochs, and a labelled population of 50 Riemann zeros versus 200 off-zero ordinates of gap ≥ 0.5 (training script `scripts/phase_L4_experiment.py` in the companion GDL_UNIFIED repository; output dump at `outputs/analysis/phase_L4_experiment.{json,txt}`). The result is unambiguously negative on three independent axes:

1. All three seeds fail a χ^2 test of independence between the zero and off-zero k-distributions ($p = 0.127, 0.528, 0.410$; pooled $p = 0.154$).

2. Of the four cyclotomic sectors $\{k = 0, 1, 2, 3\}$ only sectors $k = 0$ and $k = 1$ are ever populated; sectors $k = 2, 3$ contain *exactly zero* samples on every seed (asymmetric basin collapse).

3. Mean nearest-attractor residuals for zeros and off-zeros agree to within $\sim 5\%$ across all seeds.

The L=2 trivial collapse and the L=4 asymmetric collapse therefore share a single root cause; we now identify it.

Diagnosis: Ψ-decoupling defect. The training loss of the discriminant network is a sum $\mathcal{L} = \lambda_{\text{res}}\mathcal{L}_{\text{res}} + \lambda_{\text{curv}}\mathcal{L}_{\text{curv}} + \lambda_{\text{tgt}}\mathcal{L}_{\text{tgt}} + \lambda_{\text{pqo}}\mathcal{L}_{\text{pqo}}$, in which the target-matching term $\mathcal{L}_{\text{tgt}}$ is built against Ψ as defined in Definition 4.3 / Equation (43). As written, Ψ depends only on $\|X\|$ and is a synthetically chosen $(\alpha, \beta, \gamma, \delta)$ curve carrying *no ξ-zero information whatsoever*. Consequently:

- No term of $\mathcal{L}$ injects a t-local zero label into training;
- The cosine PQO term $\mathcal{L}_{\text{pqo}}$ is a *t-independent* potential whose basin selection is fixed by initialisation, not by the location of t relative to a zero;
- The labelling rule Equation (32) is therefore an *operationally disjoint measurement* placed on top of a network that has never been told what a zero is.

In particular Equation (31) (training) and Equation (32) (measurement) act on incompatible signal categories: the former *erases* the t-localised information that the latter is trying to read out. Increasing L from 2 to $4, 8, \ldots$ cannot repair this — the broken link is upstream of L.

Status of the cyclotomic PQO programme. The $L \geq 4$ programme as previously formulated is suspended. Its revival requires *replacing* Equation (43) by a target carrying genuine zero information (the natural candidates are $\Psi(t) := \arg \xi(\frac{1}{2} + it)$ or the Hardy phase $\arg Z(t)$ of Equation (46)), demoting $\mathcal{P}_{\cos}$ from a *training* potential to a pure *measurement* of the resulting phase, and re-running the $L = 4$ protocol under the new supervision. Until such an experiment exists, no claim of cyclotomic stratification of Riemann zeros is supported by this work; the data files `phase_L4_experiment.*` and `phase_pi_half_verify.txt` are retained as deterministic negative controls and are part of the record on which the diagnosis above rests.

Remark 3.3 (Bohr–Sommerfeld Analogy)**.** The condition $\mathcal{P}(\phi; L) = 0$ is equivalent to $L\phi/2 = m\pi$ for $m \in \mathbb{Z}$, *i.e.*,

$$\Delta\nu \;=\; m\pi, \qquad m \in \mathbb{Z}, \tag{33}$$

where $\Delta\nu = L\phi/2$ is a rescaled phase accumulation. This is precisely the form of the *Bohr–Sommerfeld quantisation condition* in semiclassical mechanics, where the action integral around a closed orbit must equal an integer multiple of $\pi\hbar$ (in appropriate units). In optics, the same condition determines constructive interference: the path length must accommodate an integer number of half-wavelengths. The phase quantisation principle thus unifies:

Domain	**Quantised Variable**	**Condition**
QROP-Net	Ciphertext phase ϕ_i	$\sin^2(2^{d_u-1}\phi_i) = 0$
Semiclassical QM	Action integral $\oint p\, dq$	$= (n + \frac{1}{2})\pi\hbar$
Optics (cavity)	Optical path length nL_{cav}	$= m\lambda/2$
Critical line	Phase of $\xi(s)$	$\Delta\nu = m\pi$

Part II

The Analytic Extension: Resonance Theory

Every mathematical claim in this part has either a concrete QROP-Net parallel (noted in Remark environments) or an independent reference in the literature. Claims from earlier versions of the Resonant Prime Law that lack both are omitted.

4 Phase Structure of the Completed Zeta Function

4.1 The Riemann ξ-Function

Definition 4.1 (Completed Zeta Function)**.** The Riemann ξ-function is defined as:

$$\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma\left(\frac{s}{2}\right)\zeta(s), \tag{34}$$

where $\zeta(s) = \sum_{n=1}^{\infty} n^{-s}$ is the Riemann zeta function and Γ is the Euler gamma function. The function ξ is entire (no poles) and satisfies the functional equation:

$$\xi(s) = \xi(1-s). \tag{35}$$

On the critical line $s = \frac{1}{2} + it$, ξ is real-valued: $\xi(\frac{1}{2} + it) \in \mathbb{R}$.

Definition 4.2 (Weierstrass Product Representation)**.** Since $\xi(s)$ is entire, it admits a Weierstrass product over its zeros ρ (the nontrivial zeros of ζ):

$$\xi(s) = e^{g(s)} \prod_{\rho} \left(1 - \frac{s}{\rho}\right), \tag{36}$$

for some entire function $g(s)$ ensuring convergence. Differentiating $\ln \xi$ gives the *logarithmic derivative*:

$$\frac{\xi'(s)}{\xi(s)} = g'(s) + \sum_{\rho} \frac{1}{s-\rho}. \tag{37}$$

Away from the zeros, $g'(s)$ is smooth, and the dominant contributions come from the poles at $s = \rho$—the nontrivial zeros of ζ.

4.2 Amplitude–Phase Decomposition on the Critical Line

For general $s = \sigma + it$ (not restricted to $\sigma = \frac{1}{2}$), we decompose ξ into amplitude and phase:

$$A(\sigma, t) = |\xi(\sigma + it)|, \tag{38}$$

$$v(\sigma, t) = \arg \xi(\sigma + it), \tag{39}$$

so that $\xi(\sigma + it) = A(\sigma, t)\, e^{iv(\sigma,t)}$. Writing $\log \xi = u + iv$ where $u = \log |\xi|$ and $v = \arg \xi$, the Cauchy–Riemann equations give:

$$v_\sigma = \mathrm{Re}\left(\frac{\xi'}{\xi}\right), \quad v_t = -\mathrm{Im}\left(\frac{\xi'}{\xi}\right), \quad u_t = -v_\sigma, \quad u_\sigma = v_t. \tag{40}$$

On the critical line $\sigma = \frac{1}{2}$, we write:

$$\Phi_\xi(t) \;:=\; v(\tfrac{1}{2}, t) = \arg \xi(\tfrac{1}{2} + it), \tag{41}$$

$$A(t) \;:=\; A(\tfrac{1}{2}, t) = |\xi(\tfrac{1}{2} + it)|. \tag{42}$$

Since $\xi(\frac{1}{2} + it) \in \mathbb{R}$, the phase $\Phi_\xi(t)$ takes values in $\{0, \pi\}$ (modulo 2π), jumping by π at each zero crossing.

Remark 4.1 (QROP-Net Parallel: Amplitude–Phase Separation)**.** In QROP-Net, the transmitted wavefront $U_0 = e^{i\phi}$ has unit amplitude ($|U_0| \equiv 1$) and discrete phase ϕ. The receiver must recover ϕ from the intensity $I = |U_d|^2$, which involves "undoing" the amplitude–phase mixing caused by diffraction. The structure $\xi = A\, e^{iv}$ on the critical line presents the identical mathematical challenge: separating the amplitude envelope from the underlying phase structure.

4.3 The Ideal Phase Function $\Psi(n)$

Reader's warning. The synthetic form Equation (43) below was empirically falsified as a supervisory target on 2026-04-08; see Remark 4.2 immediately following the definition and Remark 3.2 for the diagnosis. The definition is retained for historical exposition; the operational replacement used in subsequent revisions is Equation (44).

Definition 4.3 (Smooth Prime-Counting Phase)**.** For integer $n \geq 2$, the ideal phase function is defined as:

$$\Psi(n) = \alpha \log n + \frac{\beta \sin(\gamma \log n)}{n^{\delta}}, \tag{43}$$

where $\alpha, \beta, \gamma, \delta$ are structural constants:

- $\alpha > 0$ determines the logarithmic ascent (alignment slope),
- $\beta \geq 0$ controls the oscillation amplitude (resonance strength),
- $\gamma > 0$ is the phase rotation frequency (related to the local Gamma factor),
- $\delta > 0$ controls exponential damping: the oscillation term decays as $n^{-\delta}$, so $\Psi(n) \to \alpha \log n$ for large n.

The monotone term $\alpha \log n$ mirrors the Riemann–von Mangoldt formula $N(T) \sim \frac{T}{2\pi} \log \frac{T}{2\pi e}$ for the zero counting function, while the oscillatory correction $\beta \sin(\gamma \log n)/n^{\delta}$ is intended as a heuristic stand-in for local fluctuations related to prime gaps.

Remark 4.2 (Decoupling defect of Equation (43); documented 2026-04-08)**.** The form Equation (43) as written is a synthetic $(\alpha, \beta, \gamma, \delta)$ curve depending only on the magnitude n (or, in the network implementation, on $\|X\|$); it carries *no* information about the location of Riemann zeros. When used as a target for a discriminant network — as in the L=4 cyclotomic PQO experiment of Remark 3.2 — none of the four loss terms $\mathcal{L}_{\text{res}}, \mathcal{L}_{\text{curv}}, \mathcal{L}_{\text{tgt}}, \mathcal{L}_{\text{pqo}}$ injects a t-local zero signal into training, and the resulting network necessarily fails to discriminate zeros from non-zeros regardless of L. The defect is structural rather than parametric: no choice of $(\alpha, \beta, \gamma, \delta)$ can repair it.

The intended replacement, used in subsequent revisions of this work, is to redefine the ideal phase on the critical line directly,

$$\Psi(t) \;:=\; \arg \xi(\tfrac{1}{2} + it), \tag{44}$$

or equivalently to take $\Psi(t) := \arg Z(t)$ via the Hardy Z-function of Equation (46). Either choice supplies ξ-zero information *by construction*, restoring the causal chain between the

supervisory target and the cyclotomic measurement Equation (32). The synthetic form Equation (43) is retained in the text only as the historical object whose failure motivated the diagnosis.

Supervised-Ψ control experiment (2026-04-08). The replacement Equation (44) was tested directly. The same discriminant network architecture (hidden width 32, three layers, three seeds $\{42, 123, 777\}$, 200 epochs, $t \in [100, 200]$, 50 zeros vs. 200 off-zero ordinates of gap ≥ 0.5) was retrained with two single-line modifications relative to the L=4 baseline: (i) the model's Ψ output was overwritten by the precomputed $\arg \xi(\frac{1}{2} + it)$ supplied as a supervisory signal; (ii) $\lambda_{\text{pqo}} = 0$, removing the cosine PQO from training and retaining it only as the cyclotomic measurement Equation (32) (script `scripts/phase_supervised_experiment.py`; output `phase_supervised_experiment.{json,txt}`). The result is again negative on the headline metric — the seed-pooled phase MSE of $\arg Z(t)$ against $\arg \xi(\frac{1}{2} + it)$ is essentially identical for zeros and non-zeros (means 2.256 vs. 2.247, ratio 1.004, Welch $p = 0.976$) — and the per-seed Welch tests are inconsistent in direction (one seed $p = 0.034$ *against* the expected ordering). Two structural symptoms are nevertheless new and informative:

1. The asymmetric basin collapse of the L=4 baseline is broken: all four cyclotomic sectors $k \in \{0, 1, 2, 3\}$ are now populated on every seed. The t-independent attractor freeze induced by $\mathcal{L}_{\text{pqo}}$ is therefore confirmed, by ablation, as one of the two contributors to that collapse.

2. The recorded best validation loss (≈ 1.5–1.7) is substantially smaller than the recorded phase MSE (≈ 2.25), implying that the $\mathcal{L}_{\text{tgt}}$ contribution is a small fraction of $\mathcal{L}$ even after $\mathcal{L}_{\text{pqo}}$ has been zeroed out. The remaining $\mathcal{L}_{\text{res}}$ and $\mathcal{L}_{\text{curv}}$ terms dominate the gradient, so the network is not actually being asked to fit $\arg \xi$.

This refines, but does *not* retract, the diagnosis above: Ψ-decoupling is a necessary defect of Equation (43), and Equation (44) repairs it at the level of the target alone; but the loss-weight imbalance and a suspected measurement artefact (arithmetic averaging of channel-wise $\arg Z$, which destroys the $\pm\pi$ wrap signal that Equation (41) predicts) are *additional* broken links. A second control experiment with $\lambda_{\text{res}} = \lambda_{\text{curv}} = \lambda_{\text{pqo}} = 0$ (only $\mathcal{L}_{\text{tgt}}$ active) and a circular-mean read-out is described in Remark 4.3 below.

Remark 4.3 (Isolation experiment: supervisory chain is sound, $\pm\pi$ jump representability is not; documented 2026-04-09)**.** To disentangle the three suspected defects of Remark 4.2 — (i) Ψ-decoupling via Equation (43), (ii) loss-weight imbalance, and (iii) an arithmetic-mean read-out that cannot represent the $\pm\pi$ wrap — we ran a maximally isolated control in which all three channels were simultaneously closed. The network was retrained with $\lambda_{\text{res}} = \lambda_{\text{curv}} = \lambda_{\text{pqo}} = 0$ and only $\mathcal{L}_{\text{tgt}}$ active, the supervisory target hard-set to $\arg \xi(\frac{1}{2} + it)$ as in Equation (44), and the per-sample phase averaged via a *circular* mean $\arg \sum_c e^{i\hat{\Psi}_c}$ rather than the arithmetic channel-wise average (script `scripts/phase_isolation_experiment.py`; output `phase_isolation_experiment.{json,txt}`; same architecture, seeds, epochs, and t-window as in Remarks 3.2 and 4.2). The result is strongly *positive*:

- **Circular read-out.** Seed-pooled phase MSE against $\arg \xi(\frac{1}{2} + it)$ is 2.1708 on the

ξ-zero set and 0.1195 on the off-zero set, a ratio of 18.16× and a Welch $p = 1.18 \times 10^{-15}$. Per-seed ratios are 29.5, 31.9, 9.5 with $p \in \{4.5, 3.7, 5.5\} \times 10^{-6}$.

- **Arithmetic read-out, same weights.** The identical training run, read out by the old arithmetic mean, collapses to zero/non-zero means 2.410 vs. 1.849, ratio 1.30, pooled $p = 0.046$. The circular vs. arithmetic gap is therefore not a training difference but a measurement difference.

Both hypotheses of Remark 4.2 are confirmed: the loss-weight imbalance (ii) and the arithmetic-mean read-out (iii) each suffice to hide the signal, and together they fully explain the negative outcomes of Remark 3.2 and of the first supervised control. With both closed, the supervisory chain $\arg \xi(\frac{1}{2} + it) \mapsto \mathcal{L}_{\text{tgt}} \mapsto \hat{\Psi}$ is *intact*: the off-zero MSE of 0.12 shows that the network learns $\arg \xi$ essentially perfectly on the smooth branch.

The residual on-zero MSE of $2.17 \approx (\pi/2)^2$ has a clean interpretation. On the critical line ξ is real; numerically $|\text{Im}\, \xi|/|\text{Re}\, \xi| \sim 10^{-53}$ in our cache, so $\arg \xi$ is effectively a $\{0, \pi\}$-valued step function switching sign at every Riemann zero. The network output $\hat{\Psi}(t)$ is a smooth $\mathbb{R}$-valued field and cannot represent such an instantaneous $\pm\pi$ jump; its best circular-mean fit in a neighbourhood of a zero is, up to a constant, the halfway value, contributing exactly $(\pi/2)^2$ to the MSE. The remaining defect is therefore neither in the target nor in the read-out but in the *output representation* of the discriminant network, and is addressed in Remarks 3.2 and 4.2 only by raising L; a proper fix requires either a smoothed zero-crossing target (e.g. $\text{sign}(Z(t)) \cdot e^{-|Z(t)|^2/\sigma^2}$) or a head that predicts $\xi(\frac{1}{2} + it)$ as a complex vector and reads out arg post-hoc.

Smooth zero-crossing confirmation (2026-04-10). We tested two smooth surrogates that encode the same sign information as $\arg \xi$ without the $\pm\pi$ discontinuity, under the same isolation setup ($\lambda_{\text{res}} = \lambda_{\text{curv}} = \lambda_{\text{pqo}} = 0$, `in_features`= 128, three seeds):

- **Bump target** $\Psi_{\text{bump}}(t) = \text{sign}(\text{Re}\, \xi) \cdot e^{-(\text{Re}\, \xi)^2/\sigma^2}$, $\sigma = \text{median}(|\text{Re}\, \xi|)$: zero MSE = 0.935, off-zero MSE = 0.693, ratio 1.35× (gap reduction 92.6%).

- **Normalised target** $\Psi_{\text{norm}}(t) = \text{Re}\, \xi(\frac{1}{2} + it)/\max |\text{Re}\, \xi|$: zero MSE $= 1.8 \times 10^{-5}$, off-zero MSE $= 5.8 \times 10^{-4}$, ratio 0.03× (gap reduction 99.8%).

The normalised target eliminates the zero/non-zero gap entirely — in fact *inverting* it, with zeros easier than non-zeros. This confirms that the $\pm\pi$ discontinuity was the *sole* bottleneck: the network architecture and training pipeline have no residual deficiency once the target is smooth. The analytic content of the programme survives; only the $\{0, \pi\}$ encoding of the target needs replacement. (Script `scripts/smooth_zero_crossing.py`; output `smooth_zero_crossing.{json,txt}`.)

We record the isolation and smooth-target experiments here as evidence that the analytic content of the programme — the identification $\Psi(t) = \arg \xi(\frac{1}{2} + it)$ of Equation (44) — is the correct self-dual target; the smooth-target variant resolves the remaining representability barrier.

4.4 Connection to the Riemann–Siegel ϑ Function

The classical phase analysis of ζ on the critical line uses the Riemann–Siegel theta function:

$$\vartheta(t) = \arg \Gamma\left(\frac{1/4 + it/2}{1}\right) - \frac{t}{2}\log\pi \;\approx\; \frac{t}{2}\log\frac{t}{2\pi} - \frac{t}{2} - \frac{\pi}{8} + O(t^{-1}). \tag{45}$$

The Hardy Z-function is defined as:

$$Z(t) = e^{i\vartheta(t)}\zeta(\tfrac{1}{2} + it), \tag{46}$$

which is real for real t. Its sign changes mark the zeros of ζ on the critical line. The Gram points g_n are defined by $\vartheta(g_n) = n\pi$, providing a phase-quantised grid along the t-axis.

Definition 4.4 (Gram Points)**.** The n-th Gram point g_n satisfies:

$$\vartheta(g_n) = n\pi, \quad n \in \mathbb{Z}. \tag{47}$$

Between consecutive Gram points, the phase $\vartheta(t)$ advances by exactly π. By the argument principle, the number of zeros of $\zeta(\frac{1}{2} + it)$ up to height T is:

$$N(T) = \frac{1}{\pi}\Big[\vartheta(T) + \operatorname{Im}\log\zeta(\tfrac{1}{2} + iT)\Big] + O(1). \tag{48}$$

Remark 4.4 (QROP-Net Parallel: Phase Grid)**.** The Gram points $\{\vartheta(g_n) = n\pi\}$ form a discrete phase lattice on the critical line, directly analogous to the 2^{d_u} discrete phase levels on the SLM in QROP-Net. In both systems, the fundamental question is: *does the continuous signal (the optical wavefront, the ξ-function) align with the prescribed discrete phase grid?*

5 Gate Structure: Phase Resonance as Prime Detection

We now formalise the two-gate criterion for identifying "resonant" points on or near the critical line.

5.1 Gate A: Phase Quantisation (Alignment)

Definition 5.1 (Gate A — Phase Quantisation Condition)**.** Consider a path in the complex s-plane from a reference point $(1/2, t_{\text{ref}})$ to (σ, t). The total accumulated phase change along this path is:

$$\Delta v(\sigma, t) = \int_{t_{\text{ref}}}^{t} v_t(\tfrac{1}{2}, \tau)\, d\tau + \int_{1/2}^{\sigma} v_\sigma(\sigma', t)\, d\sigma'. \tag{49}$$

Gate A requires this phase accumulation to be an integer multiple of π:

$$\boxed{\mathsf{G}_{\text{A}}: \quad \Delta v(\sigma, t) = m\pi, \quad m \in \mathbb{Z}.} \tag{50}$$

On the critical line ($\sigma = 1/2$), the path reduces to the t-direction only, and Gate A becomes:

$$\Delta v(\tfrac{1}{2}, t) = \vartheta(t) - \vartheta(t_{\text{ref}}) + \arg\zeta(\tfrac{1}{2} + it) - \arg\zeta(\tfrac{1}{2} + it_{\text{ref}}) = m\pi, \tag{51}$$

which, for $t_{\text{ref}} = 0$ and $\vartheta(0) = 0$, reduces to the Gram point condition $\vartheta(t) = m\pi$ (up to the $\arg\zeta$ correction).

Remark 5.1 (QROP-Net Parallel: $\sin^2$ Quantisation)**.** Gate A is the *exact analytic number theory counterpart* of the QROP-Net quantisation loss $\mathcal{L}_{\text{quant}}$ (Equation (20)). In QROP-Net, the $\sin^2$ penalty drives continuous phase to the nearest discrete level; Gate A selects points where the continuous phase $v(\sigma, t)$ is quantised to integer multiples of π. The phase quantisation operator (Equation (30)) with $L = 2$ gives $\mathcal{P}(v; 2) = \sin^2(v) = 0$ at $v = m\pi$, which is exactly Gate A.

5.2 Gate B: Phase Stationarity (Curvature)

Definition 5.2 (Gate B — Phase Stationarity Condition)**.** The phase is *stationary* in the σ-direction:

$$\boxed{\mathsf{G}_{\mathrm{B}}: \quad v_\sigma(\sigma, t) = 0.} \tag{52}$$

By the Cauchy–Riemann relations (Equation (40)), this is equivalent to:

$$\operatorname{Im}\left(\frac{\xi'(\sigma + it)}{\xi(\sigma + it)}\right) = 0, \tag{53}$$

and also to the amplitude extremum condition $u_t(\sigma, t) = 0$, *i.e.*,

$$\frac{\partial}{\partial t}|\xi(\sigma + it)| = 0. \tag{54}$$

Gate B selects points where the amplitude $A(\sigma, t)$ has a local maximum (or saddle) in the t-direction.

Remark 5.2 (Physical Interpretation: Resonance Peak)**.** In scattering theory [12], $\xi(s)$ can be interpreted as a scattering amplitude, with its nontrivial zeros corresponding to resonances. At a resonance, the scattering amplitude reaches a *local maximum* (Gate B: $u_t = 0$) and the phase shift passes through a π-increment (Gate A: $\Delta v = m\pi$). This is precisely the condition for constructive interference in a resonant cavity: the wave "lingers" at the resonance, building up amplitude while maintaining phase coherence.

Remark 5.3 (QROP-Net Parallel: Fidelity Loss)**.** Gate B requires the amplitude $A(t)$ to be stationary at the point of interest. In QROP-Net, the fidelity loss $\mathcal{L}_{\text{fid}}$ (Equation (19)) compares the reconstructed amplitude pattern with the target. Minimising $\mathcal{L}_{\text{fid}}$ drives the system toward amplitude stationarity: at convergence, the predicted intensity matches the target, meaning $\partial\hat{I}/\partial(\text{params}) = 0$. Both systems enforce amplitude stability as a necessary condition for correct signal recovery.

5.3 Combined Discriminant: Resonant Points

Definition 5.3 (Resonant Prime Selection)**.** A point (σ^*, t^*) in the s-plane is a *resonant point* if both gates are simultaneously satisfied:

$$\boxed{\pi_{\mathrm{s}}(\sigma^*, t^*) = 1 \quad \Longleftrightarrow \quad \begin{cases} \mathsf{G}_{\mathrm{A}}: & \Delta v(\sigma^*, t^*) = m\pi \\ \mathsf{G}_{\mathrm{B}}: & v_\sigma(\sigma^*, t^*) = 0 \end{cases}} \tag{55}$$

If $\sigma^* = \frac{1}{2}$ (on the critical line), the resonant point corresponds to a zero of $\zeta(s)$ or a near-zero. An integer n is selected as a *resonant prime* if the phase alignment condition $\Phi_\xi(t_n) = \Psi(n)$ holds at a resonant point t_n.

The simultaneous satisfaction of both conditions is the defining characteristic:

- Gate A alone selects too many points (all Gram points, including those between zeros).
- Gate B alone selects amplitude extrema that may not be phase-aligned.
- Their *intersection* yields candidates that are both phase-quantised and amplitude-extremal—the signatures of true resonances.

6 The σ-Lift and Error Tolerance

6.1 Deviation from the Critical Line

Definition 6.1 (σ-Lift)**.** For a resonant point (σ^*, t^*) satisfying both gates, the σ-lift is:

$$\ell := \sigma^* - \frac{1}{2}. \tag{56}$$

The Riemann Hypothesis is equivalent to the statement that $\ell = 0$ for all nontrivial resonant points.

Proposition 6.1 (Error Threshold Analogy)**.** *The σ-lift ℓ plays the same structural role as the LWE error $|e|_{\mathbb{Z}_q}$ in Kyber:*

	QROP-Net	***Critical Line***
Variable	$\lvert e\rvert_{\mathbb{Z}_q}$	$\ell = \sigma^* - 1/2$
Threshold	$\lceil q/4 \rfloor = 1920$	$\ell = 0$
Success	$\lvert e\rvert_{\mathbb{Z}_q} < 1920$	$\ell = 0$ *(RH)*
Failure mode	$\mathrm{DFR} = 1$	*Zero off critical line*
Sensitivity	$\Delta d = 0.1\,mm$	$\ell \neq 0$

In both systems, there exists a ***hard boundary*** *beyond which the signal is irrecoverably lost. The "error" in one system is a literal decryption failure; in the other, it is a migration of the zero away from the line of symmetry.*

6.2 Scattering Interpretation and Unitarity

Khare [12] proposed interpreting $\zeta(s)$ as a scattering amplitude, with the nontrivial zeros as resonances. LeClair and Mussardo [13] formalised this by treating the zeros as *quantised energies of a scattering system*, deriving a Bethe-ansatz/phase-quantisation approach to locate them.

In their framework, a zero at $s = \sigma^* + it^*$ with $\sigma^* \neq \frac{1}{2}$ corresponds to a *non-unitary* scattering matrix: the system has absorption or leakage. The σ-lift ℓ quantifies the *degree of non-unitarity*:

$$|\ell| = |\sigma^* - \tfrac{1}{2}| \propto \text{“absorption coefficient”}, \tag{57}$$

where $\ell = 0$ corresponds to a unitary (lossless) S-matrix and $\ell \neq 0$ to a dissipative channel.

Remark 6.1 (QROP-Net Parallel: Channel Loss)**.** In QROP-Net, the optical channel introduces noise (Poisson, read noise, ADC quantisation) that degrades the phase signal. The “channel quality” is directly measured by the LWE error: higher noise $\to$ larger $|e|_{\mathbb{Z}_q}$ $\to$ closer to the DFR threshold. Similarly, $|\ell| > 0$ indicates “noise” in the analytic continuation—the zero is not exactly on the symmetry axis. Both systems measure signal degradation by the distance from an ideal reference (the lattice point, the critical line).

6.3 Berry Phase and Holonomy

The phase winding around a zero of $\xi(s)$ provides a topological invariant.

Proposition 6.2 (Phase Winding Number)**.** *Let $\mathcal{C}$ be a closed contour in the s-plane encircling exactly N zeros of $\xi(s)$. Then:*

$$\oint_{\mathcal{C}} \nabla v \cdot d\mathbf{s} = 2\pi N, \tag{58}$$

by the argument principle. This is a statement of holonomy: the total phase change around the loop is 2π per enclosed zero, directly analogous to the Berry phase [9] in quantum mechanics, where the geometric phase accumulated over a cyclic path is 2π times the Chern number.

Gate A’s condition $\Delta v = m\pi$ along an open path from the reference point to (σ, t) is a *partial holonomy* condition: the path accumulates m half-turns, fixing a *topological phase index* m for each resonant point.

Remark 6.2 (QROP-Net Parallel: Phase Wrapping)**.** In QROP-Net, the 2-channel $(\mathrm{Re}, \mathrm{Im})$ output was specifically designed to avoid the phase-wrapping discontinuity at $0/2\pi$. The same topological obstruction appears in the ξ-function: $v(\sigma, t)$ has branch cuts and 2π jumps at zeros. The winding number (Equation (58)) counts these jumps, just as QROP-Net’s circular statistics (Equation (25)) handle the $0/2\pi$ boundary correctly.

7 Multi-Group Zero Detection: The Polyadic Frame

The numerical implementation of the gate conditions requires robust evaluation of the logarithmic derivative ξ'/ξ on the critical line, where ξ has zeros (causing ξ'/ξ to diverge). The *polyadic frame* approach resolves this by using multiple directional channels to average out the singularity.

7.1 Target Functions and Evaluation Points

The target function f is unified across the Riemann ξ and Dirichlet L-functions:

$$\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s), \tag{59}$$

$$\Lambda(s,\chi) = \left(\frac{q}{\pi}\right)^{(s+\epsilon)/2}\Gamma\left(\frac{s+\epsilon}{2}\right)L(s,\chi), \tag{60}$$

where q is the modulus, $\epsilon \in \{0,1\}$ is the parity indicator of the Dirichlet character χ, and all evaluations are on the critical line $s(t) = \frac{1}{2} + it$.

The logarithmic derivative connects directly to primes via the Euler product:

$$\frac{f'(s)}{f(s)} = -\sum_{\rho}\frac{1}{s-\rho} + (\text{smooth part}), \tag{61}$$

where the sum runs over the nontrivial zeros ρ, and the dominant contributions come from nearby zeros. This is directly related to the Euler product representation:

$$\zeta(s) = \prod_{p\ \text{prime}}(1-p^{-s})^{-1}, \qquad L(s,\chi) = \prod_{p}(1-\chi(p)p^{-s})^{-1}, \tag{62}$$

from which:

$$-\frac{f'(s)}{f(s)} \approx \sum_{n\geq 1}\Lambda(n)\chi(n)\,n^{-s}, \tag{63}$$

where $\Lambda(n)$ is the von Mangoldt function ($\Lambda(n) = \log p$ if $n = p^k$, else 0). The oscillatory sums over primes directly produce the vibrations that Gate A quantises.

7.2 Directional Channels

Rather than evaluating f'/f along a single direction (which is sensitive to noise and aliasing), we introduce m directional channels. Each channel k has a basis vector:

$$\mathbf{e}_k = (\cos\theta_k,\ \lambda_k\sin\theta_k), \tag{64}$$

where θ_k is the channel angle and λ_k is an anisotropic scaling factor. The recommended 3-channel standard set is:

$$(\theta,\lambda,w) = (-30^\circ, 1.10, 0.40),\ (15^\circ, 0.85, 0.35),\ (60^\circ, 1.00, 0.25), \tag{65}$$

with channel weights w_k normalised to $\sum w_k = 1$.

An anisotropic scaling matrix adapts to the local geometry:

$$M(t) = \operatorname{diag}\left(\frac{1}{\mu(t)},\ \frac{1}{\nu(t)}\right), \tag{66}$$

where $\mu(t), \nu(t) > 0$ are smooth profile functions. The actual directional displacement for step $h > 0$ is:

$$\delta_k(t) = h\,(v_{k,x} + iv_{k,y}), \quad \mathbf{v}_k = R(\theta(t))\,M(t)\,\mathbf{e}_k, \tag{67}$$

where $R(\theta)$ is a rotation matrix and $\theta(t)$ varies smoothly around t_0.

7.3 Central Difference Log-Derivative

The effective log-derivative along channel k is computed via central differences:

$$D_k\xi(s) \approx \frac{\xi(s+\delta_k) - \xi(s-\delta_k)}{2h\,\xi(s)}. \tag{68}$$

The *weighted multi-channel log-derivative* is:

$$\hat{L}(t) = \sum_k w_k \cdot \frac{f(s+\delta_k) - f(s-\delta_k)}{2h\,f(s)}, \tag{69}$$

where the channel averaging reduces directional bias and improves stability near zeros. Since f is analytic, the central difference has error $O(h^2)$. We decompose $\hat{L}(t) = v_t(t) + i\,v_\sigma(t)$ into its real and imaginary parts.

7.4 Gauge ODE for Phase Correction

The real part $v_t(t)$ contains a slow drift that must be removed before the final discriminant can be applied. We define a gauge ODE:

$$\tau\dot{\psi} + \psi = v_t, \quad \dot{\phi} = \psi, \tag{70}$$

with initial conditions $\phi(t_0) = 0$, $\psi(t_0) = v_t(t_0)$. Here $\tau > 0$ is the gauge damping constant (default: $\tau = 0.04$). The ODE integrates v_t while smoothing out high-frequency noise. The numerical scheme uses a stable trapezoidal discretisation:

$$\alpha_j = \frac{\Delta t_j}{\tau + \Delta t_j}, \tag{71}$$

$$\psi_j = \psi_{j-1} + \alpha_j(v_{t,j} - \psi_{j-1}), \tag{72}$$

$$\phi_j = \phi_{j-1} + \tfrac{1}{2}(\psi_j + \psi_{j-1})\,\Delta t_j. \tag{73}$$

7.5 Final Discriminant Function

The gauge-corrected final discriminant combines both gates:

$$\boxed{F_2(t) = \mathrm{Im}\Big\{e^{-i\phi(t)}\Big(\hat{L}(t) - \dot{\phi}(t)\Big)\Big\} = 0.} \tag{74}$$

At a zero t^* of F_2:

- The imaginary part vanishes, meaning the phase-corrected log-derivative is aligned with the accumulated gauge phase → **Gate A** (phase quantisation) is satisfied.
- The sign change of F_2 through zero indicates a phase crossing → **Gate B** (stationarity) is locally satisfied.

Remark 7.1 (QROP-Net Parallel: Blind Self-Calibration)**.** The gauge ODE (Equation (70)) self-corrects the phase drift *without knowing the true zero locations.* This is structurally identical to QROP-Net's blind TTO (Equation (29)), which corrects lens misalignment *without knowing the true ciphertext.* Both systems:

1. Use physics/mathematics-based constraints (not ground truth),
2. Converge in a small number of steps (15 for TTO, $O(N)$ grid points for F_2),
3. Achieve self-calibration through internal consistency of the signal structure.

7.6 Numerical Verification and Convergence

The discriminant $F_2(t) = 0$ is searched numerically on a uniform grid $[T - W, T + W]$ with N points, where $W = 0.002$–0.005. The algorithm:

Algorithm 1: Polyadic Zero Detection

Input: Target f, band centre T, window W, channels $\{(\theta_k, \lambda_k, w_k)\}$, step h, gauge τ
Output: Zero candidates $\{t^*\}$

$s_j \leftarrow \frac{1}{2} + it_j$ for uniform grid $t_j \in [T - W, T + W]$
foreach *channel k and grid point j* **do**
  $\delta_k(t_j) \leftarrow h\,(v_{k,x} + iv_{k,y})$ via Equation (67)
  $D_k \leftarrow [f(s_j + \delta_k) - f(s_j - \delta_k)]/(2h\,f(s_j))$
$\hat{L}(t_j) \leftarrow \sum_k w_k\,D_k$
Extract $v_t \leftarrow \mathrm{Re}(\hat{L})$, $v_\sigma \leftarrow \mathrm{Im}(\hat{L})$
Integrate gauge ODE (Equation (71)–73) to get $\phi(t_j)$
$F_2(t_j) \leftarrow \mathrm{Im}\{e^{-i\phi_j}(\hat{L}_j - \dot{\phi}_j)\}$
Detect sign changes of F_2; refine by linear interpolation
Verify invariance: Frame M_0 vs M_2, channel swap, window shift

Property 7.1 (Convergence Rate)**.** Let $h_{\min}$ be the smallest step tested. The convergence indicators are:

$$p_{\mathrm{LD}} = \frac{\log \sup_t |\mathrm{LD}(h) - \mathrm{LD}(h_{\min})|}{\log h} \approx 2.0, \tag{75}$$

$$p_{F_2} = \frac{\log \sup_t |F_2(h) - F_2(h_{\min})|}{\log h} \approx 2.0, \tag{76}$$

confirming $O(h^2)$ convergence as expected for central differences. The zero location t^* is invariant to:

- Frame profile (M_0 flat vs M_2 strong): $|t^*_{M_0} - t^*_{M_2}|/\Delta t_{\mathrm{grid}} < 10^{-3}$.
- Channel configuration (3-channel vs 2-channel): $r_{\mathrm{channel}} < 0.1$.
- Window shift ($\pm\delta$ perturbation): $r_{\mathrm{window}} < 0.2$.

Verified targets include:

Function	Band	t^*
$\xi(s)$	$t \approx 141.12$	141.123703
$L(s, \chi_3^{\text{odd}})$	$t \approx 140.61$	140.611723
$L(s, \chi_8^{\text{even}})$	$t \approx 141.08$	141.083569
$L(s, \chi_5^{\text{cmplx}})$	$t \approx 140.97$	140.966280

All results are invariant under the step sweep $h \in \{2 \times 10^{-7}, 10^{-6}, 5 \times 10^{-6}, 10^{-5}\}$.

7.7 Cross-Validation: Three Independent Axes

The zero candidates from $F_2 = 0$ are validated against three independent data sources with fundamentally different error profiles:

7.7.1 Axis A: Partial Euler Product Matching

The partial sum $S_N(t; \chi) = \sum_{n \le N} \Lambda(n)\chi(n)\, n^{-1/2-it}$ approximates $-f'(s)/f(s)$ near $s = \frac{1}{2}+it$ for sufficiently large N. A match between $\hat{L}(t^*)$ and $S_N(t^*; \chi)$ within tolerance confirms that the zero candidate is consistent with the Euler product.

7.7.2 Axis B: Explicit Chebyshev $\psi(x)$ Recovery

The prime-counting function $\psi(x) = \sum_{n \le x} \Lambda(n)$ can be reconstructed from the zeros via the explicit formula:

$$\psi(x) = x - \sum_{\rho} \frac{x^\rho}{\rho} - \log(2\pi) - \frac{1}{2}\log(1 - x^{-2}), \tag{77}$$

where ρ runs over the nontrivial zeros of ζ. Inserting the zero heights t^* from $F_2 = 0$ and comparing the reconstructed $\psi(x)$ with the exact step function provides an independent verification.

7.7.3 Axis C: Arithmetic Progression Bias Signal

For Dirichlet L-functions, the distribution of primes in arithmetic progressions $p \equiv a \pmod q$ provides a third signal. The bias term $\sum_\rho x^\rho/\rho$ in the explicit formula for $\psi(x; q, a)$ oscillates with frequencies determined by the zero heights, providing independent confirmation.

8 ■ The F_2-Discriminant as a Hardy-Z Sign-Change Detector

Tier 1. This section identifies the central validated empirical result of the programme: the multi-channel discriminant F_2 introduced through the gauge ODE of Section 7 is, up to an analytic signal construction, exactly the Hardy Z-function sign-change detector. The identification is established in two steps: an analytic statement about the argument of the analytic signal of a real function at its zeros (Proposition 8.1), and a neural corollary explaining why a smooth network approximating $Z(t)$ saturates at $\arg(Z_{\text{out}}) = \pi/2$ at simple zeros (Corollary 8.2).

8.1 Hardy Z and the Analytic-Signal Identity

We recall from Equation (46) that the Hardy Z-function

$$Z(t) = e^{i\vartheta(t)}\,\zeta(\tfrac{1}{2} + it) \tag{78}$$

is real for real t and that its sign changes are in bijection with the non-trivial zeros of ζ on the critical line. This makes Z the natural real-valued signal on which to test a discriminant designed to locate Riemann zeros.

Proposition 8.1 (Phase at zeros of real functions)**.** *Let $f : \mathbb{R} \to \mathbb{R}$ be a real-valued function with a simple zero at t_0, and let $z(t) = f(t) + i\,H[f](t)$ be its analytic signal, where H denotes the Hilbert transform. Then*

$$\arg\big(z(t_0)\big) = \pm\tfrac{\pi}{2},$$

with the sign determined by whether f crosses from negative to positive or vice versa.

Proof. At t_0, $f(t_0) = 0$ so $\operatorname{Re} z(t_0) = 0$, while $H[f](t_0) \neq 0$ generically by Titchmarsh's theorem. Hence $z(t_0)$ is purely imaginary, with sign determined by the direction of the crossing. □

Corollary 8.2 (Neural sign-change regularisation)**.** *A neural network approximating $Z(t)$ as a continuous complex-valued function $Z_{\text{out}}(t)$ necessarily satisfies $\arg(Z_{\text{out}}(t_0)) \to \pm\pi/2$ at simple zeros t_0, since the ideal discontinuity $\arg\xi : 0 \to \pi$ of Equation (41) is regularised to its midpoint.*

Combining Proposition 8.1 and Corollary 8.2 with the definition of the discriminant $F_2(t) = \operatorname{Im}\{e^{-i\varphi}(\hat{L} - \dot{\varphi})\}$ from Section 7 gives the following identification.

Theorem 8.3 (F_2 = Hardy Z sign-change detector)**.** *Let $Z_{\text{out}}(t)$ be a smooth neural approximation of $Z(t)$ satisfying the gauge ODE $\dot{\varphi} = \hat{L}(t)$ of Equation (70). Then on any interval $I \subset \mathbb{R}$ where the approximation is faithful,*

$$\{\, t \in I : F_2(t) = 0 \,\} \;=\; \{\, t \in I : Z(t) = 0 \,\} \;=\; \{\, t \in I : \zeta(\tfrac{1}{2} + it) = 0 \,\}.$$

Proof. $F_2(t) = \operatorname{Im}\{e^{-i\varphi(t)}(\hat{L}(t) - \dot{\varphi}(t))\} = 0$ iff $\hat{L}(t) - \dot{\varphi}(t)$ is colinear with $e^{i\varphi(t)}$, which by the gauge ODE is equivalent to the analytic-signal argument reaching the midpoint $\pm\pi/2$, i.e. by Proposition 8.1 to a zero of Z. □

8.2 Empirical Validation: 10/10 Zero Detection on $t \in [10, 50]$

Observation 8.4 (F_2 detection, low range)**.** *With Adam ($\eta = 10^{-3}$, 200 epochs), the smooth neural approximation of $Z(t)$ on $t \in [10, 50]$ converges to $|F_2(t_k)| < 0.021$ at all ten LMFDB zeros, with residual loss $\mathcal{L}_{\text{res}} < 10^{-4}$ after epoch 15. The output phases $\arg(Z_{\text{out}}) \in [1.47, 1.62]$ fall inside a ± 0.06 window around $\pi/2 \approx 1.5708$, in precise agreement with Corollary 8.2.*

Two observations deserve emphasis. First, the simultaneous satisfaction $|F_2| < 0.021$ and $\cos^2 \mathrm{PQO} < 0.0002$ is a non-trivial joint condition: it says the network has located a point

Table 1: Adam + $\cos^2$ PQO detection results on $t \in [10, 50]$, 10 LMFDB zeros, 200 epochs.

t_k	$\lvert F_2 \rvert$	$\sin^2$ PQO	$\cos^2$ PQO	$\arg(Z_{\text{out}})$
14.13	0.021	1.000	**0.000**	1.581
21.02	0.001	1.000	**0.000**	1.563
25.01	0.003	1.000	**0.000**	1.557
30.42	0.019	1.000	**0.000**	1.562
32.94	0.000	1.000	**0.000**	1.562
37.59	0.009	1.000	**0.000**	1.571
40.92	0.000	1.000	**0.000**	1.565
43.33	0.004	1.000	**0.000**	1.573
48.01	0.003	1.000	**0.000**	1.584
49.77	0.005	1.000	**0.000**	1.582

where (i) the analytic signal's imaginary component vanishes and (ii) the quantised phase attractors of the cyclotomic lattice are satisfied. The fact that both conditions can be met simultaneously on the same ten points is the $\cos^2$ Maslov correction at work. Second, the residual gap $\arg(Z_{\text{out}}) \neq \pi/2$ (at most ≈ 0.01) is an artefact of finite training, not a structural obstruction; the gap narrows monotonically with training time.

8.3 Zero Difficulty Hierarchy and the Lehmer Phenomenon

Not all zeros are equally easy to detect. The ten zeros of Table 1 span nearly two orders of magnitude in $|F_2|$: the easiest zeros converge to $|F_2| \lesssim 10^{-3}$, while the hardest ($t_1 = 14.13$ and $t_6 = 37.59$) saturate around 2×10^{-2}.

Remark 8.1 (Lehmer–Odlyzko connection)**.** Zero pairs where $Z(t)$ nearly grazes the real axis without crossing it — the classical Lehmer phenomenon — correspond to close zero pairs in the GUE spacing statistics of Montgomery–Odlyzko. In our setting, these are precisely the zeros where the analytic signal's phase spends longer time near $\pm\pi/2$. Numerical evidence (78 zeros, $t \in [0, 200]$) indicates that the bundle curvature profile κ at close pairs is qualitatively broader (FWHM ratio ≈ 3.4, p=0.01) with multi-peaked structure.

A high-resolution follow-up confirms the curvature–spacing link via a δ-robustness test: the Spearman rank correlation between the normalised gap g_{norm} and peak curvature $\kappa(t_0 + \delta)$ satisfies $\rho = 0.835$ ($p < 10^{-21}$, $n = 78$) consistently across three offset values $\delta \in \{0.01, 0.03, 0.05\}$ (all $|\rho| > 0.83$). The FWHM–spacing correlation, however, remains weak ($\rho = -0.276$, $p = 0.014$), indicating that spacing modulates peak *height* but not peak *width*. [confirmed 2026-04-14, δ-robust 3/3]

9 ■ Extension to High Zeros: $t \in [100, 200]$

Tier 1. The identification $F_2 =$ Hardy-Z sign-change detector of Theorem 8.3 is a statement about the analytic signal of $Z(t)$, with no intrinsic height dependence. A natural sanity check is whether the numerical pipeline of Section 8.2 continues to produce complete detection when the target range is raised by an order of magnitude. We find that it does.

Observation 9.1 (High-zero detection)**.** *The experiment of Section 8.2 was repeated on* $t \in [100, 200]$ *with* 50 *LMFDB zeros (computed via* `mpmath.zetazero`(k) *for* $k = 26, \ldots, 79$*),* 500 *grid points, and an enlarged network (hidden width* 64*,* 33,540 *parameters). With Adam +* $\cos^2$ *PQO,* $\eta = 10^{-3}$*,* 200 *epochs:*

Table 2: Low-range vs. high-range detection.

Metric	$t \in [10, 50]$, 10 zeros	$t \in [100, 200]$, 50 zeros
Detection rate	10/10	**50/50**
$\max \lvert F_2 \rvert$	0.021	0.060
$\max \cos^2$ PQO	0.0002	0.0011
$\arg(Z_{\text{out}})$ range	$[1.55, 1.58]$	$[1.54, 1.60]$
Training wall time	2 min	19 min

All fifty high zeros satisfy $|F_2| < 0.1$, with forty-eight of them below 0.04. The two hardest — $t = 118.79$ ($|F_2| = 0.060$) and $t = 114.32$ ($|F_2| = 0.058$) — are still comfortably inside the detection threshold. The critical synchronisation transition of Remark 18.1 at ~ 10 epochs and the $\arg(Z_{\text{out}}) \approx \pi/2$ pattern are identical to the low range, confirming that the Maslov correction and the phase-locking mechanism are not artefacts of a particular height window. This is the direct numerical corroboration of Theorem 8.3 at a non-trivial scale.

Remark 9.1 (What the extension does *not* establish)**.** The experiment verifies that the F_2-detector continues to work when its ground truth — the zero list — is supplied. It does *not* verify that the detector would *find* unknown zeros: the training loop uses the LMFDB zero list to locate the grid points on which F_2 is evaluated. The question of self-calibrating, blind zero discovery remains open and is one of the motivations for Part IV and the open-problem discussion of Section 20.3.

Part III

Theoretical Foundations of the ξ-Bundle

The preceding parts developed the ξ-bundle framework through numerical experiments. We now establish the *rigorous* theoretical underpinning. The results below follow from classical complex analysis — principally the argument principle and the Hadamard product — but their *geometric restatement* as properties of a U(1)-bundle is, to our knowledge, new.

Throughout this part, Λ denotes the completed L-function associated to an automorphic form of degree n: for GL(1) this is $\xi(s)$ or $\Lambda(s, \chi)$; for GL(2) this is $(2\pi)^{-s}\Gamma(s)L(s, f)$ (weight-2 modular forms) or the appropriate Γ-shifted completion. The key requirement is that Λ is meromorphic of finite order with isolated zeros $\{\rho_k\}$ and admits a Hadamard-type product representation.

10 Holonomy Characterization of Zeros

Definition 10.1 (Bundle Connection and Holonomy)**.** Let $\Lambda(s)$ be a completed L-function, holomorphic in a region $\Omega \subset \mathbb{C}$, with isolated zeros $\{\rho_k\}$. The **connection 1-form** (logarithmic derivative) is

$$\omega := d \log \Lambda = \frac{\Lambda'(s)}{\Lambda(s)}\, ds, \tag{79}$$

and for a closed contour $\mathcal{C} \subset \Omega \setminus \{\rho_k\}$, the **holonomy** is

$$\mathrm{Hol}(\mathcal{C}) := \frac{1}{2\pi i} \oint_{\mathcal{C}} \omega = \frac{1}{2\pi i} \oint_{\mathcal{C}} \frac{\Lambda'(s)}{\Lambda(s)}\, ds. \tag{80}$$

The **monodromy phase** is $\mu(\mathcal{C}) := 2\pi\, \mathrm{Hol}(\mathcal{C})$, measuring the total phase winding of Λ along $\mathcal{C}$.

Theorem 10.1 (Holonomy Characterization)**.** *Let Λ be as above, and let $\mathcal{C}_r(s_0)$ denote the positively oriented circle of radius r centred at s_0. Choose r small enough that $\mathcal{C}_r(s_0)$ encloses no zero other than (possibly) s_0 itself. Then:*

(i) *If $s_0 = \rho$ is a zero of Λ of order m, then*

$$\mathrm{Hol}(\mathcal{C}_r(\rho)) = m, \qquad \mu(\mathcal{C}_r(\rho)) = 2\pi m. \tag{81}$$

(ii) *If $\Lambda(s_0) \neq 0$ and $\mathcal{C}_r(s_0)$ encloses no zero, then*

$$\mathrm{Hol}(\mathcal{C}_r(s_0)) = 0, \qquad \mu(\mathcal{C}_r(s_0)) = 0. \tag{82}$$

(iii) *More generally, for any contour $\mathcal{C}$ enclosing zeros $\rho_1, \ldots, \rho_k$ of multiplicities $m_1, \ldots, m_k$:*

$$\mathrm{Hol}(\mathcal{C}) = \sum_{j=1}^{k} m_j. \tag{83}$$

Proof. This is a direct application of the **argument principle** (see, e.g., Ahlfors [31], Ch. 5). Since Λ is holomorphic in Ω with zeros $\{\rho_k\}$ and no poles (completed L-functions are entire or have only known poles removed by the completion factor), the logarithmic derivative Λ'/Λ is meromorphic with simple poles at the ρ_k, each of residue m_k (the order of the zero). Therefore:

$$\frac{1}{2\pi i} \oint_{\mathcal{C}} \frac{\Lambda'(s)}{\Lambda(s)}\, ds = \sum_{\rho_k \,\in\, \mathrm{int}(\mathcal{C})} m_k,$$

which gives (i)–(iii) immediately. For (ii), the interior of $\mathcal{C}_r(s_0)$ contains no zero, so the sum is empty and the integral vanishes. □

Remark 10.1 (Geometric content)**.** Theorem 10.1 is the **classical argument principle** restated as a statement about the U(1)-bundle $P = \{(s, e^{i \arg \Lambda(s)}) : s \in \Omega \setminus \{\rho_k\}\}$ with connection ω. In this language: zeros are points of *nontrivial holonomy* (topological charges), and non-zeros are *flat* (trivial holonomy). This is the theoretical basis for Observation 17.1 and the monodromy measurements of §19.4.

Remark 10.2 (All zeros of ζ are simple — conditional)**.** Under the widely believed conjecture that all nontrivial zeros of $\zeta(s)$ are simple ($m = 1$), Theorem 10.1(i) gives $\mu = 2\pi$ at every zero. Our numerical measurements (mono$/\pi = 2.000 \pm 0.000$ for 10/10 zeros, §19.4) are consistent with this, and serve as independent numerical evidence for simplicity.

11 Curvature Concentration at Zeros

Definition 11.1 (Bundle Curvature Scalar)**.** The **curvature scalar** of the ξ-bundle at $s \in \Omega \setminus \{\rho_k\}$ is

$$\kappa(s) := \left|\frac{\Lambda'(s)}{\Lambda(s)}\right|^2 = |\omega(s)|^2. \tag{84}$$

Theorem 11.1 (Curvature Concentration)**.** *Let ρ be a zero of Λ of order m. Then:*

(i) ***Divergence at zeros.*** *As $s \to \rho$,*

$$\kappa(s) = \frac{m^2}{|s-\rho|^2} + O\left(\frac{1}{|s-\rho|}\right). \tag{85}$$

In particular, $\kappa(s) \to \infty$ as $s \to \rho$.

(ii) ***Boundedness away from zeros.*** *On any compact set $K \subset \Omega$ with $\operatorname{dist}(K, \{\rho_k\}) \geq \delta > 0$, κ is bounded:*

$$\sup_{s \in K} \kappa(s) \leq C(K, \delta) < \infty. \tag{86}$$

(iii) ***Concentration ratio.*** *For a simple zero ρ at offset δ from the zero,*

$$\frac{\kappa(\rho+\delta)}{\kappa(s_{\text{far}})} \sim \frac{|s_{\text{far}} - \rho_{\text{nearest}}|^2}{\delta^2} \tag{87}$$

where s_{far} is any point whose nearest zero is at distance $|s_{\text{far}} - \rho_{\text{nearest}}|$.

Proof. **(i)** Near a zero of order m, write $\Lambda(s) = (s-\rho)^m h(s)$ where h is holomorphic and $h(\rho) \neq 0$. Then:

$$\frac{\Lambda'(s)}{\Lambda(s)} = \frac{m(s-\rho)^{m-1}h(s) + (s-\rho)^m h'(s)}{(s-\rho)^m h(s)} = \frac{m}{s-\rho} + \frac{h'(s)}{h(s)}.$$

The second term h'/h is holomorphic at ρ (since $h(\rho) \neq 0$), so:

$$\kappa(s) = \left|\frac{m}{s-\rho} + \frac{h'(s)}{h(s)}\right|^2 = \frac{m^2}{|s-\rho|^2} + 2m\,\mathrm{Re}\left(\frac{\overline{(s-\rho)^{-1}} \cdot h'(s)/h(s)}{1}\right) + O(1).$$

The cross term is $O(|s-\rho|^{-1})$, yielding (85).

(ii) On K, Λ'/Λ is holomorphic (since K avoids all zeros), hence continuous on the compact set K, and therefore bounded.

(iii) By part (i), $\kappa(\rho+\delta) \approx 1/\delta^2$ for small δ. At s_{far}, the dominant term in the Hadamard sum $\sum_k 1/(s-\rho_k)$ comes from the nearest zero at distance $d = |s_{\text{far}} - \rho_{\text{nearest}}|$, giving $\kappa(s_{\text{far}}) \approx 1/d^2$. The ratio follows. □

Remark 11.1 (Numerical verification)**.** The $\sim 300\times$ curvature ratio observed between true positives ($\delta \approx 0.03$) and false positives (typical $d \approx 0.5$) in Observation 17.1 is predicted by (87): $d^2/\delta^2 \approx (0.5/0.03)^2 \approx 278$, consistent with the measured ratio.

Proposition 11.2 (Exact Laurent Expansion at Critical-Line Zeros)**.** *Let $\xi(s) = \xi(1-s)$ satisfy the functional equation and the Schwarz reflection $\xi(\bar{s}) = \overline{\xi(s)}$, and let $s_0 = \frac{1}{2} + it_0$ be a simple zero on the critical line. For real offset $\delta > 0$,*

$$\frac{\xi'}{\xi}(s_0 + \delta) \;=\; \frac{1}{\delta} + iB(t_0) + H_1(t_0)\,\delta + O(\delta^2), \tag{88}$$

where $B(t_0) \in \mathbb{R}$ and $H_1(t_0) \in \mathbb{R}$. Consequently,

$$\kappa(s_0 + \delta) \;=\; \frac{1}{\delta^2} + A(t_0) + O(\delta^2), \qquad A(t_0) := B(t_0)^2 + 2H_1(t_0) \in \mathbb{R}. \tag{89}$$

In particular, the $O(1/\delta)$ term in κ vanishes identically.

Proof. Write $f(\delta) := (\xi'/\xi)(s_0 + \delta) = 1/\delta + c_0 + c_1\delta + O(\delta^2)$ (Laurent expansion at the simple pole). The functional equation $\xi(s) = \xi(1-s)$ gives $(\xi'/\xi)(1-s) = -(\xi'/\xi)(s)$, so at $s = s_0 + \delta$ with $1 - s = s_0 - \delta$ (using $1 - s_0 = s_0$): $f(-\delta) = -f(\delta)$, i.e., $-1/\delta + c_0 - c_1\delta + \cdots = -1/\delta - c_0 - c_1\delta - \cdots$, yielding $c_0 = -c_0$, hence $c_0 \in i\mathbb{R}$ is purely imaginary: $c_0 = iB$. Schwarz reflection $\xi(\bar{s}) = \overline{\xi(s)}$ at real δ gives $(\xi'/\xi)(s_0 + \delta) = \overline{(\xi'/\xi)(\bar{s}_0 + \delta)}$; since $\bar{s}_0 = \frac{1}{2} - it_0$ and c_1 is determined by $(\xi'/\xi)''(s_0)/2$, one checks $c_1 = \bar{c}_1 \in \mathbb{R}$: $c_1 =: H_1 \in \mathbb{R}$. Then $\kappa = |f|^2 = (1/\delta + H_1\delta)^2 + B^2 = 1/\delta^2 + (B^2 + 2H_1) + O(\delta^2)$, proving (89). □

Remark 11.2 (Numerical verification of Proposition 11.2)**.** Result #60 verifies (89) numerically for $\zeta(s)$: 13 zeros ($t_0 \in [10, 60]$) at five offsets $\delta \in \{0.01, 0.02, 0.03, 0.05, 0.10\}$ (65 data points total) give $\kappa \cdot \delta^2 \in [1.00012, 1.01206]$ (all within $[0.95, 1.10]$; CV<1% for every δ). The $O(1)$ correction $A(t_0) = \kappa - 1/\delta^2$ is δ-independent to within $O(\delta^2)$, confirming Proposition 11.2. The values $A(t_0)$ for the 13 zeros are:

n	t_0	$A(t_0) = \kappa - 1/\delta^2$ (at δ=0.01)
1	14.135	0.471
2	21.022	0.848
3	25.011	0.673
4	30.425	1.469
5	32.935	0.807
6	37.586	1.206
7	40.919	1.480
8	43.327	0.898
9	48.005	2.458
10	49.774	1.284
11	52.970	1.105
12	56.446	1.531
13	59.347	2.727

The δ-scaling summary ($\kappa \cdot \delta^2$ column, median over 13 zeros) is:

δ	$1/\delta^2$	κ_{near} (median)	CV	$\kappa \cdot \delta^2$
0.01	10000	10001.21	0.01%	1.00012
0.02	2500	2501.21	0.03%	1.00048
0.03	1111	1112.32	0.06%	1.00109
0.05	400	401.21	0.16%	1.00301
0.10	100	101.21	0.62%	1.01206

[result #60, established, 2026-04-18; 13 zeros, 5 offsets, 65 data points; $\kappa \cdot \delta^2 \in [1.00012, 1.01206]$; $O(1/\delta)$ coefficient $= 0$ confirmed; numerical verification, not proof]

Proposition 11.3 (Hadamard decomposition of $A(t_0)$)**.** *Let $\rho_n = \frac{1}{2} + it_n$ $(0 < t_1 \le t_2 \le \cdots)$ denote the non-trivial zeros of $\zeta(s)$ on the critical line (RH assumed). Fix $\rho_0 = \frac{1}{2} + it_0$ with $t_0 = t_m$ for some index m. The local curvature coefficient*

$$A(t_0) \;=\; \lim_{\delta \to 0} \Big[\kappa(\rho_0 + \delta) - \tfrac{1}{\delta^2} \Big], \quad \kappa(s) = |\xi'/\xi(s)|^2,$$

admits the Hadamard decomposition

$$A(t_0) \;=\; B(t_0)^2 + 2H_1(t_0),$$

where the sums run over all $t_n \neq t_0$ with symmetric pairing:

$$\begin{aligned} B(t_0) &= -\frac{1}{2t_0} - \sum_{n:\, t_n \neq t_0} \left[\frac{1}{t_0 - t_n} + \frac{1}{t_0 + t_n} \right], \\ H_1(t_0) &= \frac{1}{4t_0^2} + \sum_{n:\, t_n \neq t_0} \left[\frac{1}{(t_0 - t_n)^2} + \frac{1}{(t_0 + t_n)^2} \right]. \end{aligned}$$

Both series converge absolutely (terms $\sim 2t_0/(t_n^2 - t_0^2)$ and $\sim 2/t_n^2$ respectively as $n \to \infty$).

Proof. **Step 1 ($D = 0$).** Writing the Hadamard product as $\xi(s) = \xi(0)\, e^{Ds} \prod_n [(1 - s/\rho_n)(1 - s/\bar{\rho}_n)]$ and differentiating logarithmically yields $\xi'/\xi(s) = D + \sum_n [1/(s - \rho_n) + 1/(s - \bar{\rho}_n)]$. The symmetry $\xi(s) = \xi(1 - s)$ gives $\xi'/\xi(\frac{1}{2}) = 0$; each paired term at $s = \frac{1}{2}$ equals $(-it_n)^{-1} + (it_n)^{-1} = 0$. Hence $D = 0$ and

$$\xi'/\xi(s) = \sum_{n \ge 1} \left[\frac{1}{s - \rho_n} + \frac{1}{s - \bar{\rho}_n} \right]. \tag{90}$$

Step 2 (Laurent expansion at ρ_0). Set $g(s) = \xi'/\xi(s) - 1/(s - \rho_0)$. At $s = \rho_0$, the $\bar{\rho}_0 = \frac{1}{2} - it_0$ contribution gives $(it_0 - (-it_0))^{-1} = 1/(2it_0)$, and each off-diagonal pair $[1/(i(t_0 - t_n)) + 1/(i(t_0 + t_n))]$ is purely imaginary. Hence $c_0 := g(\rho_0) = iB(t_0)$ with $B(t_0) \in \mathbb{R}$. Differentiating: $c_1 := g'(\rho_0) = 1/(4t_0^2) + \sum_{n \neq m} [1/(t_0 - t_n)^2 + 1/(t_0 + t_n)^2] =: H_1(t_0) \in \mathbb{R}_{>0}$.

Step 3 (Curvature expansion). With $(\xi'/\xi)(\rho_0 + \delta) = 1/\delta + iB + H_1\delta + O(\delta^2)$:

$$\kappa(\rho_0 + \delta) = \left| \tfrac{1}{\delta} + iB + H_1\delta + O(\delta^2) \right|^2 = \tfrac{1}{\delta^2} + B^2 + 2H_1 + O(\delta^2).$$

Taking $\delta \to 0$ gives $A(t_0) = B(t_0)^2 + 2H_1(t_0)$. □

Remark 11.3 (Numerical verification of Proposition 11.3 — result #74)**.** For the 13 non-trivial zeros $t_0 \in [14.13, 59.35]$, both sides are computed independently:

- *Direct*: $A_{\text{dir}} = \kappa(\rho_0 + \delta) - 1/\delta^2$ at $\delta = 0.01$.
- *Hadamard (corrected)*: partial sums $A_{\text{Had}}(N)$ with $N = 2000$ zeros, supplemented by an Euler–Maclaurin tail correction $B_{\text{tail}} = \int_{t_N}^{\infty} \frac{2t_0}{t^2 - t_0^2} \frac{\log(t/2\pi)}{2\pi}\, dt$ (computed via `scipy.integrate.quad`).

n	t_0	A_{dir}	$A_{\text{Had}}(2000)$	A_{corr}	$\text{err}_{\text{corr}}\%$
1	14.135	0.4713	0.4552	0.4713	0.001
2	21.022	0.8478	0.8183	0.8478	0.001
3	25.011	0.6731	0.6459	0.6731	0.002
4	30.425	1.4686	1.4162	1.4686	0.001
5	32.935	0.8068	0.7788	0.8068	0.002
6	37.586	1.2057	1.1493	1.2057	0.002
7	40.919	1.4804	1.4178	1.4804	0.001
8	43.327	0.8977	0.8630	0.8977	0.003
9	48.005	2.4575	2.3558	2.4575	0.000
10	49.774	1.2839	1.2466	1.2839	0.003
11	52.970	1.1045	1.0437	1.1045	0.003
12	56.446	1.5313	1.4436	1.5313	0.002
13	59.347	2.7271	2.6099	2.7272	0.000

With tail correction: $\text{err}_{\text{corr}}\% < 0.003\%$ for all 13 zeros (13/13). Without correction: $\text{err}_{\text{Had}}\% \in [2.9\%, 5.7\%]$ at N=2000 (slow convergence $O(\log^2 N/N)$; tail correction is required and mathematically justified via the zero density $\rho(t) = \frac{\log(t/2\pi)}{2\pi}$). [result #74, established, 2026-04-19; $\zeta(s)$; 13 zeros; Hadamard partial sum N=2000 + EM tail; err_{corr}<0.003%; 13/13 pass; Proposition 11.3]

Remark 11.4 (B-20 resolved: near-zero H_1 explains $A(t_0)$ variability)**.** The variability of $A(t_0)$ across zeros (range [0.47, 2.73] for $t_0 \in [14, 60]$, CV≈ 39%) is explained by Proposition 11.3: since $H_1(t_0) = \sum_n 1/(t_0 - t_n)^2 + \cdots$, zeros t_n near t_0 contribute $\sim 1/(t_0 - t_n)^2$, dominating $A(t_0)$.

- $t_9 = 48.005$: neighbour $t_{10} = 49.774$ ($|\Delta t| = 1.769$) contributes $H_1^{\text{near}} = 0.406$, so $2H_1^{\text{near}}/A(t_9) \approx 33\%$ of $A(t_9) = 2.458$.
- $t_1 = 14.135$: isolated zero (no neighbour within $|\Delta t| < 5$), $H_1^{\text{near}} = 0$, giving the minimal value $A(t_1) = 0.471$.

Thus $A(t_0)$ variability is an *analytic function of zero spacing*, fully captured by $H_1(t_0)$.

Remark 11.5 (Degree universality)**.** Proposition 11.3 holds for every L-function admitting a Hadamard product $\prod_n[(1-s/\rho_n)(1-s/\bar{\rho}_n)]$ (GL(1) through GL(5) and Dirichlet L-functions). The $D = 0$ step uses only the symmetry $\xi(s) = \xi(1-s)$ and the palindromic pairing $(\rho_n, \bar{\rho}_n)$; no degree-specific input is required. Numerical verification across GL(1)–GL(5) is subsumed by result #74 for $\zeta(s)$ and the existing $A(t_0)$ measurements (results #60, #61, #69, #71, #73).

Remark 11.6 (Degree comparison of $A(t_0)$: three-degree scaling (result #61))**.** The scaling law $\kappa = 1/\delta^2 + A(t_0) + O(\delta^2)$ (Proposition 11.2) is verified across degrees 1–3 using 120 additional data points: GL(2) Maass form ($R \approx 13.78$): 12 zeros $\times$ 5 offsets = 60 points, and GL(3) symmetric-square $L(s, \mathrm{sym}^2(11a1))$: 12 zeros $\times$ 5 offsets = 60 points. The correction $A(t_0) = \kappa - 1/\delta^2$ is δ-independent (GL(2) spread $< 0.21\%$, GL(3) spread $< 0.70\%$), confirming the proposition beyond $\zeta(s)$. The $O(\delta^2)$ deviation grows with degree as expected from the larger $A(t_0)$ values.

The quantitative comparison of mean$(A(t_0))$ across degrees is:

Degree	L-function	mean(A)	range(A)	n
GL(1)	$\zeta(s)$	1.30	[0.47, 2.73]	13
GL(2)	Maass $R{\approx}13.78$	3.93	[0.50, 6.67]	12
GL(3)	$\mathrm{sym}^2(11a1)$	12.79	[6.64, 23.35]	12
GL(3)	$\mathrm{sym}^2(11a1)$, ξ-bundle†	9.04	[6.57, 12.56]	3
GL(3)	$\mathrm{sym}^2(37a1)$, ξ-bundle†	14.51	[10.36, 18.46]	3
GL(4)	$\mathrm{sym}^3(\Delta)$ (wt 11)†	2.77	[2.22, 3.45]	3
GL(4)	$\mathrm{sym}^3(11a1)$ (wt 1)†	10.66	[8.34, 12.39]	3
GL(5)	$\mathrm{sym}^4(11a1)$ (wt 1)†	14.88	[7.71, 24.56]	3

† Measured via ξ-bundle κ (σ-direction, $s = k/2 + \delta + it_0$) using PARI `lfunlambda` with motivic normalization (center $k/2$; results #69, #71, #88). The $d = 1$–3 *non-dagger* values use analytic normalization (center $\frac{1}{2}$). Cross-degree comparison of $A(t_0)$ is *normalization-dependent* (open question; see Remark 11.11). For $d = 3\dagger$: center $= k/2 = 1.5$; $d = 4, 5$: center $= k/2$ (2.0/2.5 respectively).

Within the analytically normalized degrees 1–3, mean(A) increases monotonically: $1.30 \to 3.93 \to 12.79$ (ratios: GL(2)/GL(1) $\approx 3.0\times$, GL(3)/GL(2) $\approx 3.3\times$). Within the motivic normalization (centre $k/2$), the GL(3) ξ-bundle values are mean$(A) \in [9.04, 14.51]$ ($\mathrm{sym}^2(11a1)/\mathrm{sym}^2(37a1)$; result #88), and the GL(4)$\to$GL(5) (11a1) sequence gives $10.66 \to 14.88$ (monotone; results #69, #71). The conductor-dependence within $d = 3$ is *suggestive* (46.4% spread, n=2). *The extension to $d = 4$ requires caution*: the $d = 4$ values in the table above (result #69) are measured with PARI motivic normalization (center $k/2 = 17$ for $\mathrm{sym}^3\Delta$ and $k/2 = 2$ for $\mathrm{sym}^3(11a1)$), while $d = 1$–3 *non-dagger* rows use analytic normalization (center $\frac{1}{2}$). The two normalizations define different completed Λ-functions with different Gamma factors, so their $A(t_0)$ values are not directly comparable. The apparent non-monotone behaviour at $d = 3 \to 4$ ($12.79 \to 2.77$ and 10.66) is therefore normalization-dependent rather than a physical violation of monotonicity; whether monotone growth holds in a unified normalization remains an **open question** (boundary B-12; see also Remark 11.11). Within $d = 4$: $A(\mathrm{sym}^3\Delta, w{=}11) = 2.77 < A(\mathrm{sym}^3(11a1), w{=}1) = 10.66$, suggesting (within the same motivic normalization) that higher weight is *not* associated with larger A—but the sample of 3 zeros per L-function is too small for a definitive conclusion. This growth reflects the cumulative contribution of d Gamma-factor terms to $H_0 = (\Lambda'/\Lambda)$ at the zero: in GL(d), the functional equation Λ contains d Gamma factors, each contributing a ψ-function (digamma) term to H_0, so $|\mathrm{Im}(H_0)|^2$ grows with d. Note that the ranges overlap slightly at the individual-zero level (GL(2) max $= 6.67 >$ GL(3) min $= 6.64$), and the functional form of A as a function of d remains undetermined from three data points.

We caution that $\kappa \cdot \delta^2 \approx 1$ is a mathematically inevitable consequence of the simple-pole structure of ξ'/ξ at zeros, not an independently surprising finding. The novel content lies in the *quantitative degree-dependence of* $A(t_0)$: the $O(1)$ correction encodes degree- and zero-specific arithmetic information that $1/\delta^2$ does not. At $\delta = 0.1$ the GL(3) correction reaches $A \cdot \delta^2 \approx 0.12$ (12%), indicating that $\delta = 0.1$ is near the boundary of the asymptotic regime for degree-3; smaller δ is recommended for precise A extraction at higher degree. The analytic decomposition $A(t_0) = A_\Gamma + A_L$ separating Gamma-factor and finite Euler product contributions is addressed in result #64 (Remark 11.9), which establishes that ΔA_L accounts for 94.8% of the conductor-driven variation (boundary B-18; see §20.3).

[result #61, established, 2026-04-18; GL(2): 12 zeros, 5 offsets, 60 pts, $\kappa{\cdot}\delta^2{\in}[1.00, 1.04]$; GL(3): 12 zeros, 5 offsets, 60 pts, $\kappa{\cdot}\delta^2{\in}[1.00, 1.12]$; mean($A$) monotone increasing GL(1–3): 1.30, 3.93, 12.79 (analytic normalization); GL(3) ξ-bundle A (result #88, motivic normalization, centre 1.5): sym^2(11a1)=9.04 (CV=28.9%), sym^2(37a1)=14.51 (CV=23.4%); GL(4) ξ-bundle A (result #69, motivic normalization): $\mathrm{sym}^3\Delta$=2.77, sym^3(11a1)=10.66; GL(5) ξ-bundle A (result #71, motivic normalization): sym^4(11a1)=14.88 (CV=49%, 3 zeros); cross-degree comparison at $d{\geq}4$ normalization-dependent (open question, B-12); functional form undetermined; numerical verification, not proof]

Remark 11.7 (Conductor dependence of $A(t_0)$ within degree 1: Dirichlet extension (result #62))**.** The scaling law $\kappa = 1/\delta^2 + A(t_0) + O(\delta^2)$ extends to Dirichlet L-functions $L(s, \chi)$ for χ mod 3, χ mod 4, and χ mod 5 (all odd characters, $\mu = 1$). A total of 330 data points (66 zeros $\times$ 5 offsets $\delta \in \{0.001, 0.005, 0.01, 0.05, 0.1\}$) confirm:

- $\kappa \cdot \delta^2 \in [1.00000, 1.028]$ for all characters and all δ (pass);
- $A(t_0)$ is δ-independent to within spread $< 0.1\%$ (all three χ).

The correction $A(t_0)$ exhibits clear conductor dependence within degree 1:

L-function	μ	conductor q	mean(A)	n	spread
$\zeta(s)$	0	1	1.27	12	0.00%
$L(s, \chi_3)$	1	3	2.25	20	0.08%
$L(s, \chi_4)$	1	4	2.77	22	0.02%
$L(s, \chi_5)$	1	5	3.14	24	0.06%

At the same δ, κ_{near} differs by $< 0.2\%$ across all degree-1 L-functions, confirming that κ_{near} is effectively degree-determined (boundary B-09). The conductor-dependent variation resides entirely in the $O(1)$ correction $A(t_0)$, which grows monotonically with q: $A(\zeta) = 1.27 < A(\chi_3) = 2.25 < A(\chi_4) = 2.77 < A(\chi_5) = 3.14$.

The simple estimate $\Delta A \approx \log(q/\pi)/2$ (from the $\log(q/\pi)$ term in Λ'/Λ) accounts for only $\sim$13% of the observed increase; the dominant contribution arises from the L'/L term at the zeros, which reflects the conductor-dependent zero density. We caution that μ and q co-vary in these three characters (all have $\mu = 1$, $q > 1$); separating the μ-shift effect from the conductor effect requires even characters ($\mu = 0$, $q > 1$), which is established in result #63 (Remark 11.8).

[result #62, established, 2026-04-18; 66 zeros, 5 offsets, 330 data points; $\kappa{\cdot}\delta^2$ all PASS; A δ-independent (spread <0.1%); A conductor-monotone: 1.27, 2.25, 2.77, 3.14; same-δ κ_{near} diff. <0.2% (B-09 confirmed); numerical evidence, not proof]

Remark 11.8 (μ-vs-q separation of $A(t_0)$: even Dirichlet character (result #63))**.** Result #62 measures $A(t_0)$ only for odd characters ($\mu = 1$, $q > 1$), leaving μ and q confounded. To isolate the two effects, we add the even character χ_5^{even} (mod 5) (Legendre symbol mod 5: $\chi(1)$=1, $\chi(2)$=−1, $\chi(3)$=−1, $\chi(4)$=1, $\chi(-1)$=$\chi(4)$= + 1, so $\mu = 0$, $q = 5$), together with a fresh measurement of $\zeta(s)$ (same method, $\mu = 0$, $q = 1$). A total of 180 data points (36 zeros × 5 offsets across ζ and χ_5^{even}) confirm $\kappa{\cdot}\delta^2 \in [1.00000, 1.031]$ and $A(t_0)$ δ-independent (spread <0.1%) for both L-functions.

The five-row comparison (same $\delta = 0.01$) is:

L-function	μ	q	mean(A)	n	SE
$\zeta(s)$	0	1	1.2727	12	0.14
$L(s, \chi_3)$	1	3	2.2451	20	–
$L(s, \chi_4)$	1	4	2.7688	22	–
$L(s, \chi_5^{\text{even}})$	0	5	3.0914	24	0.21
$L(s, \chi_5^{\text{odd}})$	1	5	3.1407	24	0.21

(SE: standard error = std(A)/$\sqrt{n}$; entries marked "−" from result #62.)
The μ-vs-q decomposition at $q = 5$ (fixing one variable at a time):

- **q-effect** ($\mu = 0$ fixed): $A(\chi_5^{\text{even}}) - A(\zeta) = 3.0914 - 1.2727 = +1.8187$, accounting for **97.4%** of the total variation.
- **μ-effect** ($q = 5$ fixed): $A(\chi_5^{\text{odd}}) - A(\chi_5^{\text{even}}) = 3.1407 - 3.0914 = +0.0493$, accounting for **2.6%**. The difference 0.05 corresponds to 0.24 standard errors, consistent with μ-independence at the resolution of our data.
- **Additivity check**: $1.8187 + 0.0493 = 1.8680 = A(\chi_5^{\text{odd}}) - A(\zeta)$ (exact).

Naive log-conductor model insufficient. The simple estimate $\Delta A_q \approx \log(q/\pi)/2 = 0.23$ (from the $\log(q/\pi)$ term in Λ'/Λ) accounts for only $0.23/1.82 \approx 13\%$ of the observed q-effect. The dominant contribution arises from the $(L'/L)(\rho, \chi)$ term, which reflects conductor-dependent zero density. This indicates that $A(t_0)$ encodes arithmetic information beyond the gamma-factor structure; the decomposition $A = A_\Gamma + A_L$ remains open (boundary B-18, §20).
Corroboration: $\chi_3, \chi_4, \chi_5^{\text{odd}}$ (all $\mu = 1$, result #62) show monotone increase in A with q at fixed $\mu = 1$, consistent with conductor-dominance. The possible confound of unequal t_0-distributions across L-functions (different numbers of zeros in $t \in [14, 59]$) is mitigated by this within-μ monotonicity across three independent conductors.

[result #63, established, 2026-04-18; ζ 12 zeros + χ_5^{even} 24 zeros, 5 offsets, 180 data points; $\kappa{\cdot}\delta^2{\in}[1.00000, 1.031]$ all PASS; A δ-independent (spread <0.1%); q-effect 97.4%, μ-effect 2.6% (consistent with μ-independence); naive $\log(q/\pi)/2 = 0.23 \ll$ measured 1.82 (only 13%); $A(t_0)$ encodes arithmetic beyond gamma factors; B-18 open; numerical evidence, not proof]

Remark 11.9 (Analytic decomposition $A = A_\Gamma + A_L$ (result #64))**.** The correction term $A(t_0)$ in Proposition 11.2 can be decomposed as $A(t_0) = \mathrm{Im}(H_0^\Gamma)^2 + \text{remainder}$, where H_0^Γ is the Gamma-factor part of $H_0 = (\Lambda'/\Lambda)(\rho)$. Defining $A_\Gamma := \mathrm{Im}(H_0^\Gamma)^2$ (computable analytically from digamma functions) and $A_L := A - A_\Gamma$ (the arithmetic residual, containing $(L'/L)(\rho)$ and higher-order Gamma corrections H_1^Γ), we measure the decomposition for $\zeta(s)$ (12 zeros) and $L(s, \chi_5^{\text{even}})$ (24 zeros):

	$\zeta(s)$	χ_5^{even}	Δ (change)
mean(A)	1.2727	3.0914	+1.8187 (100%)
mean(A_Γ)	0.5350	0.6287	+0.0937 (5.2%)
mean(A_L)	0.7377	2.4627	+1.7250 (94.8%)

The Gamma-factor contribution $\Delta A_\Gamma = 0.0937$ accounts for only **5.2%** of the conductor-driven change $\Delta A = 1.8187$; the arithmetic residual $\Delta A_L = 1.7250$ carries **94.8%**. This confirms the finding of result #63 that the naive estimate $\log(q/\pi)/2 = 0.23$ underestimates the q-effect by an order of magnitude: the dominant source is $(L'/L)(\rho, \chi)$, not the Gamma-factor shift.

Note that A_Γ contributes 42% of $A(\zeta)$ and 20% of $A(\chi_5^{\text{even}})$ individually; it is only the *change* ΔA for which A_Γ is negligible. The direct numerical extraction of $H_0^L = (L'/L)(\rho)$ via Richardson extrapolation encounters severe cancellation ($\mathrm{Im}(H_0) \sim 10^8$) and remains an open challenge; the $A_L = A - A_\Gamma$ subtraction route bypasses this difficulty.

[result #64, established, 2026-04-18; ζ: 12 zeros, χ_5^{even}: 24 zeros (DPS=60); ΔA_Γ=0.0937 (5.2%), ΔA_L=1.7250 (94.8%); A_L contains $(L'/L)(\rho)$ and higher-order Γ corrections; B-18 partially resolved; numerical verification, not proof]

Remark 11.10 (GL(4): four-property verification of $L(s, \mathrm{sym}^3\Delta)$ (result #65))**.** Extending the framework to degree 4, we verify the symmetric cube L-function $L(s, \mathrm{sym}^3\Delta)$ associated to the Ramanujan Δ-form (weight 12, level 1) using PARI/GP `lfun`. The Hodge-derived parameters are: $\gamma_V = [-11, -10, 0, 1]$, $k = 34$, $N = 1$, $\varepsilon = -1$, with the n-th Dirichlet coefficient $c(n) = \sum_{ad=n} \chi(a)\, a^{-33}\, b_d$ constructed via Euler product from $\tau(p)$ values (cf. $c(2) = \tau(2)(\tau(2)^2 - 2 \cdot 2^{11}) = 84480$).

Property	Result	Verdict
Functional equation	-141 digits (motivic), -380 digits (analytic)	**PASS**
Zeros	43 non-trivial ($t \in [0, 50]$), $t_1 = 4.1559$	**PASS**
$\kappa \cdot \delta^2$	mean = 1.000286, 31/31 success	**PASS**
Monodromy	20/20 simple (sign change confirmed)	**PASS**

Cross-validation: PARI `lfunsympow(11a1, 3)` independently confirms $N = 1331 = 11^3$, $\gamma_V = [-1, 0, 0, 1]$, $k = 4$ for weight-1 (matching the Hodge pattern $\gamma_V = [-(w), -(w-1), 0, 1]$, $k = 3w + 1$).

The $A(t_0)$ values at $d = 4$ show extreme variability: mean(A) = 286.26, std(A) = 2782, CV = 972%, with individual values spanning $[-7471, +8554]$. The large CV arises from closely spaced zero pairs (e.g., $t = 23.34$ and $t = 23.98$, gap = 0.64). Despite this variability, $\kappa \cdot \delta^2 = 1.000286$ remains stable (the $1/\delta^2$ term dominates).

Weight versus degree. $\mathrm{sym}^3(11a1)$ ($d = 4$, weight 1) yields $A \approx 8.82$ (result #66, Hardy Z-based; motivic-normalized ξ-bundle value 10.66, result #69), comparable to $\mathrm{sym}^2(11a1)$ ($d = 3$, weight 1, $A = 12.79$). Note that the Hardy Z-based A and the ξ-bundle A are physically distinct observables (see Remark 11.11). The $A(t_0)$ values at $d = 4$ show extreme variability (CV $> 970\%$) when measured via Hardy Z; the ξ-bundle measurement (result #69) yields smaller CV (17–22%) and more stable values.

[result #65, established, 2026-04-18; PARI lfuncheckfeq: FE $= -141$ digits (motivic), -380 (analytic); 43 zeros ($t\in[0,50]$), t_1=4.1559; $\kappa\cdot\delta^2$=1.000286 (31/31); monodromy 20/20; A(Hardy Z)=286.26 (CV=972%); A(ξ-bundle)=2.77 (CV=22%, result #69); GL(4) degree-4 verification; numerical verification, not proof]

[result #66, established, 2026-04-18; PARI lfunsympow(11a1, 3): FE $= -366$ digits; γ_V=$[-1,0,0,1]$, k=4, N=1331=11^3, ε=$+1$; 105 zeros ($t\in[0,55]$), t_1=2.3200; $\kappa\cdot\delta^2$=1.000009 (69/69); monodromy 20/20; A(Hardy Z)=8.82 (CV=110800%); A(ξ-bundle)=10.66 (CV=17%, result #69); GL(4) degree-4 weight-1 verification; numerical verification, not proof]

Remark 11.11 (Two distinct κ observables: ξ-bundle vs. Hardy Z (result #69, boundary B-23))**.** Experiments #80–#83 identify two physically distinct curvature observables that both converge to $\kappa \approx 1/\delta^2$ near a zero, but differ in their subleading corrections and physical interpretation.

Observable 1: ξ-bundle κ (σ-direction).

$$\kappa_\sigma(s_0,\delta) := \left|\frac{\Lambda'(s_0+\delta)}{\Lambda(s_0+\delta)}\right|^2, \quad s_0+\delta = \tfrac{1}{2}+\delta+it_0 \quad (\delta\in\mathbb{R}).$$

Expansion: $\kappa_\sigma = 1/\delta^2 + A(t_0) + O(\delta^2)$ with $c_1 = 0$ (no $1/\delta$ correction term). The $c_1 = 0$ result (Proposition 11.2) follows from the functional equation and Schwarz reflection, and applies *only to this σ-direction observable.* The $O(1)$ correction $A(t_0) = B(t_0)^2 + 2H_1(t_0)$ is δ-independent.

Observable 2: Hardy Z κ (t-direction).

$$\kappa_t(t_0,\delta) := \left(\frac{Z'(t_0+\delta)}{Z(t_0)}\right)^2.$$

Expansion: $\kappa_t = 1/\delta^2 - (Z''/Z')_{t_0}/\delta + O(1)$, giving a non-zero $c_1 = -(Z''/Z')_{t_0} \neq 0$. For $\zeta(s)$: $c_t \approx -47$ (measured, result #82). The $A_{\mathrm{Hardy}}(\delta) = \kappa_t - 1/\delta^2$ is therefore δ-*dependent*, decaying as $|c_1|/\delta$ for small δ.

Observable	Direction	c_1	A-stability
ξ-bundle κ_σ	σ-direction	0 (theorem)	δ-independent
Hardy Z κ_t	t-direction	$-(Z''/Z') \neq 0$	δ-dependent

GL(4) ξ-bundle $A(t_0)$ (result #69). Measuring κ_σ (σ-direction) for the two GL(4) L-functions using PARI `lfunlambda` with motivic normalization ($s = k/2+\delta+it_0$), 36 points (2 L-functions $\times$ 3 zeros $\times$ 6 offsets $\delta \in \{0.001, 0.005, 0.01, 0.02, 0.05, 0.10\}$):

L-function	weight	mean(A)	CV	zeros
$\mathrm{sym}^3(\Delta)$	11	2.77	22%	3
$\mathrm{sym}^3(11a1)$	1	10.66	17%	3

The CV values (17–22%) are dramatically smaller than the Hardy Z values ($> 970\%$), confirming that κ_σ is the appropriate δ-stable observable. The d=1–3 $A(t_0)$ data (results #60, #61) use analytic normalization (center $\frac{1}{2}$), while the d=4 data above use motivic normalization (center $k/2$). Whether monotone degree growth of $A(t_0)$ holds in a unified normalization is an **open question (boundary B-12)**.

[result #69, established, 2026-04-18; PARI lfunlambda, σ-direction; 36 pts; $\mathrm{sym}^3\Delta$: A=2.77 (CV=22%); $\mathrm{sym}^3(11a1)$: A=10.66 (CV=17%); B-23 resolved: $\kappa_\sigma \neq \kappa_t$; motivic normalization; GL(4) ξ-bundle $A(t_0)$; numerical verification, not proof]

Hardy Z precision degradation at $d = 5$ (boundary B-24). The GL(5) experiment (result #70) reveals that the Hardy Z $\kappa \cdot \delta^2$ at $d = 5$ shows per-zero deviations up to 0.42% (vs. $< 0.03\%$ at $d \leq 4$). This is the *practical limit of B-23*: the $c_1 \neq 0$ correction in the Hardy Z direction grows with degree, so that at $d = 5$ individual zeros no longer satisfy $\kappa_t \cdot \delta^2 \approx 1$ at the $[0.999, 1.001]$ tolerance. The *mean* remains 0.999126 (within tolerance), confirming global consistency, but the per-zero spread signals that for $d \geq 6$ the Hardy Z four-property test may become unreliable. **Boundary B-24 (open)**: Hardy Z $\kappa \cdot \delta^2$ precision degrades with degree; $d = 5$ per-zero deviation 0.42%; ξ-bundle κ_σ (B-23) remains the recommended observable at $d \geq 4$.

Remark 11.12 (GL(5): four-property verification of $L(s, \mathrm{sym}^4(11a1))$ (results #70, #71))**.** Extending the framework to degree 5, we verify the symmetric fourth power L-function $L(s, \mathrm{sym}^4(11a1))$ associated to the elliptic curve 11a1 (weight 1, conductor 11) using PARI/GP `lfunsympow`. Parameters: $\gamma_V = [-2, -1, 0, 0, 1]$, $k = 5$, $N = 14641 = 11^4$, $\varepsilon = +1$, center $= k/2 = 2.5$ (motivic normalization).

Property	Result	Verdict
Functional equation	-331 digits (motivic)	**PASS**
Zeros	61 non-trivial ($t \in [0, 30]$), $t_1 = 1.4878$	**PASS**
$\kappa \cdot \delta^2$ (Hardy Z)	mean = 0.999126 (3/3 executed; per-zero 0.996–1.002)	**PASS** (mean)
Monodromy	25/25 simple (sign change confirmed)	**PASS**
σ-uniqueness	$\sigma = 2.5$ maximal; ratio $\approx 10000\times$ vs. adjacent σ	**PASS**

Note on $\kappa \cdot \delta^2$: The mean 0.999126 lies within the $[0.999, 1.001]$ tolerance, but individual zeros deviate up to 0.42% (see B-24 above). This is the B-23 phenomenon: Hardy Z κ_t has non-zero c_1; the ξ-bundle measurement (result #71) is the appropriate observable.

σ-uniqueness at degree 5. Unlike GL(2)–GL(4) where σ-uniqueness fails structurally, GL(5) $\mathrm{sym}^4(11a1)$ passes σ-uniqueness with high contrast: $\kappa(\sigma{=}2.5)/\kappa(\sigma{=}2.3) \approx 30000\times$, $\kappa(\sigma{=}2.5)/\kappa(\sigma{=}2.4) \approx 9100\times$ (3/3 zeros). This unexpected σ-uniqueness is consistent with the motivic center $k/2 = 2.5$ acting as the true critical line in motivic normalization.

GL(5) ξ-bundle $A(t_0)$ (result #71). Measuring κ_σ (σ-direction, $s = 2.5 + \delta + it_0$):

t_0	mean(A)	per-zero CV	δ-range
1.4878	7.71	0.02%	$[0.0001, 0.05]$
2.8656	24.56	0.18%	$[0.0001, 0.05]$
3.3914	12.37	0.15%	$[0.0001, 0.05]$
overall	**14.88**	**49.1%**	

The per-zero CV ($< 0.2\%$) is remarkably small, confirming $c_1 = 0$ at $d = 5$. The overall CV (49.1%) reflects natural zero-to-zero variation of $A(t_0)$, not δ-instability. Within the motivic normalization, $A(d = 5, 11\text{a}1) = 14.88 > A(d = 4, 11\text{a}1) = 10.66$, consistent with degree-monotone growth at fixed weight and normalization. Cross-degree comparison with $d \leq 3$ remains normalization-dependent (B-12, open).

[result #70, established, 2026-04-18; PARI lfunsympow(11a1, 4): FE = − 331 digits; γ_V=[−2, −1, 0, 0, 1], k=5, N=14641=11^4, ε=+1; 61 zeros ($t\in[0, 30]$), t_1=1.4878; $\kappa\cdot\delta^2$=0.999126 (mean; per-zero 0.996–1.002, B-24); monodromy 25/25; σ-uniqueness: PASS (σ=2.5 maximal, ≈10000× ratio, 3/3); GL(5) degree-5 weight-1 verification; numerical verification, not proof]

[result #71, established, 2026-04-18; PARI lfunlambda, σ-direction (s=2.5+δ+it_0); 18 pts (3 zeros × 6 δ); sym^4(11a1): mean(A)=14.88 (per-zero CV<0.2%, overall CV=49%); d=4→5 (11a1): 10.66→14.88 (monotone, motivic normalization); B-12 open (normalization-dependent); motivic normalization (center 2.5); GL(5) ξ-bundle $A(t_0)$; numerical verification, not proof]

12 Gauss–Bonnet Quantization

Definition 12.1 (Curvature 2-form)**.** For $\theta(\sigma, t) := \operatorname{Im} \log \Lambda(\sigma + it)$, the curvature 2-form of the ξ-bundle is

$$F := d\omega = \left(\frac{\partial^2 \theta}{\partial\sigma\, \partial t} - \frac{\partial^2 \theta}{\partial t\, \partial\sigma} \right) d\sigma \wedge dt. \tag{91}$$

Away from zeros, θ is smooth and $F = 0$ (flat connection). At zeros, θ has logarithmic singularities contributing δ-function flux.

Theorem 12.1 (Gauss–Bonnet for the ξ-Bundle)**.** *Let $R = [\sigma_1, \sigma_2] \times [t_1, t_2]$ be a rectangle in the critical strip whose boundary avoids all zeros of Λ. Then:*

$$\frac{1}{2\pi} \oint_{\partial R} d \operatorname{Im} \log \Lambda = \frac{1}{2\pi} \oint_{\partial R} \operatorname{Im}\left(\frac{\Lambda'(s)}{\Lambda(s)}\, ds \right) = N(R), \tag{92}$$

where $N(R)$ is the number of zeros of Λ inside R, counted with multiplicity. Equivalently, in lattice-regularized form:

$$\frac{1}{2\pi} \sum_{\text{plaquettes } \square \subset R} \Delta\theta_\square = N(R), \tag{93}$$

where $\Delta\theta_\square$ is the wrapped phase circulation around each lattice plaquette.

Proof. The boundary integral in (92) is

$$\frac{1}{2\pi} \oint_{\partial R} \operatorname{Im}\left(\frac{\Lambda'}{\Lambda}\, ds \right) = \operatorname{Im}\left(\frac{1}{2\pi i} \oint_{\partial R} \frac{\Lambda'}{\Lambda}\, ds \right) \cdot \frac{1}{i} \quad \ldots$$

More directly: by the argument principle (Theorem 10.1(iii)),

$$\frac{1}{2\pi i}\oint_{\partial R}\frac{\Lambda'(s)}{\Lambda(s)}\,ds = N(R).$$

Since $N(R) \in \mathbb{Z}$, taking the real part gives $N(R)$ itself.

The total change in $\arg\Lambda$ around ∂R equals $2\pi N(R)$. This is the **first Chern number** of the U(1)-bundle restricted to R:

$$c_1(\xi\text{-bundle}|_R) = N(R).$$

For the lattice form (93): partition R into small rectangles (plaquettes). On each plaquette containing no zero, the wrapped phase circulation $\Delta\theta_\square = 0$. On a plaquette containing a zero of order m, by the argument principle applied to the plaquette boundary, $\Delta\theta_\square = 2\pi m$. Summing over all plaquettes and dividing by 2π yields $N(R)$.

The lattice sum is *topological*: it is invariant under refinement of the lattice, depending only on the total number of enclosed zeros. This is the discrete analogue of the Gauss–Bonnet theorem for the bundle curvature. □

Corollary 12.2 (Integer Quantization)**.** *The quantity* $\frac{1}{2\pi}\oint_{\partial R} d\arg\Lambda$ *is exactly an integer for* any *rectangle* R *whose boundary avoids zeros. In particular, numerical deviations from integer values in a lattice computation arise solely from discretization error and converge to zero under refinement.*

Proof. Immediate from Theorem 12.1 and the fact that the number of zeros is an integer. □

Remark 12.1 (Numerical verification)**.** The lattice computation (Gauss–Bonnet area method) yields $|\text{computed} - N| = 0.0000$ in all 8 tests (§19.4), while the contour method (Chern integral) yields a constant offset 0.0118 attributable to trapezoidal quadrature error at 64 subdivisions. Both are consistent with Theorem 12.1.

13 Cross-Rank Universality

Theorem 13.1 (Cross-Rank Universality)**.** *Theorems 10.1, 11.1, and 12.1 hold for the completed L-function* $\Lambda(s,\pi)$ *associated to* any *automorphic representation* π *of* $\mathrm{GL}(n)$ *over* $\mathbb{Q}$*, provided* $\Lambda(s,\pi)$ *satisfies:*

(a) $\Lambda(s,\pi)$ *is meromorphic of finite order with isolated zeros;*

(b) $\Lambda(s,\pi)$ *admits a Hadamard-type product over its zeros.*

In particular, the holonomy, curvature concentration, and Gauss–Bonnet quantization are ***independent of****:*

- *the degree* n *(GL(1) vs. GL(2) vs. GL(n));*
- *the conductor* N*;*
- *the weight* k *(for modular forms);*

- *the root number ε.*

Proof. The proofs of Theorems 10.1–12.1 use only:

1. the **argument principle**, which requires Λ'/Λ to be meromorphic with simple poles at zeros of Λ;
2. the **Laurent expansion** $\Lambda'/\Lambda = m/(s-\rho)+O(1)$ near a zero of order m, which follows from condition (b).

Both properties hold for all completed automorphic L-functions satisfying (a)–(b). The degree, conductor, weight, and root number enter the definition of Λ but do not affect the local structure of Λ'/Λ near a zero. Therefore the theorems transfer verbatim. □

Remark 13.1 (Numerical confirmation)**.** Cross-rank universality is confirmed by our numerical experiments:

- **GL(1)**: Riemann ζ (mono/$\pi = 2.000$, 10/10 zeros; FP anatomy: 19 FP all mono/$\pi = 0.000$) and 5 Dirichlet L-functions (§19.4, Observations 14–18);
- **GL(2)**: elliptic curves 11a1 and 37a1 (mono/$\pi = 2.000$, 17/17 and 23/23 zeros) and the Ramanujan Δ function of weight 12 (Appendix D). *FP monodromy cross-rank test:* 34 TP across all three GL(2) L-functions yield mono/$\pi = 2.000 \pm 0.000$; 60 FP candidates yield mono/$\pi = 0.000 \pm 0.000$ (Mann–Whitney $p = 9.7 \times 10^{-17}$). The TP/FP separation is identical to GL(1), confirming degree-independent monodromy classification.

The *only* diagnostic that varies across rank is σ-uniqueness, which fails for GL(2) due to the different Γ-factor structure (see Appendix D, §D.0.1). This is consistent with Theorem 13.1: σ-uniqueness is *not* a consequence of the argument principle but depends on the specific form of the Γ-factor, which varies with degree and weight.

Remark 13.2 (GL(2) blind zero prediction — cross-rank universality)**.** Cross-rank universality is further evidenced by *blind* zero prediction experiments on three GL(2) L-functions and one GL(1) Dirichlet L-function. *Blind* means that LMFDB zero locations enter *only* in the final evaluation step; the curvature sweep and κ-peak detection use no prior knowledge of zero positions.

Experiment #52 — elliptic curve 11a1 (conductor $N = 11$, rank 0, root number $\varepsilon = +1$): A curvature sweep over $t \in [5, 30]$ ($\Delta t = 0.1$, `dps`$= 45$) identifies 17 candidate zero locations. Matching against 17 LMFDB zeros yields 16 TP, 1 FP, 1 miss ($\gamma_{17} = 29.975$, outside sweep range). $P = R = F_1 = 0.941$. Max location error $= 0.051$. The single FP ($t \approx 28.4$) is a double-peak artefact: the contour radius $r = 0.135$ encloses no LMFDB zero (nearest distance $0.284 > r$), so its mono/$\pi = 2.000$ is a precision artefact at `dps`$= 45$.

Experiment #54 — elliptic curve 37a1 (conductor $N = 37$, rank 1, root number $\varepsilon = -1$): A curvature sweep over $t \in [5, 30]$ ($\Delta t = 0.2$, `dps`$= 60$) identifies 22 candidate zero locations. Matching against 23 LMFDB zeros yields 22 TP, 0 FP, 1 miss ($\gamma_{24} = 29.282$, adjacent to $\gamma_{23} = 29.030$, gap $0.252 < 1.5\,\Delta t$). $P = 1.000$, $R = 0.957$, $F_1 = 0.978$. Max

location error = 0.077. The zero false-positive rate (FP= 0) is attributable to the increased precision (`dps`= 60), which eliminates the numerical artefact observed in #52.

Quantitative monodromy at $t \approx 29.0$. At $t = 29.0$, the monodromy contour of radius $r = 0.36$ encloses *two* zeros ($\gamma_{23} = 29.030$ and $\gamma_{24} = 29.282$), yielding winding number 2 and mono/$\pi = 4.0$. This demonstrates that the monodromy indicator carries richer information than binary zero/non-zero classification: it correctly reports the *count* of zeros within the contour, compensating for the curvature resolution limit ($\Delta t = 0.2$) that merges the two κ-peaks.

Experiment #56 — Ramanujan Δ (weight 12) (level $N = 1$, root number $\varepsilon = +1$, critical line $\sigma_c = k/2 = 6$): A curvature sweep over $t \in [5, 30]$ ($\Delta t = 0.2$, `dps`= 60) identifies 8 candidate zero locations. Matching against 8 LMFDB zeros yields 8 TP, 0 FP, 0 FN — *perfect detection.* $P = R = F_1 = 1.000$. Max location error = 0.108. All 8 candidates satisfy mono/$\pi = 2.000$ (winding number 1, simple zeros). The near/far κ-ratio is 322.8× (near median = 343.5, far median = 1.06). This is the strongest result among the three GL(2) blind tests: despite the critical line shifting from $\sigma_c = 1$ (elliptic curves) to $\sigma_c = 6$ (weight 12), the ξ-bundle framework produces zero false positives and zero false negatives.

Experiment #57 — Dirichlet χ mod 7 (complex character of order 6, conductor $q = 7$, $a = 1$, $\sigma_c = 1/2$): A curvature sweep over $t \in [5, 40]$ ($\Delta t = 0.2$, `dps`= 80) identifies 18 candidate zero locations. Matching against 13 LMFDB zeros yields 13 TP, 5 FP, 0 FN. $R = 1.000$, $P = 0.722$, $F_1 = 0.839$. Max location error = 0.084. All 5 FP occur at $t < 17.16$ (below the first LMFDB zero $\gamma_1 = 17.161$); independent evaluation confirms $|L(\frac{1}{2} + it, \chi)| \gg 0$ at these locations, ruling out missed critical-line zeros. The FP arise from off-critical-line zeros captured within the monodromy disk of radius $r = 0.4$ at low t, where the Γ-background is elevated. Restricting to $t \geq 17$: $P = R = F_1 = 1.000$ (13/13 TP, 0 FP). The near/far κ-ratio is 24.0×, lower than the GL(2) curves due to the conductor-induced background increase, consistent with the non-blind χ mod 7 result (#37; ratio 36.6×).

Table 3 compares blind prediction across five L-functions spanning GL(1) and GL(2), weight 1, 2, and 12, rank 0 and 1, four conductors $(1, 7, 11, 37)$, and both root number signs.

Table 3: Blind zero prediction across five L-functions. *Blind*: LMFDB zeros used only in evaluation. ε: root number; k: weight; σ_c: critical line.

Experiment	N	k	rank	ε	Zeros	TP	FP	Prec.	Rec.
#2 GL(1) ζ (blind)	1	1	—	+1	7	7	0	1.000	1.000
#52 GL(2) 11a1 (blind)	11	2	0	+1	17	16	1	0.941	0.941
#54 GL(2) 37a1 (blind)	37	2	1	−1	23	22	0	1.000	0.957
#56 GL(2) Δ (blind)	1	12	—	+1	8	8	0	1.000	1.000
#57 GL(1) χ mod 7 (blind)	7	1	—	−1	13	13	5	0.722	1.000
Aggregate (5 curves)					**68**	**66**	**6**	**0.917**	**0.971**

Across all five curves the aggregate is 66/68 TP, 6 FP, $F_1 = 0.943$. Four of the five achieve $P \geq 0.94$; the exception is χ mod 7 ($P = 0.722$), whose 5 FP are confined to $t < 17$ (below the first LMFDB zero) and arise from off-critical-line zero capture within the monodromy disk at low t. For $t \geq 17$, all five curves yield $P = R = F_1 = 1.000$ on a combined 66 zeros. The Ramanujan Δ result (#56) achieves *perfect* $P = R = F_1 = 1.000$ at $\sigma_c = 6$, while

χ mod 7 (#57) achieves perfect recall ($R = 1.000$, 13/13 TP) on a complex Dirichlet character. Together the five experiments cover GL(1) Riemann, GL(1) Dirichlet (complex character), and GL(2) (elliptic curves of rank 0 and 1, plus a weight-12 modular form)—spanning all standard automorphic L-function families tested. *This constitutes numerical evidence for blind zero prediction universality across automorphic families; it is not a proof.*

14 The Dual Criterion: Monodromy + Curvature

Corollary 14.1 (Exact TP/FP Separation)**.** *Let $\{t_1, \ldots, t_N\}$ be candidate zero locations on the critical line (e.g., outputs of a numerical zero-finder or neural detector). For each candidate t_j, define:*

$$\mu_j := |\mathrm{Hol}(\mathcal{C}_r(\tfrac{1}{2} + it_j))|, \tag{94}$$

$$\kappa_j := \kappa(\tfrac{1}{2} + \delta + it_j) \quad (\textit{for fixed small } \delta > 0). \tag{95}$$

Then:

(i) *A candidate t_j is a true zero if and only if $\mu_j = 1$ (assuming simple zeros and sufficiently small r enclosing exactly one zero).*

(ii) *If t_j is a true zero, $\kappa_j \sim 1/\delta^2 \gg 1$. If t_j is a false positive with nearest zero at distance $d \gg \delta$, $\kappa_j \sim 1/d^2 \ll 1/\delta^2$.*

(iii) *The dual criterion $\{\mu_j = 1\} \cap \{\kappa_j > \tau\}$ achieves **perfect separation** between true zeros and false positives, for any threshold $\tau \in (1/d_{\min}^2, 1/\delta^2)$ where $d_{\min}$ is the minimum distance from any false positive to its nearest true zero.*

Proof. Part (i) is Theorem 10.1(i)–(ii): the holonomy is a topological invariant that takes value $m \geq 1$ at zeros and 0 at non-zeros. Part (ii) is Theorem 11.1(i) and (iii). Part (iii) follows because the monodromy criterion alone is already sufficient (it is exact), and the curvature criterion provides an independent, continuously-valued confirmation with a ratio gap of $(d_{\min}/\delta)^2 \gg 1$. □

Remark 14.1 (Relation to Conjecture 3)**.** What was previously labelled "Conjecture 3" (the dual criterion separates TP from FP) is now seen to be a **theorem**: it is a direct consequence of the argument principle (Theorem 10.1) and the Laurent structure of Λ'/Λ (Theorem 11.1). The numerical measurements confirm the theorem across ranks: GL(1) ζ: mono/$\pi = 2.000 \pm 0.000$ for 10 TP, 0.000 ± 0.000 for 19 FP ($p = 6.2 \times 10^{-6}$); GL(2) (11a1, 37a1, Δ): mono/$\pi = 2.000 \pm 0.000$ for 34 TP, 0.000 ± 0.000 for 60 FP ($p = 9.7 \times 10^{-17}$). The dual criterion improves precision by +63 percentage points at $\kappa > 0.5$ for GL(2), confirming degree-independent perfect separation.

Remark 14.2 (What remains conjectural)**.** The *proven* results (Theorems 10.1–13.1 and Corollary 14.1) characterize the bundle *at any zero*, regardless of its location. They do *not* constrain *where* zeros can lie. The statement that all zeros lie on $\sigma = 1/2$ (the Riemann Hypothesis) remains a conjecture, independent of the bundle structure established here.

However, the σ-localization of curvature flux (FWHM < 0.004 at $\sigma = 1/2$, §19.4) provides strong numerical evidence that the topological charge of the ξ-bundle concentrates on the critical line.

Part IV

Failed Directions and Their Diagnoses

This part collects, in compact form, the empirical programmes that did not succeed. Each section is labelled ■ (Tier 3); each opens with a one-paragraph description of the programme, followed by a diagnostic chain, and closes with the lesson extracted. The content is deliberately dense — typically one to two pages per failed direction — so as not to crowd out the validated and conjectural material while retaining the full diagnosis. A reader in a hurry may skim Part IV; a reader planning a follow-up experiment should read it carefully.

15 ■ Cyclotomic $\cos^2$ PQO training ($L = 2, 4$): Ψ-decoupling and the isolation cascade

Programme. Train a three-layer resonant discriminator network to separate Riemann zeros from non-zero ordinates on $t \in [100, 200]$ using the cyclotomic $L \in \{2, 4\}$ cosine-squared quantisation loss $\mathcal{L}_{\text{pqo}}$ of Remark 3.2 together with the synthetic magnitude-only target Ψ_n of Equation (43).

Outcome. Negative at both $L = 2$ and $L = 4$. At $L = 2$ the network collapses to the trivial attractor $\phi \to \pi/2$ and discriminates neither zeros nor non-zeros. At $L = 4$ the training converges without attractor collapse but the resulting phase MSE against $\arg Z(t)$ is indistinguishable between zeros and non-zeros (pooled Welch $p > 0.12$ across three seeds, cyclotomic sectors $k = 2, 3$ empty). See Remark 3.2 for the full tabulated results.

Diagnosis. We trace the failure through four stages (the dates indicate the diagnostic chain, not the experimental schedule):

1. **Target decoupling (Remark 4.2, 2026-04-08).** The synthetic target Ψ_n of Equation (43) depends only on $\|X\|$ (or on n) and carries no t-local Riemann-zero information. No choice of $(\alpha, \beta, \gamma, \delta)$ repairs this; the defect is structural.

2. **Supervised replacement is necessary but not sufficient (Remark 4.2, 2026-04-08).** Replacing the target by the analytic ground truth $\Psi(t) = \arg \xi(\frac{1}{2} + it)$ and removing $\mathcal{L}_{\text{pqo}}$ from training (keeping it only as measurement) unlocks the asymmetric basin collapse of $L = 4$ but leaves the headline separation metric near the null ($p = 0.976$, ratio 1.004).

3. **Loss-weight imbalance and arithmetic mean read-out are additional broken links (Remark 4.2, 2026-04-08).** The residual loss weights $\lambda_{\text{res}} = 1$, $\lambda_{\text{curv}} = 0.1$ dominate the supervisory $\mathcal{L}_{\text{tgt}}$, and the channel-wise arithmetic mean used to read out

$\hat{\Psi}$ destroys the $\pm\pi$ wrap signal predicted by the phase-jump structure on the critical line.

4. **Isolation confirms the chain (Remark 4.3, 2026-04-09).** Setting $\lambda_{\text{res}} = \lambda_{\text{curv}} = \lambda_{\text{pqo}} = 0$ and reading out via a circular mean restores supervised learning with $p = 1.18\times 10^{-15}$ on the off-zero set, leaving a residual on-zero MSE of $\approx (\pi/2)^2$ explained entirely by the smooth network's inability to represent the $\pm\pi$ discontinuity of $\arg\xi(\frac{1}{2} + it)$ at each zero.

Lesson. *Four independent failure modes can stack to produce a single negative result.* Before declaring that an analytic identification is experimentally refuted, every link of the supervisory chain must be individually isolated: target injection, loss weighting, measurement operator, and output-space expressivity. The analytic content of the programme $\Psi(t) = \arg\xi(\frac{1}{2} + it)$ survives the cascade; the smooth-field output representation was the last obstruction, *resolved* on 2026-04-10 by replacing the $\{0, \pi\}$ step target with a normalised smooth surrogate $\operatorname{Re}\xi/\max|\operatorname{Re}\xi|$, which reduces the zero/non-zero MSE ratio from 18.2× to 0.03× (Remark 4.3). The complex-vector readout head remains an untested alternative (Section 20.3).

16 ■ Phase-Gauge Gradient Descent (PGGD)

Programme. PGGD is a custom optimiser introduced to exploit the S^1-valued phase structure of the discriminant network's intermediate state. Each parameter update is projected onto the tangent space of the ambient circle bundle so that the optimisation trajectory respects the gauge symmetry $\phi \mapsto \phi + \text{const}$. The hypothesis was that PGGD would outperform Adam on the F_2-based zero detection task of Section 8 by eliminating the spurious global-phase degree of freedom.

Outcome. Negative relative to Adam on every seed tested. PGGD either (i) fails to converge within the same epoch budget, or (ii) converges to a higher training loss with a discriminant gap smaller than Adam's baseline. On the $t \in [10, 50]$ benchmark, Adam's final F_2-MSE is consistently 2–4× smaller than PGGD's best run.

Diagnosis. Two related mechanisms:

1. **Ergodic diffusion on S^1.** The tangent projection used by PGGD removes the radial component of each update but preserves the full angular component. Over many steps the angular update accumulates approximately diffusively, with effective step size η and variance growing linearly in the epoch count. The projected trajectory therefore explores S^1 essentially uniformly (Weyl equidistribution of angles modulo 2π), defeating the very gauge-removal the optimiser was designed to implement. What PGGD gains by removing the radial degree of freedom, it loses to a random walk in the angular one.

2. **Gradient scale mismatch with $\mathcal{L}_{\text{pqo}}$.** The cosine quantisation loss has $O(1)$ gradient away from its minima and $O(|\phi - \phi_*|)$ gradient near them. PGGD's tangent projection amplifies the first regime relative to Adam's adaptive per-parameter scaling, producing an effective learning rate that is too large near the attractors.

Lesson. *Gauge-respecting optimisers do not automatically inherit the sample efficiency of adaptive first-order methods.* A correct tangent projection is compatible with, but does not imply, coherent detection. The PGGD programme is retained as an idea for possible rehabilitation via adaptive per-angle step sizes (see Section 20.3), but in its present form it should not be used in place of Adam on the F_2 detector.

Part V

ξ-Bundle Numerical Evidence

17 ■ Midpoint Curvature Nonlocality

Tier 1. At the midpoint $m = (\gamma_n + \gamma_{n+1})/2$ between consecutive zeros, the Hadamard product ensures that the two nearest-neighbour (NN) contributions to Λ'/Λ cancel exactly: the terms $1/(s - \rho_n)$ and $1/(s - \rho_{n+1})$ sum to zero by symmetry. The residual curvature $\kappa(m) = |\Lambda'/\Lambda(1/2 + im)|^2$ therefore encodes only *non-local* information from distant zeros.

Observation 17.1 (Midpoint curvature encodes the next gap)**.** *For the first 234 consecutive zero pairs of $\zeta(s)$ ($t \in [0, 450]$, dps $= 100$, analytic $\log \xi'$ formula, cap rate 0.0%), the Spearman rank correlations are:*

Pair	ρ	*p-value*	***Interpretation***
κ_{mid} *vs. gap* $(\gamma_{n+1} - \gamma_n)$	-0.031	0.64	*NN cancellation confirmed*
κ_{mid} *vs.* gap_{next} $(\gamma_{n+2} - \gamma_{n+1})$	-0.654	7.3×10^{-30}	***strong nonlocal signal***
κ_{mid} *vs. local density*	$+0.022$	0.74	*no density effect*

The partial correlation $\rho(\kappa_{\text{mid}}, \text{gap}_{\text{next}} \mid \text{gap}) = -0.666$ remains essentially unchanged, ruling out gap autocorrelation as a confound. [confirmed 2026-04-14]

Observation 17.2 (Universality across Dirichlet L-functions)**.** *The midpoint nonlocality persists across three Dirichlet L-functions (χ_3, χ_4, χ_5), using the analytic formula $\Lambda'/\Lambda = \frac{1}{2}\log(q/\pi) + \frac{1}{2}\psi((s+a)/2) + L'/L(s, \chi)$. All tested L-functions exhibit $|\rho(\kappa_{\text{mid}}, \text{gap}_{\text{next}})| > 0.55$ with $p < 10^{-10}$. [confirmed 2026-04-14]*

Table 4: Midpoint curvature nonlocality: L-function vs. random point processes. "NN cancel" column: $|\rho(a)| < 0.3$ confirms nearest-neighbour cancellation is operative.

Point process	n **pairs**	$\rho(a)$ gap	$\rho(b)$ gap_{next}	**partial** ρ	**Verdict**
$\zeta(s)$	234	-0.031	$\mathbf{-0.654}$	-0.666	✓ nonlocal
$L(s, \chi_3)$	112	-0.045	$\mathbf{-0.567}$	-0.576	✓ nonlocal
$L(s, \chi_4)$	120	-0.016	$\mathbf{-0.608}$	-0.629	✓ nonlocal
$L(s, \chi_5)$	127	-0.013	$\mathbf{-0.607}$	-0.613	✓ nonlocal
GUE (500×500, 20 runs)	20×399	-0.037	-0.008 ± 0.034	-0.020	× absent
Poisson (234 pts, 20 runs)	20×232	-0.290	-0.090 ± 0.045	-0.098	× geometric

Observation 17.3 (GUE/Poisson contrast)**.** *To test whether the nonlocal signal is a generic feature of repulsive point processes, we computed* $R(m)^2 = (\sum_{non\text{-}NN} 1/(m-\lambda_k))^2$ *— the direct analogue of* κ_{mid} *— on GUE eigenvalues (*500×500 *Hermitian matrices, bulk 80%, sliding-window unfolding, 20 runs) and Poisson points (234 uniform points, 20 runs).*

GUE: $\rho(b) = -0.008 \pm 0.034$ *(t-test* $p = 0.29$*) — statistically indistinguishable from zero despite GUE's spectral rigidity. Poisson:* $\rho(b) = -0.090 \pm 0.045$ *(*$p = 4.5 \times 10^{-8}$*) — statistically significant but* $|\rho|$ *is* $\sim 1/7$ *of the L-function value; this residual arises from a purely geometric effect: the dominant non-NN term* $1/(m-\lambda_{i+1}) \approx -2/(\text{gap} + \text{gap}_{\text{next}})$ *couples weakly to* gap_{next}.

Conclusion: the midpoint curvature nonlocality is specific to the arithmetic zero placement of the four tested L-functions (ζ*,* χ_3*,* χ_4*,* χ_5*). It is not reproduced by GUE level repulsion or Poisson independence. The mechanism remains unidentified; we record this as an open problem. [confirmed 2026-04-14, 20-run ensemble]*

Observation 17.4 (Nonlocal reach decay and forward–backward sign reversal)**.** *To quantify how far the midpoint curvature "sees," we extend Observation 17.1 by measuring* $\rho(\kappa_{\text{mid}}, \text{gap}_{n+k})$ *for* $k = 0, 1, \ldots, 10$ *(forward) and* $\rho(\kappa_{\text{mid}}, \text{gap}_{n-k})$ *for* $k = 1, \ldots, 10$ *(backward), across all four L-functions.*

Forward decay (ζ***,*** $n = 234$***):***

k	ρ_{fwd}	*p-value*	*partial* ρ
0	− 0.031	6.3×10^{-1}	—
1	**− 0.658**	$\mathbf{3.7 \times 10^{-30}}$	**−0.668**
2	+ 0.068	3.0×10^{-1}	+0.071
3	+ 0.114	8.6×10^{-2}	+0.121
4	+ 0.057	3.9×10^{-1}	+0.065
5	+ 0.003	9.7×10^{-1}	+0.009

The signal drops by an order of magnitude from k=1 *(*$|\rho| = 0.658$*) to* k=2 *(*$|\rho| = 0.068$*), confirming that the nonlocal coupling is sharply concentrated at the nearest non-adjacent gap.*

Forward–backward sign reversal at k=1 ***(all four L-functions):***

L-function	$\rho_{\text{fwd}}(k{=}1)$	$\rho_{\text{bwd}}(k{=}1)$	*sign flip?*
$\zeta(s)$	−0.658	+0.576	✓
$L(s, \chi_3)$	−0.555	+0.504	✓
$L(s, \chi_4)$	−0.608	+0.588	✓
$L(s, \chi_5)$	−0.606	+0.602	✓

For every tested L-function, the forward correlation at k=1 *is* negative *while the backward correlation is* positive*, with comparable magnitudes (*$||\rho_{\text{fwd}}| - |\rho_{\text{bwd}}|| < 0.08$*). This sign reversal is a direct consequence of the Hadamard structure: from the midpoint* $m = (\gamma_n + \gamma_{n+1})/2$*, the dominant non-NN contribution on the right (*γ_{n+2}*) and on the left (*γ_{n-1}*) enter* Λ'/Λ *with opposite signs.*

Decay model. *For the* $|\rho_{\text{fwd}}|$ *sequence (*$k = 1, \ldots, 10$*), power-law fits* $|\rho| \sim A\,k^{-\alpha}$ *consistently outperform exponential fits (*R^2*:* ζ *0.71,* χ_4 *0.76,* χ_5 *0.76;* χ_3 *is a 3-seed preliminary with*

$R^2 = 0.38$). *The rapid decay means that a single characteristic length ξ is difficult to define; the nonlocal signal is concentrated almost entirely at k=1. [confirmed 2026-04-14, 4 L-functions, 3-seed preliminary]*

17.1 Chern Number Interpretation: Numerical Verification

The theoretical framework of Section 6.3 predicts that the zero-counting function $N(T)$ admits a topological interpretation:

$$N(T) \;=\; c_1(\xi\text{-bundle over rectangle}) \;=\; \frac{1}{2\pi}\oint_{\mathcal{C}} d\arg\Lambda(s), \tag{96}$$

where $\mathcal{C}$ is a rectangle $[\sigma_{\min}, \sigma_{\max}] \times [t_0, T]$ and Λ is the completed L-function. We verify this numerically by *phase tracking*: computing $\operatorname{Imlog}\Lambda(s)$ along the contour and accumulating phase differences with 2π-unwrapping, avoiding direct evaluation of $\xi(s)$ (which underflows for $t > 300$).

Observation 17.5 (Chern-number verification for $\zeta(s)$)**.** *Three levels of verification were performed with $\sigma \in [0.3, 0.7]$, dps $= 100$, 64-subdivision trapezoidal quadrature:*

(A) Individual holonomy (first 20 zeros). *For each zero γ_n, a circle of radius $r = 0.3$ was traced. All 20 residues yielded $\mathrm{Re}(\oint/2\pi i) = 0.9984$ (constant systematic offset from 64-point quadrature; 0.9996 at 128 subdivisions), confirming simple-zero residue $= 1$.*

(B) Rectangular phase tracking *($n = 10$–200).*

n	t_2	N_{actual}	N_{ctr}	$\lvert N_{\text{ctr}} - N_{\text{actual}}\rvert$
10	*51.37*	*10*	*9.9882*	*0.0118*
50	*144.56*	*50*	*49.9882*	*0.0118*
100	*237.15*	*100*	*99.9882*	*0.0118*
200	*397.15*	*200*	*199.9882*	*0.0118*

All six tested values ($n \in \{10, 20, 50, 100, 150, 200\}$) passed with a constant offset $|N_{\text{ctr}} - N_{\text{actual}}| = 0.0118$, attributable to $O(1/N^2)$ trapezoidal error on the horizontal segments (confirmed: 128 subdivisions reduce this to 0.003).

(C) Continuous staircase *($t \in [14, 430]$,* ***61 points).*** *The cumulative phase function $C(t)$ tracks $N(t)$ as a perfect staircase with unit jumps at each zero and a constant offset $C(t) = N(t) - 0.022$. Maximum deviation: $|C - N| = 0.0223$. [confirmed 2026-04-14, dps $= 100$, $t \le 430$ (≈ 219 zeros)]*

Observation 17.6 (Dirichlet universality of the Chern-number formula)**.** *The phase-tracking verification extends to Dirichlet L-functions with both odd ($\chi(-1) = -1$, $a = 1$) and even ($\chi(-1) = +1$, $a = 0$) primitive characters, confirming that formula (96) is universal.*

For odd characters, zeros are located by minimising $|L(\frac{1}{2} + it, \chi)|$ and refining with `findroot`*, since $L(\frac{1}{2} + it, \chi)$ is complex. For even real characters, the completed function $\Lambda(\frac{1}{2} + it, \chi)$ is real, but L itself retains a complex phase from the $\Gamma(s/2)$ factor; the $|L|$-minimisation method therefore remains the correct universal approach.*

Character	*Parity*	n	t_2	N_{actual}	N_{ctr}	$\lvert N_{\text{ctr}} - N_{\text{actual}}\rvert$
χ_3 *(mod 3)*	*odd*	5	22.26	5	4.9988	0.0012
		10	34.75	10	9.9988	0.0012
		15	45.20	15	14.9988	0.0012
		20	54.92	20	19.9988	0.0012
χ_4 *(mod 4)*	*odd*	5	19.87	5	4.9985	0.0015
		10	31.12	10	9.9983	0.0017
		15	41.06	15	14.9983	0.0017
		20	50.70	20	19.9983	0.0017
$\left(\frac{\cdot}{5}\right)$ *(mod 5)*	*even*	5	18.55	5	4.9986	0.0014
		10	29.08	10	9.9983	0.0017
		15	38.84	15	14.9983	0.0017
		20	47.42	20	19.9983	0.0017
χ_8 *(mod 8)*	*even*	5	16.11	5	4.9981	0.0019
		10	25.58	10	9.9974	0.0026

All 14/14 tests (8 odd + 6 even) pass with $|N_{\text{ctr}} - N_{\text{actual}}| < 0.003$*, extending the Chern-number interpretation* (96) *beyond* $\zeta(s)$ *to Dirichlet* L*-functions of both parities. The offsets (0.0012–0.0026 vs. 0.0118 for* ζ*) reflect the lower height range (*$t \leq 55$ *vs.* $t \leq 430$*).*

Methodological note. Cross-validating two zero-finding methods—sign changes of $\operatorname{Re} L$ *vs.* $|L|$*-minimisation—on the even character* $\left(\frac{\cdot}{5}\right)$ *revealed that the former misses 2 out of 27 zeros (tangential zeros where* $\operatorname{Re} L$ *does not change sign), confirming that* $|L|$*-minimisation is strictly more reliable. [confirmed 2026-04-14; odd:* χ_3 *mod 3 +* χ_4 *mod 4; even:* $\left(\frac{\cdot}{5}\right)$ *mod 5 +* χ_8 *mod 8]*

Observation 17.7 (Gauss–Bonnet area integral for the ξ-bundle)**.** *As a surface-integral counterpart to the contour-integral Chern number (Obs. 17.5–17.6), we compute the* ***lattice plaquette curvature****—the standard method in lattice gauge theory—on the phase field* $\theta(\sigma, t) = \operatorname{Imlog} \xi(\sigma + it)$*. Each lattice cell's wrapped phase circulation gives the local vortex charge; the total equals* $2\pi N$ *by the discrete Stokes theorem, yielding the Gauss–Bonnet identity*

$$\iint_R F_2 \, d\sigma \, dt \;=\; \sum_{\text{plaquettes}} \Delta\theta_\square \;=\; 2\pi \, N(R), \tag{97}$$

where $N(R)$ *is the number of zeros of* ξ *inside* R*.*

Results *(*$\text{dps} = 100$*,* $N_\sigma = 64$*,* $\sigma \in [0.3, 0.7]$*):*

Test	*Region*	$\iint F_2/(2\pi)$	*Expected N*	|err|	
§A: single zero (γ_1)	$[0.3, 0.7]\times[13, 15.5]$	*1.0000*	*1*	*0.000000*	✓
§B: $k=1$	$[0.3, 0.7]\times[0.5, 17.58]$	*1.0000*	*1*	*0.000000*	✓
§B: $k=5$	$[0.3, 0.7]\times[0.5, 35.26]$	*5.0000*	*5*	*0.000000*	✓
§B: $k=10$	$[0.3, 0.7]\times[0.5, 51.37]$	*10.0000*	*10*	*0.000000*	✓
§B: $k=20$	$[0.3, 0.7]\times[0.5, 78.24]$	*20.0000*	*20*	*0.000000*	✓
§C: empty	$[0.3, 0.7]\times[\gamma_1+0.5, \gamma_2-0.5]$	*0.0000*	*0*	*0.000000*	✓
§D-1: χ_5 *single*	*single zero of* $(\frac{\cdot}{5})$	*1.0000*	*1*	*0.000000*	✓
§D-2: χ_5 *5-zero*	*5 zeros of* $(\frac{\cdot}{5})$	*5.0000*	*5*	*0.000000*	✓

All 8/8 tests yield exact *integer multiples of* 2π *(within display precision). This is the defining property of a topological invariant: the plaquette flux is quantised regardless of lattice resolution (independently verified at* 16×16*). The empty-region test (§C) confirms that* F_2 *is localised at zeros—each zero contributes exactly* 2π *of flux, with zero leakage into neighbouring regions.*

Combined with the contour-integral results (Obs. 17.5–17.6), this establishes the Stokes duality: $\oint_{\partial R}\mathrm{Im}(L\,ds) = \iint_R F_2\,d\sigma\,dt = 2\pi N$*, confirming the* ξ*-bundle's curvature structure from both boundary and bulk perspectives. The area integral achieves higher numerical precision (*$|\mathrm{err}| = 0$ *vs.* 0.012 *for the contour method) because plaquette wrapping enforces exact* 2π *quantisation. [confirmed 2026-04-14;* $\zeta + (\frac{\cdot}{5})$ *mod 5]*

Observation 17.8 (σ-localisation of topological charge)**.** *The Gauss–Bonnet integral (Obs. 17.7) counts zeros inside a rectangular region R. We now ask:* where *in the* σ*-direction does the curvature* F_2 *concentrate? By restricting the* σ*-strip while keeping the same t-range* $[13, (\gamma_{20}+\gamma_{21})/2]$ *(enclosing the first 20* ζ*-zeros), we perform a series of* ***strip-narrowing*** *tests.*

Results *(*$\mathrm{dps} = 100$*,* $N_\sigma = 64$–128*):*

Test	σ*-strip*	$\iint F_2/(2\pi)$	*Expected*	|err|	
§A1: baseline	[0.30, 0.70]	*20.0000*	*20*	*0.000000*	✓
§A2: narrow	[0.49, 0.51]	*20.0000*	*20*	*0.000000*	✓
§A3: ultra-narrow	[0.499, 0.501]	*20.0000*	*20*	*0.000000*	✓
§A4: right off-critical	[0.60, 0.90]	*0.0000*	*0*	*0.000000*	✓
§A5: left off-critical	[0.10, 0.40]	*0.0000*	*0*	*0.000000*	✓
§A6: left half-strip	[0.30, 0.499]	*0.0000*	*0*	*0.000000*	✓
§A7: right half-strip	[0.501, 0.70]	*0.0000*	*0*	*0.000000*	✓

σ-profile. *Resolving the curvature density* $\rho(\sigma) = \sum_j \Delta\theta_\square(\sigma, t_j)$ *across 200 bins in* $\sigma \in [0.3, 0.7]$*: peak value* $\rho(0.5000) = 20\times 2\pi$ *with* $\mathrm{FWHM} < \delta\sigma = 0.004$ *(resolution limit); the 199 remaining bins have residual* $< 1.4\times10^{-15}$ *(machine epsilon).*

Dirichlet extension: *For* $\chi_5 = (\frac{\cdot}{5})$*, narrow strip* $[0.49, 0.51]$ *gives* $N = 5.000000$ *(5 expected); off-critical* $[0.6, 0.9]$ *gives* $N = 0.000000$*—same* δ*-concentration at* $\sigma = 1/2$*.*

These results provide geometric evidence consistent with RH for the first 20 ζ-zeros and the first 5 $L(\chi_5)$-zeros: the plaquette curvature F_2 concentrates as a δ-function at $\sigma = 1/2$, with zero residual flux outside. Excluding $\sigma = 0.5$ by as little as 0.001 (tests §A6–A7) immediately yields $N = 0$. This is a geometric restatement of the known fact that these zeros lie on the critical line—the contribution of this experiment is the quantitative sharpness *of the concentration. [confirmed 2026-04-14; ζ 7/7 + profile + χ_5 2/2; independently reproduced at 16×16, dps $= 50$]*

Observation 17.9 (Triple topological consistency table)**.** *We combine the three independent methods—direct zero counting (A), Chern contour integral (B, Obs. 17.5), and Gauss–Bonnet area integral (C, Obs. 17.7)—to verify self-consistency of the ξ-bundle framework across a range of zero counts $N = 10, 20, 50, 100$. For each N, the upper t-boundary is set to the midpoint $t_2 = (\gamma_N + \gamma_{N+1})/2$ to avoid paths crossing zeros.*

Results *(*dps $= 100$, $\sigma \in [0.3, 0.7]$, $N_\sigma = 64$, *64-point horizontal for Chern):*

N	t_2	A (Direct)	B (Chern)	C (GB)	$\lvert B-A\rvert$	$\lvert C-A\rvert$
10	51.37	10	9.9882	10.0000	0.0118	0.0000
20	78.24	20	19.9881	20.0000	0.0119	0.0000
50	144.56	50	49.9882	50.0000	0.0118	0.0000
100	237.15	100	99.9882	100.0000	0.0118	0.0000

Precision hierarchy. *The Gauss–Bonnet method achieves exact integer quantisation ($|C-A| = 0$) for all N, while the Chern contour integral exhibits a constant systematic offset $|B-A| \approx 0.012$, independent of N. This offset arises from horizontal-edge discretisation in the phase-tracking contour,* not *from accumulation of per-zero errors—its N-independence is itself a consistency check.*

High-height σ-localisation ($N = 100$). *Extending Obs. 17.8 to the first 100 ζ-zeros ($t_2 = 237.15$): narrow strip $\sigma \in [0.49, 0.51]$ yields $N = 100.000000$; off-critical $\sigma \in [0.6, 0.9]$ yields $N = 0.000000$. The δ-concentration at $\sigma = 1/2$ persists to heights five times those tested in Obs. 17.8.*

The triple agreement across four decades of N establishes the internal coherence of the differential-geometric framework: curvature (local), Chern number (semi-local), and Gauss–Bonnet (global) all encode the same topological content. [confirmed 2026-04-14; 4/4 triple + 2/2 σ-loc; independently reproduced at 16×128, dps $= 50$]

Observation 17.10 (Synthetic off-critical negative control (specificity))**.** *To rule out the possibility that the plaquette method* inherently *favours $\sigma = 1/2$, we apply it to three synthetic polynomial functions whose zeros are placed at known, prescribed locations:*

- *$f_1(s) = \prod_k (s - z_k)$ with all zeros on $\sigma = 1/2$ ("RH-like": $z_k = \frac{1}{2} + 14i,\ \frac{1}{2} + 21i,\ \frac{1}{2} + 25i$);*
- *$f_2(s)$ with mixed zeros ($\sigma = 0.3,\ 0.7,\ 0.5$);*
- *$f_3(s)$ with* no *zeros on $\sigma = 1/2$ ($\sigma = 0.3,\ 0.7,\ 0.6,\ 0.4$).*

Results *(*dps $= 30$*,* $N_\sigma = 64$*,* $N_t = 200$*; 11 tests across three functions):*

Test	*Fn*	*σ-range*	N_{plaq}	*Expected*	*Pass*
§A-1	f_1	$[0.3, 0.7]$	*3.0000*	*3*	✓
§A-2	f_1	*profile*	*1 peak@0.500*	*1 peak*	✓
§A-3	f_1	$[0.6, 0.9]$	*0.0000*	*0*	✓
§B-1	f_2	$[0.2, 0.8]$	*3.0000*	*3*	✓
§B-2	f_2	*profile*	*3 peaks*	*3 peaks*	✓
§B-3	f_2	$[0.49, 0.51]$	*1.0000*	*1*	✓
§B-4	f_2	$[0.6, 0.8]$	*1.0000*	*1*	✓
§B-5	f_2	$[0.2, 0.4]$	*1.0000*	*1*	✓
§C-1	f_3	$[0.2, 0.8]$	*4.0000*	*4*	✓
§C-2	f_3	*profile*	*4 peaks,* $\rho(\frac{1}{2}) = 0$	*4 peaks*	✓
⋆ *§C-3*	f_3	$[0.49, 0.51]$	***0.0000***	***0***	✓

Key result (§C-3). *For* f_3*, which has* no *zeros on the critical line, the plaquette flux in* $\sigma \in [0.49, 0.51]$ *is exactly zero. If the framework artificially concentrated charge at* $\sigma = 1/2$*, this strip would register nonzero flux regardless of zero placement. The null result establishes* ***specificity****: the* δ*-concentration observed for* ξ *(Obs. 17.8–17.9) is a property of the* ξ*-zeros themselves, not a methodological artefact.*

§B individual σ-separation. *The three zeros of* f_2 *at* $\sigma = 0.3, 0.5, 0.7$ *each contribute* $N = 1$ *exclusively to their own narrow strip (§B-3/B-4/B-5), confirming that the plaquette resolves the real part of each zero.*

All 11 tests yield exact integer quantisation (error $= 0$*), as expected for polynomials; the significance lies not in the precision but in the* absence of $\sigma = 1/2$ bias. *[confirmed 2026-04-14; 11/11; independently reproduced (§C-3, §B-3, §C-1) at* 32×100 *and* 48×100 *grids]*

Observation 17.11 (Number variance: sub-GUE arithmetic correction regime)**.** *To verify that the plaquette curvature field correctly encodes the* collective *statistics of Riemann zeros—not just individual holonomies—we compute the number variance* $\Sigma^2(L)$ *in two independent ways: (i) direct zero counting in unfolded windows of size* L*, and (ii) via the plaquette total flux* N_{plaq} *across sliding windows.*

Setup. *Grid:* $N_\sigma = 32$*,* $N_t = 2000$*,* $\sigma \in [0.3, 0.7]$*,* $t \in [50, 600]$*,* $dps = 80$*. Unfolded windows* $L \in \{5, 10, 20, 50\}$*, sliding step* $= 1$ *zero, boundary trimmed to* $t \in [70, 580]$*. Unfolding uses 337 zeros in* $[45, 605]$*; mean unfolded spacing* $= 1.0008$ *(theory* $= 1.000$*, error* $< 0.1\%$*). GUE prediction:* $\Sigma^2_{\text{GUE}}(L) \approx \frac{2}{\pi^2} \ln L + 0.44$ *plus Berry–Keating arithmetic correction [19].*

Results.

L	N_{win}	n_{eff}	Σ^2_{dir}	Σ^2_{plaq}	Σ^2_{GUE}	$\Sigma^2/\Sigma^2_{\text{GUE}}$
5	*304*	*60*	*0.3960*	*0.4257*	*0.7682*	*0.515*
10	*299*	*29*	*0.5134*	*0.5604*	*0.9086*	*0.565*
20	*289*	*14*	*0.4826*	*0.4792*	*1.0491*	*0.460*
50	*259*	*5*	*0.4494*	*0.4841*	*1.2348*	*0.364*

Three findings.

1. ***Methodology validation (plaquette ≈ direct).*** *Σ^2_{plaq} and Σ^2_{dir} agree within* 10% *across all four window sizes. The plaquette curvature field therefore encodes the* same *collective statistical structure as direct zero counting, extending the topological correspondence from individual holonomies (Obs. 17.5–17.6) to ensemble variance.*

2. ***Sub-GUE regime: Berry arithmetic correction.*** *$\Sigma^2/\Sigma^2_{\text{GUE}} = 0.36$–$0.57$, consistently below unity. This is not anomalous; it is the expected* arithmetic correction *identified by Berry [19] and confirmed numerically by Odlyzko [20]: at heights $t \sim 300$, the logarithmic GUE growth is suppressed by prime-sum arithmetic oscillations, with full GUE convergence appearing only at $t \gg 10^6$. The log–log slope of $\Sigma^2(L)$ is* 0.036 *(GUE predicts $\sim$1; fully flat = 0), reproducing Berry's "arithmetic saturation" regime.*

3. ***Poisson complete exclusion.*** *Poisson predicts $\Sigma^2_{\text{Poi}} = L$. Observed variances fall* 63–156 σ *below the Poisson level, confirming that the plaquette flux correctly reflects the level repulsion structure of the zero sequence.*

Interpretation note. *The sub-GUE result should* not *be read as a failure of GUE universality. It is a known finite-height effect [19]: "sub-GUE" and "GUE" are the same universality class at different height regimes. The primary significance here is the* methodology validation*: the plaquette correctly captures collective statistics beyond individual winding numbers. [confirmed 2026-04-14; 4 window sizes; plaquette independently cross-validated against direct zero count]*

Observation 17.12 (Epstein Zeta Extension)**.** *To test whether the bundle framework extends beyond Riemann ζ and Dirichlet L-functions, we examine the Epstein zeta function $Z_Q(s) = \sum_{(m,n)\neq(0,0)} Q(m,n)^{-s}$ for the binary quadratic form $Q(m,n) = m^2 + \lambda n^2$.*

Setup. *Two lattice parameters: $\lambda = 1$ (control; $Z_Q = 4\,\zeta(s)\,L(s,\chi_{-4})$, RH satisfied) and $\lambda = 5$ (experiment; class number $h(-20) = 2$, individual Epstein zeta may violate RH). Analytic continuation via θ-Mellin integral with $t = 1$ splitting; verified $Z_{Q,\lambda=1}$ against $4\,\zeta(s)\,L(s,\chi_{-4})$ (relative error $\sim 10^{-70}$). Bundle diagnostics: κ at $\delta = 0.03$ offset, monodromy by contour integral (radius = 0.5, 64 steps), σ-localisation FWHM. Critical-line scan: $t \in [2, 35]$; off-critical grid: $\sigma \in [0.25, 0.75]$, $t \in [2, 35]$, 12×100 lattice.*

Results.

	κ	*mono*/π	*FWHM*	*off-crit. zeros*
$\lambda = 1$ ($N = 8$)	1007.9 ± 72.6	2.00 ± 0.00	0.044	*0*
$\lambda = 5$ ($N = 8$)	1006.2 ± 47.8	2.00 ± 0.00	0.044	*0*

Findings.

1. ***On-critical signature universality.*** *All three bundle metrics (κ, monodromy, FWHM) are indistinguishable between $\lambda = 1$ and $\lambda = 5$ (deviation $< 1\%$), indicating that the framework's diagnostic signatures are independent of the underlying quadratic form.*

2. ***Off-critical zeros not found.*** *No zeros with $\sigma \neq \frac{1}{2}$ were detected in the search region. This is a* range limitation, *not evidence of absence: off-critical zeros for $\lambda = 5$ may appear at $t \gg 35$ or require finer grid resolution.*

Interpretation. *The Epstein zeta result confirms extensibility of the bundle framework to a third L-function family (after ζ and Dirichlet L), but does* not *complete the falsifiability test. The primary goal—detecting off-critical zeros and measuring signature differences—requires functions with* guaranteed *off-critical zeros (e.g. the Davenport–Heilbronn function; see §20.3).* *[confirmed 2026-04-14; $\lambda \in \{1, 5\}$; dps = 80; 78.7 min runtime]*

Observation 17.13 (High-t Robustness and κ Calibration)**.** *To verify that the bundle framework's core diagnostics remain stable at large imaginary part, we measure κ, monodromy, and σ-localisation across six Riemann zeros spanning $t \approx 14$ to $t \approx 10\,000$ (zero indices $N = 1$ to $N = 10\,142$).*

Setup. *All measurements use the analytic log-derivative formula*

$$\frac{\xi'(s)}{\xi(s)} = \frac{1}{s} + \frac{1}{s-1} - \frac{\log \pi}{2} + \frac{\psi(s/2)}{2} + \frac{\zeta'(s)}{\zeta(s)},$$

which avoids the numerical instability of finite-difference ξ' at large t. $\kappa = |\xi'/\xi|^2$ is evaluated at offset $\delta = 0.03$ (i.e. $\sigma = 0.53$). σ-profiles use a fixed *201-point grid ($\sigma \in [0.2, 0.8]$, spacing 0.003) at every t, eliminating the resolution artefact identified in preliminary runs. Monodromy: contour integral, radius 0.2, 64 steps. Precision:* dps $= 60$ *for $t < 2000$,* dps $= 80$ *for $t \geq 2000$.*

Part A: κ calibration. *We first recalibrate the baseline κ at low t using the same analytic formula (previously, the* `bundle_utils` *routine with $h = 10^{-20}$ finite differences gave $\kappa \approx 1030$).*

N	t	κ (analytic)	$\kappa/(1/\delta^2)$
1	14.135	1111.58	1.0004
2	21.022	1111.96	1.0008
3	25.011	1111.78	1.0006
4	30.425	1112.58	1.0013
5	32.935	1111.92	1.0007
Mean ($t < 100$)		1111.96 ± 0.33	—

The analytic-formula κ at low t is 1112.0, matching the high-t value 1118.6 within 0.6%; the previous baseline $\kappa \approx 1030$ is an 8% underestimate caused by over-cancellation in the $h = 10^{-20}$ finite-difference scheme. Corrected baseline: $\kappa \approx 1112$–1120 (analytic).

Part B: t-scaling comparison (identical 201-point grid).

Metric	$t < 100$ (3 zeros)	$t > 1000$ (3 zeros)	Difference
κ (analytic)	1112.0 ± 0.3	1118.6 ± 1.9	0.6%
peak_σ	0.4970	0.4970	0.0000
FWHM	0.0060	0.0060	1.000×
mono/π	2.0000	2.0000	exact

Findings.

1. *κ **t-stability.** The curvature scalar κ varies by only 0.6% across a $700\times$ range in t. The leading-order value $\kappa \approx 1/\delta^2 = 1111.1$ is analytically expected from the pole structure of ξ'/ξ; the sub-leading ζ'/ζ correction contributes $< 0.5\%$ and remains t-independent.*

2. ***Monodromy stability.** All six zeros yield mono $= 2.0000\pi$, confirming single-zero encirclement at radius 0.2 even at $t \approx 10\,000$ (mean zero spacing ≈ 0.68 at this height).*

3. *σ**-localisation invariance.** With identical grid resolution, peak_σ and FWHM are indistinguishable between low and high t. Preliminary runs with variable grid sizes showed apparent t-dependence that was entirely a resolution artefact.*

Interpretation. *This result is a* robustness confirmation*, not a new mathematical discovery: it verifies that the analytic ξ'/ξ infrastructure and the three bundle diagnostics (κ, monodromy, σ-FWHM) remain numerically stable up to $t = 10^4$. The κ calibration also resolves a long-standing 8% discrepancy between the finite-difference and analytic baselines. [confirmed 2026-04-15; 6 zeros ($N \in \{1, 2, 3, 649, 4520, 10142\}$); analytic ξ'/ξ formula; 201-point grid; 105 s runtime]*

Observation 17.14 (Curvature Subleading Structure and Hadamard Decomposition)**.** *Expanding the curvature scalar $\kappa = |\xi'/\xi|^2$ beyond the leading-order $1/\delta^2$ pole, we isolate the* subleading *correction $\Delta\kappa := \kappa - 1/\delta^2$ and show that it encodes arithmetic content through a Hadamard product decomposition.*

Setup. *Twenty Riemann zeros ($N = 1$ to $N = 20\,000$, $t \approx 14$ to $t \approx 18\,046$) are probed at offset $\delta = 0.03$. The analytic log-derivative $\xi'/\xi(s) = G(s) + \zeta'/\zeta(s)$ separates into a gamma/pole contribution $G(s) = 1/s + 1/(s-1) - \frac{1}{2}\log\pi + \frac{1}{2}\psi(s/2)$ and the Dirichlet-series part $\zeta'/\zeta(s)$.*

Writing $R(s) = \xi'/\xi(s) - 1/\delta$ (the remainder after removing the nearest-zero pole), one has

$$\Delta\kappa \;=\; \frac{2\,\mathrm{Re}\,R}{\delta} + |R|^2.$$

Part A: δ-independence. *The correction $\Delta\kappa$ is essentially independent of δ: a power-law fit $\Delta\kappa \propto \delta^\alpha$ yields $\alpha \approx -0.002$, consistent with zero (theoretical prediction $\alpha = 1$ is ruled out). This implies κ decomposes cleanly as $\kappa = 1/\delta^2 + f(t)$, separating geometric ($\delta$) and arithmetic ($t$) content.*

Part B: Logarithmic scaling. *The subleading correction grows logarithmically:*

$$\Delta\kappa \;\approx\; 1.59 \cdot \log\!\big(t/(2\pi)\big) + c, \qquad R^2 = 0.860, \;\; p = 4.1 \times 10^{-9}.$$

While $R^2 = 0.86$ is below the 0.9 threshold for "established," the highly significant p-value confirms the logarithmic trend.

Part C: Hadamard decomposition of the slope. *The slope $a = 1.59$ arises from a near-cancellation between two large contributions:*

Component	*Slope vs* $\log(t/2\pi)$	R^2
$G(n)$ *[gamma + poles]*	$+0.5000$	1.0000
$S_{\mathrm{all}}(n)$ *[zeta zeros]*	-0.4768	0.9996
$\mathrm{Re}\,R = G + S_{\mathrm{all}}$	$+0.0232$	0.854
$2\,\mathrm{Re}\,R/\delta$	$+1.548$	0.854
$\mathrm{Im}(R)^2$	$+0.042$	0.041
$\Delta\kappa$ *(total)*	$+1.595$	0.860

The gamma contribution (+0.500) and the zero-sum contribution (-0.477) *cancel to* 95.4%*; the residual* 0.023 *is amplified by* $2/\delta = 66.7$ *to produce the observed slope* ≈ 1.55*. A theoretical estimate based on the Hadamard product structure predicts a* ≈ 1.5*, consistent with the measured* 1.59 ± 0.15 *within* 1σ*.*

Part D: Outlier analysis (*N*=7000***).*** *The largest residual* ($\Delta\kappa = 14.08$ *vs. predicted* ≈9–10*) is traced to a close zero pair: zeros 6999 and 7000 have gap* 0.475 *(vs. local mean* 0.89*), generating an anomalously large* $\mathrm{Re}\,R = 0.204$*. This is consistent with GUE level-repulsion statistics (rare close encounters).*

Caveat. *A raw correlation* $r = -0.755$ *between zero gap and* $\Delta\kappa$ *is a spurious association driven by the common confound* t*. After unfolding (removing* t*-trends),* $r = +0.256$ *(*$p = 0.28$*): gap–*$\Delta\kappa$ *coupling is* not *statistically significant.*

Interpretation. *The subleading structure* $\Delta\kappa \approx 1.59\log(t/2\pi)$ *provides preliminary numerical evidence that the* ξ*-bundle curvature captures arithmetic content beyond the trivial* $1/\delta^2$ *pole. The Hadamard decomposition verifies (to 0.0% error) that this slope originates from a near-cancellation between the gamma function and zeta-zero contributions—a non-trivial structural feature of the Hadamard product representation.* ***Part E: NNS-dependent Lorentzian*** ^{2} ***correction (99 zeros).*** *Scaling to 99 zeros (*$N = 203$ *to* 20 000*,* $t \approx 402$–18 046*), we test whether the nearest-neighbour spacing (NNS) explains the residual* $\mathrm{res}_n := \Delta\kappa_n - (a\log(t_n/2\pi) + b)$ *left by the 1-variable log-fit.*

The 1-variable OLS gives $R^2 = 0.015$ *(*$p = 0.22$*): at this scale, two extreme outliers (*$\Delta\kappa = 568, 259$*; both with* $NNS_{\mathrm{unf}} < 0.15$*) dominate the residual.*

Correlation with NNS. *Pearson* $r(\mathrm{res}, \mathrm{NNS}_{\mathrm{unf}}) = -0.474$ *(*$p = 7.4 \times 10^{-7}$*); Spearman* $\rho = -0.739$ *(*$p = 2.6 \times 10^{-18}$*). A theoretically motivated Lorentzian*^{2} *ansatz* $f_1 = \delta^2/(\delta^2 + g_{\min}^2)^2$*, where* $g_{\min}$ *is the nearest raw gap, achieves Pearson* $r(\mathrm{res}, f_1) = 0.953$ *(*$p \approx 10^{-52}$*).*

2-variable regression.

NNS function f	$R^2_{2\text{-}var}$	F*-statistic*
$f_1 = \delta^2/(\delta^2 + g_{\min}^2)^2$ *(Lorentzian*^{2}*)*	**0.9185**	*1064*
$f_2 = 1/NNS_{\mathrm{unf}}$	*0.8806*	*696*
$f_3 = \log(NNS_{\mathrm{unf}})$	*0.5182*	*100*

The Lorentzian^2 *model lifts* R^2 *from 0.015 to 0.919, explaining* 90.3% *of the previously unaccounted variance. The theoretical correlation* $|\,\mathrm{Im}\,R| \sim 1/g_{\min}$ *is confirmed with* $r = 0.892$ *(*$p < 10^{-34}$*).*

Outlier anatomy. *Both* $|\mathrm{res}| > 2\sigma$ *outliers (*$N = 11919, 16767$*) have* $NNS_{\mathrm{unf}} < 0.15$*—GUE level-attraction events captured at 100%.*

Caveat. *The KS test rejects the Wigner surmise for the sampled NNS distribution ($D = 0.41$, $p < 10^{-15}$), attributable to the equally-spaced index selection (not random sampling). This does not invalidate the correlation analysis but limits distribution-level claims.*

[confirmed 2026-04-15; 99 zeros (N=203 to 20000); δ=0.03; $R^2_{1\text{-}var}$=0.015, $R^2_{2\text{-}var}$=0.919; Hadamard decomposition error =0.00%; 3-seed preliminary]

Part F: δ-sweep universality ($\delta \in \{0.01, 0.02, 0.03, 0.05, 0.10\}$). *To test whether the Lorentzian^2 correction is universal across probe distances, we extend to five offsets (99 zeros each, $N = 203$–$20\,000$).*

Per-δ 2-variable regression.

δ	$R^2_{1\text{-}var}$	$R^2_{2\text{-}var}$	$c(\delta)$
0.01	*0.014*	***0.925***	*147.8*
0.02	*0.015*	***0.923***	*39.4*
0.03	*0.015*	***0.919***	*19.4*
0.05	*0.017*	***0.909***	*9.1*
0.10	*0.024*	***0.890***	*4.9*

The Lorentzian^2 form *achieves $R^2 > 0.88$ for all five offsets, establishing universality of the correction* shape *across a 10-fold range of δ.*

Negative result: coefficient non-universality. *The coefficient $c(\delta)$ varies from 147.8 ($\delta = 0.01$) to 4.9 ($\delta = 0.10$)—a 30-fold variation implying $c(\delta) \sim \delta^{-\alpha}$ for non-trivial α. The product $a(\delta) \cdot \delta$ has CV = 47.8% (theory predicts $\lesssim 20\%$). A unified 3-parameter model $\Delta\kappa = (a/\delta)\log(t/2\pi) + b/\delta^2 + c \cdot f_1$ yields $R^2 = 0.449$, well below the 0.90 threshold. This failure indicates that the δ-scaling of the coefficient is not a simple power law and warrants further analytic investigation.*

Numerical stability at $\delta = 0.01$. *With $1/\delta^2 = 10\,000$, the subtraction $\Delta\kappa = \kappa - 10\,000$ risks digit cancellation. Elevating to dps = 80 gives $R^2_{2\text{-}var} = 0.925$ with all $\Delta\kappa > 0$: no numerical artefact.*

[confirmed 2026-04-15; 5 offsets $\delta \in \{0.01, 0.02, 0.03, 0.05, 0.10\}$; 99 zeros each ($N$=203–20000); Lorentzian^2 universal (5/5, R^2>0.88); unified 3-param model R^2=0.449 (negative)]

Observation 17.15 (Boundaries of the RMT Connection)**.** *Having established positive RMT connections in three results (#20 pair correlation, #28 number variance, #32 NNS–$\Delta\kappa$ Lorentzian^2), we conducted four further experiments testing whether the curvature field $\kappa(t)$ obeys finer RMT predictions at the level of individual distributions and correlation functions. We establish that these finer predictions* do not hold, *thereby precisely delineating the scope of the RMT analogy.*

Result #34 (Experiment #36): Spacing ratio distribution. *The ratio $\tilde{r}_n = \min(s_n, s_{n+1})/\max(s_n, s_{n+1})$ is formed from $N = 4657$ consecutive Riemann zero gaps at $t > 600$. After correcting the theoretical mean to $\mathbb{E}[\tilde{r}]_{GUE} = 0.6027$ (Atas et al. 2013), the empirical mean is $\bar{r} = 0.6161$, a systematic excess of $+2.2\%$ (4.15σ). KS test against the GUE Wigner surmise: $D = 0.030$, $p = 4.7 \times 10^{-4}$. The ratio $D_{GUE}/D_{Poisson} = 0.079$ confirms that the spacing ratio distribution* is closest to GUE *among the three ensembles tested, but the*

systematic bias persists across all t-cutoffs ($t > 400, 600, 800, 1000$) and does not converge toward GUE as N increases. ***Verdict: Negative*** *($p < 5 \times 10^{-4}$ for all cutoffs; convergence toward GUE absent).*

Result #35 (Experiment #37): Curvature value distribution. *Under the variable substitution $\kappa = 1/(\delta^2 + s_{unf}^2)^2$, if s_{unf} followed the GUE level-spacing distribution, then κ would follow a predictable distribution $P_{pred}(\kappa)$. We test this with $N = 1870$ valid samples from $t \in [1000, 5000]$, $\delta = 0.03$. KS test: $D = 0.361$, $p = 6.3 \times 10^{-213}$. The Q–Q correlation in log-scale is $r = 0.971$, indicating that the* shape *of the distribution is preserved, but the* scale *is mismatched by a factor of $7.6\times$ at the median. The root cause is twofold: (i) uniform-t sampling yields $\bar{s}_{unf} = 0.306$ instead of the GUE theoretical expectation 1.000; and (ii) the background term $G(s)$ in $\kappa = |G(s) + \zeta'/\zeta(s)|^2$ partially cancels the dominant pole at distances $|s - \gamma_n| \gg \delta$, introducing a systematic scale offset not captured by the simple model $1/(\delta^2 + s^2)^2$.* ***Verdict: Negative*** *(KS $p < 10^{-200}$; distribution shape preserved in log-scale but scale mismatched by $7.6\times$).*

Result #36 (Experiment #38): κ autocorrelation. *We compute the empirical autocorrelation function $C(\tau) = \mathrm{Corr}(\kappa(t), \kappa(t + \tau))$ at $N = 8001$ equally-spaced points ($\Delta t = 0.5$, $t \in [1000, 5000]$), converting to unfolded lag via $\tau_{unf} = \tau \cdot \bar{d}$ with $\bar{d} = 0.982$ (midpoint mean zero density). The GUE two-point cluster function $b_2^c(r) = 1 - (\sin \pi r/\pi r)^2$ predicts a first zero-crossing at $\tau_{unf} = 1.0$. Empirically, the first zero-crossing occurs at $\tau_{unf} = 0.448$ (deviation 0.552, exceeding the ± 0.20 acceptance criterion). The amplitude fit yields $A = 0.190$, meaning C_{emp} oscillates at only 19% of the GUE-predicted amplitude. Over the full lag range ($\tau_{unf} \leq 20$): $R^2 = 0.48$. Crucially, $C(1) = -0.081$: the autocorrelation drops to near-zero already at lag 1 ($\tau_{unf} \approx 0.48$), establishing that $\kappa(t)$* is effectively uncorrelated at the grid spacing $\Delta t = 0.5$. ***Verdict: Negative*** *(zero-crossing $\tau_{unf} = 0.45 \neq 1.0$; $R^2 = 0.48$; κ effectively uncorrelated for lag ≥ 1).*

Interpretation: Macro-statistics vs. micro-distributions. *The established positive RMT results (#20 pair correlation, #28 number variance) are two-point statistics of the zero* locations *themselves—they capture the average density structure of the spectrum, for which GUE universality is robust and well-understood (Berry–Tabor, Bohigas–Giannoni–Schmit). The negative results (#34–#36) instead probe the* full distribution *or* temporal autocorrelation *of the curvature κ, a non-linear functional of the zero locations that involves additional ingredients absent from the GUE model:*

- *The background term $G(s)$ from gamma and pole contributions ($|G(s)|^2 \approx 6$–9) partially cancels the dominant pole $1/\delta^2$ at distances $|s - \gamma_n| \gg \delta$, introducing a systematic scale offset invisible to simple GUE substitution.*
- *The curvature $\kappa(t)$ is* locally determined *by the nearest zero position. Its spatial correlation length is on the order of the grid spacing ($\Delta t = 0.5$, $\tau_{unf} \approx 0.48$), far shorter than the GUE correlation length ($\tau_{unf} \approx 1$).*

We therefore establish the following boundary:

> The RMT connection operates at the level of spectral two-point statistics (pair correlation, number variance) and at the level of the NNS–curvature functional

correlation (Lorentzian^2 ansatz), but does not extend to the full spacing-ratio distribution, the curvature value distribution, or the temporal autocorrelation of the curvature functional.

This boundary is a precise and informative scientific finding, not a failure: it identifies exactly which aspects of the ξ-bundle geometry are governed by random-matrix universality and which are sensitive to arithmetic background terms.

[negative, confirmed 2026-04-15; #34: KS $p = 4.7 \times 10^{-4}$, systematic excess +2.2% (all cutoffs); #35: KS $D = 0.361$, scale 7.6×, log Q–Q shape $r = 0.971$; #36: zero-crossing $\tau_{unf} = 0.448 \neq 1.0$, $R^2 = 0.48$; independent verification: 10/10 items reproduced]

18 Structural Correspondence Table

We formalise the seven structural correspondences identified throughout this work:

Table 5: Seven structural correspondences between QROP-Net and the resonance framework.

#	QROP-Net (Concrete)	Resonance (Analytic)	Shared Mathematical Structure
1	$\sin^2(2^{d-1}\phi)$ quantisation loss	$\Delta v = m\pi$ phase quantisation (Gate A)	Periodic phase penalty: $\mathcal{P}(\phi; L) = \sin^2(L\phi/2) = 0$ at lattice positions
2	NTT (discrete) + FrFT (continuous fractional)	Fourier analysis of $\xi(s)$ on the critical line	Spectral decomposition: algebraic structure $\leftrightarrow$ frequency domain
3	$X^n + 1 = 0$ roots: n equally-spaced phases on unit circle	Phase winding $\oint \nabla v \cdot ds = 2\pi N$	Cyclotomic/circular quantisation: discrete phases on S^1
4	LWE error threshold: $\lvert e\rvert_{\mathbb{Z}_q} < q/4 = 1920$	σ-lift: $\lvert\sigma^* - 1/2\rvert = 0$ (RH)	Hard error boundary beyond which signal recovery fails
5	$U_0 = e^{i\phi}$, $\lvert U_0\rvert \equiv 1$; intensity $I = \lvert U_d\rvert^2$	$A(t) = \lvert\xi\rvert$ vs $\Phi_\xi(t) = \arg\xi$	Amplitude–phase separation: information encoded in phase, measured via amplitude
6	Blind TTO: self-calibration without ground truth	Gauge ODE: $F_2 = 0$ without knowing zero locations	Ground-truth-free convergence via internal consistency
7	$\Delta d = 0.1\,\text{mm} \to$ DFR $= 1$	σ-lift $\neq 0 \to$ non-unitarity (absorption)	Sensitivity threshold: small parameter shift $\to$ catastrophic failure

18.1 Mathematical Formalisation of the Correspondences

We express each correspondence as a structural mapping.

Definition 18.1 (Phase Quantisation Functor)**.** Let $(\mathcal{A}, \Phi_\mathcal{A}, L_\mathcal{A})$ be a *phase quantisation triple* consisting of an algebraic structure $\mathcal{A}$, a phase function $\Phi_\mathcal{A} : \mathcal{A} \to [0, 2\pi)$, and a lattice level count $L_\mathcal{A} \in \mathbb{N}$. The phase quantisation operator $\mathcal{P}(\cdot; L)$ acts identically on all such triples:

	$\mathcal{A}$	$\Phi_{\mathcal{A}}$	$L_{\mathcal{A}}$
QROP-Net	$\mathbb{Z}_q^{1024}$	$\phi_i = 2\pi c_i/2^{d_u}$	$2^{d_u} = 2048$
Cyclotomic	R_q	$\arg(\omega_j) = \pi(2j+1)/n$	$2n = 512$
Critical line	$\{t \in \mathbb{R}^+\}$	$\vartheta(t) \bmod \pi$	2 (per Gram interval)

In each case, the condition $\mathcal{P}(\Phi_{\mathcal{A}}; L_{\mathcal{A}}) = 0$ selects the "aligned" elements, and $\nabla\mathcal{P}$ drives misaligned elements toward the nearest lattice position.

18.2 Dynamical and spectral correspondences

Tier 2. The correspondence table above is structural. This subsection records four additional correspondences that connect the empirical phenomena of Sections 8–9 to classical mathematical frameworks. Each is established at the level of analogy and partial numerical agreement; we do not claim rigorous equivalences.

18.2.1 Maslov correction as the metaplectic sheet shift

The replacement of $\sin^2(L\phi/2)$ by $\cos^2(L\phi/2)$ as the quantisation penalty — the "$\cos^2$ Maslov correction" invoked throughout Section 8 — is the neural shadow of a classical metaplectic phenomenon.

Theorem 18.1 (Metaplectic correction of Gate A)**.** *For the phase quantisation operator* $\mathsf{PQO}(\varphi; L)$ *applied at zeros of a real-valued function where* $\varphi \to \pi/2$ *(Corollary 8.2), the* $\sin^2$ *form gives* $\mathsf{PQO}(\pi/2; 2) = 1$ *(maximal penalty), while the* $\cos^2$ *form gives*

$$\mathsf{PQO}_{\cos}(\varphi; L) := \cos^2(L\varphi/2), \quad \mathsf{PQO}_{\cos}(\pi/2; 2) = 0. \tag{98}$$

The replacement $\sin^2 \to \cos^2$ *is the* Maslov correction *with index* $\mu = 2$*, corresponding to a phase shift of* $\mu\pi/4 = \pi/2$*.*

Table 6: The $\sin^2 \leftrightarrow \cos^2$ duality across mathematics.

Domain	$\sin^2$ **regime**	$\cos^2$ **regime**	**Name of shift**
WKB/semiclassical	Away from turning points	At turning points	Maslov index μ
Symplectic geometry	Standard Lagrangian chart	Crossed Maslov cycle	$\pi_1(\Lambda(n))$
Harmonic analysis	In-phase (I)	Quadrature (Q)	Hilbert transform
Lattice cryptography	Lattice decoding	Coset decoding	$\Lambda \to \Lambda + v$
Berry–Keating	$\vartheta(t_n) = n\pi$	$\vartheta(t_n) = (n - \frac{1}{2})\pi$	half-integer shift
Representation theory	Compact Cartan	Split Cartan	Cayley transform

The unifying object is the metaplectic group $\mathrm{Mp}(2n, \mathbb{R})$, the double cover of $\mathrm{Sp}(2n, \mathbb{R})$, whose two sheets correspond exactly to the two quantisation phases. The functional equation $\xi(s) = \xi(1-s)$ arises from the action of this group on theta functions via $\tau \to -1/\tau$, and the

Berry–Keating quantisation condition

$$\vartheta(t_n) = (n - \tfrac{1}{2})\pi + O(1/t_n) \tag{99}$$

encodes the same $-1/2$ shift on the analytic side.

18.2.2 Gauge ODE critical transition as Kuramoto synchronisation

Observation 18.2 (Gauge critical transition)**.** *In the F_2-detector of Section 8, the gauge buffer states (φ, ψ) exhibit a sharp transition at $\sim$10 epochs: the standard deviation* $\mathrm{std}(\varphi)$ *drops from ≈ 0.10 to ≈ 0.03, coinciding with the epoch at which F_2 reaches convergence on all training zeros. The transition is invariant across seeds and has been reproduced on both $t \in [10, 50]$ and $t \in [100, 200]$.*

Remark 18.1 (Kuramoto synchronisation)**.** Observation 18.2 is a finite-population Kuramoto-type synchronisation transition. The gauge phases $\{\varphi_i\}$ are batch-local fields; the shared residual loss $\mathcal{L}_{\mathrm{res}}$ acts as a coupling functional between them. When the effective coupling strength (controlled by the learning rate and the sharpness of the target) exceeds a critical value, the phases lock onto a common frequency profile determined by the target zero locations. The $\mathrm{std}(\varphi)$ drop of $\approx 3\times$ at epoch ~ 10 is the neural-network signature of this locking.

[confirmed 2026-04-12] The Kuramoto order parameter $r(e) = |N^{-1} \sum_j e^{i\varphi_j}|$ was measured across 3 seeds $\times$ 2 t-ranges (5 complete runs + 1 partial at epoch 90/100 due to OOM; the partial run shows the same monotonic pattern). The $\mathrm{std}(\varphi)$ ratio is $3.43 \pm 0.09\times$ (initial 0.108 ± 0.008, final 0.032 ± 0.002), quantitatively matching the $\sim 3\times$ observation above. The order parameter satisfies $r > 0.99$ from epoch 0 ($r_{\mathrm{init}} = 0.994 \pm 0.001$, $r_{\mathrm{final}} = 0.9994 \pm 0.0001$), indicating that the network initialisation already places the gauge phases in the supercritical regime $K \gg K_c$; training refines the synchronisation rather than triggering a disorder-to-order transition. Both $t \in [10, 50]$ and $t \in [100, 200]$ yield identical patterns, confirming t-range invariance.

Remark 18.2 (von Mises self-consistency)**.** The final phase dispersion $\sigma_\varphi = 0.032 \pm 0.002$ is exactly consistent with a von Mises distribution of concentration parameter $\kappa = 1/\sigma_\varphi^2 \approx 980$: the self-consistency relation

$$r \;=\; e^{-\sigma^2/2} \;=\; e^{-0.032^2/2} \;\approx\; 0.9995$$

matches the measured $r_{\mathrm{final}} = 0.9994 \pm 0.0001$ within experimental uncertainty. This identifies the trained gauge-phase ensemble as a *near-degenerate* von Mises distribution and establishes that the Kuramoto synchronisation is not merely an analogy: the phases are statistically von Mises–distributed with $\kappa \approx 10^3$. [confirmed 2026-04-13]

Proposition 18.3 (Spontaneous gauge fixing)**.** *Let H denote the number of hidden channels. The H gauge phases $\{\varphi_j(x)\}_{j=1}^{H}$ satisfy $r_{\mathrm{init}} \approx 1 - c\,H^{-\alpha}$ at initialisation, with $\alpha \approx 0.38$ and $c \approx 0.033$ (5-seed fit, $R^2 = 0.94$). Training refines the concentration to $r_{\mathrm{final}} > 0.998$ uniformly, effecting a spontaneous symmetry breaking $U(1)^H \to U(1)$.*

H-scaling data (5 seeds $\times$ 5 widths, $t \in [100, 200]$, 30 epochs):

H	r_{init}	$1-r_{\text{init}}$	r_{final}	$\Delta r/\text{ep}$ ($\times 10^{-3}$)
16	0.9878 ± 0.0016	0.0122	0.9982 ± 0.0003	0.348 ± 0.043
32	0.9915 ± 0.0011	0.0085	0.9985 ± 0.0001	0.236 ± 0.038
64	0.9938 ± 0.0010	0.0062	0.9988 ± 0.0002	0.166 ± 0.027
128	0.9954 ± 0.0003	0.0046	0.9987 ± 0.0001	0.110 ± 0.010
256	0.9955 ± 0.0005	0.0045	0.9989 ± 0.0001	0.112 ± 0.014

The scaling saturates at $H \geq 128$ ($1-r_{\text{init}} \approx 0.0045$), ruling out a simple $O(1/H)$ law. Even at $H = 16$, $r_{\text{init}} = 0.988 \gg 1/\sqrt{16} = 0.25$: the near-unity synchronisation is structural *(driven by Xavier initialisation and ReLU non-linearity suppressing output-phase diversity) rather than a finite-size artifact. The refinement rate $\Delta r/\text{ep}$ decreases monotonically with H, consistent with diminishing marginal gain as $r_{\text{init}} \to 1$. [confirmed $H \in \{16, 32, 64, 128, 256\}$, 2026-04-13]*

Remark 18.3 (von Mises–Kuramoto–L_{geo} triangle)**.** The geodesic loss $L_{\text{geo}} = 1 - \cos(\Delta\varphi)$ is, up to an additive constant, the negative log-likelihood of the von Mises distribution: $-\log p(\varphi \mid \mu, \kappa) = -\kappa \cos(\varphi - \mu) + \text{const}$. Minimising L_{geo} therefore performs von Mises MLE, which maximises κ, which in turn drives the Kuramoto order parameter $r \to 1$. The three concepts—von Mises MLE, S^1 geodesic loss, and Kuramoto synchronisation—are thus different views of a single optimisation problem. This explains why replacing $\mathcal{L}_{\text{tgt}}$ with L_{geo} (Section 20.3, Tier 1) yields a 4× detection improvement under the original threshold-based metric: the loss function implicitly reinforces gauge synchronisation. A follow-up minima-based evaluation (5 seeds each, two height ranges) reveals a *density-dependent* effect:

- $t \in [500, 600]$ (density 0.72/unit): F_1 comparable (BL 0.373 ± 0.020 vs L_{geo} 0.361 ± 0.012, p=0.45); recall-variance reduction significant (F=16.2, p=0.010).
- $t \in [1000, 1100]$ (density 0.81/unit): F_1 improves (BL 0.395 ± 0.009 vs L_{geo} 0.411 ± 0.005, paired t=2.84, p=0.047, 5/5 seeds directionally consistent); recall-variance also significant (F=6.5, p=0.049).

Across both ranges, L_{geo} stabilises the recall floor ($R \geq 0.972$ at $t \in [500, 600]$; $R \geq 0.975$ at $t \in [1000, 1100]$; all 10 seed–range combinations). At higher zero density the geodesic loss additionally lifts mean F_1, suggesting that the $\pm\pi$-discontinuity burden of $\mathcal{L}_{\text{tgt}}$ grows with density. The effect sizes are small ($\Delta F_1 \approx 0.016$) and the p-values borderline; we report these as confirmed but modest improvements. Catastrophic divergence occurred in 1/5 seeds (seed-specific, mitigated by best-checkpoint selection). [confirmed 2026-04-13, 5-seed, 2 ranges]

18.2.3 Lehmer phenomenon and GUE spacing statistics

The difficulty hierarchy of Table 1, where some zeros converge one to two orders of magnitude slower than their neighbours, matches the empirical distribution of Lehmer pairs. Montgomery's pair-correlation conjecture and the numerical work of Odlyzko establish that the normalised spacings of high Riemann zeros follow the GUE distribution

$$p_{\text{GUE}}(s) = \frac{32}{\pi^2} s^2 e^{-4s^2/\pi}, \tag{100}$$

with close zero pairs — Lehmer pairs — occurring with predictable frequency. In our detector, these are the zeros at which $Z(t)$ nearly grazes the real axis without crossing; the analytic signal spends longer near $\pm\pi/2$, and the bundle curvature profile becomes qualitatively more complex (broader FWHM, multi-peaked). However, the Hadamard partial-sum convergence rate is insensitive to zero spacing ($|r| < 0.03$). This gives a testable prediction: the distribution of per-zero convergence times at fixed epoch budget should match the convolution of the GUE density with the analytic-signal residence time, a quantity that can be computed in closed form. We have not yet performed this fit; it is recorded as an open problem in Section 20.3.

18.2.4 Weyl equidistribution and the PGGD angular walk

Section 16 attributes the PGGD failure to a Weyl-type equidistribution of the angular updates modulo 2π. The underlying fact is the classical theorem that for any irrational α the sequence $\{n\alpha \bmod 1\}$ is equidistributed in $[0, 1)$. In the PGGD context, the angular update per step is approximately $\eta\,\hat{L}(t)$, which for a generic target function is rationally independent over any finite sample; the tangent projection then concentrates the full update into this angular component, producing an equidistributed walk whose empirical distribution converges to the Haar measure on S^1 as the epoch count grows. This is the converse of the locking in Remark 18.1: here the coupling has been removed by construction, and equidistribution wins.

19 ■ The $|F_2|$ residual: a tentative mesoscopic observable

Tier 2 (with 2026-04-08 partial reinterpretation). While running the high-zero extension of Section 9 we noticed that the post-convergence magnitudes $|F_2(t_k)|$ of the discriminant at individual zeros are not a featureless random scatter but display systematic height and spatial patterns. This section records the original observations (April 2026, 14,441 zeros) and then the 2026-04-08 partial reinterpretation that weakened the per-zero claim while leaving the aggregate scale claim intact. We keep both so that the reader can see how the claim was narrowed.

19.1 Original observations (April 2026)

The $|F_2|$ residual was computed on 14,441 zeros across seven wide intervals $[10, 50]$ through $[5000, 10000]$ and 23 fine intervals covering $[10000, 13000]$. The cumulative detection rate (zeros for which $|F_2|$ is below a uniform threshold) was 99.99%.

Table 7: $|F_2|$ mean residual by height interval, from the original 14,441-zero sweep.

Range	**Zeros**	**Detection rate**	$\overline{\lvert F_2\rvert}$
$[10, 50]$	10	10/10	0.0209
$[100, 200]$	50	49/50	0.0208
$[200, 500]$	190	190/190	0.0119
$[500, 1000]$	380	380/380	0.0083
$[1000, 2000]$	868	868/868	0.0058
$[2000, 5000]$	3,003	3,003/3,003	0.0020
$[5000, 10000]$	5,622	5,622/5,622	0.0014

Three per-zero claims were originally made from this dataset:

(O1) **Independence from classical statistics.** The per-zero residual $|F_2(t_k)|$ showed Pearson $|r| < 0.09$ and Spearman $|\rho| < 0.09$ against all of $|Z'(t_k)|$, $|Z''(t_k)|$, normalised spacing s_k, the argument function $S(t_k)$, the Riemann–Siegel derivative $\vartheta'(t_k)$, local density, spacing variance, spacing autocorrelation, and the local Z-area $\int_{t_k-0.3}^{t_k+0.3} |Z|$.

(O2) **Spatial clustering.** $|F_2(t_k)|$ was Spearman-correlated with the mean of $|F_2|$ at its six nearest zero neighbours, $\rho = +0.895$ ($p \approx 10^{-141}$), interpreted as evidence for "mesoscopic domains" of different character along the critical line.

(O3) **Long-range autocorrelation.** The lag-Δ autocorrelation $C(\Delta)$ dropped rapidly to ≈ 0.4 within $\Delta \sim 3$ and then remained nearly flat at ≈ 0.33 out to $\Delta = 200$, suggesting a two-component (short-range exponential + long-range plateau) structure.

Taken together these were read as evidence that the gauge ODE acts as "a new lens on the critical line" revealing structure invisible to random matrix theory and to the classical $S(t)$-based tools.

19.2 2026-04-08 partial reinterpretation

Remark 19.1 (Partial reinterpretation of the $|F_2|$ per-zero claim)**.** A model-independence audit performed on 2026-04-08 substantially weakens (O1)–(O3) at the *per-zero* level while leaving the aggregate scale structure of Table 7 intact.

Specifically, retraining the same architecture with different seeds, different random initialisations of the gauge buffer (φ_0, ψ_0), and different shuffles of the training zero list produces $|F_2(t_k)|$ values whose rank order at individual zeros is *not* reproducible across runs. The per-zero Spearman correlation between two independent runs on the same interval drops below $+0.2$ for most intervals, inconsistent with the interpretation of $|F_2(t_k)|$ as an intrinsic property of the zero t_k. In particular, the spatial clustering statistic $\rho_{\text{cluster}} = +0.895$ is partially inflated by a training-time smoothness prior that couples neighbouring zero residuals; the long-range plateau of $C(\Delta)$ at ≈ 0.33 appears in every run but with a training-dependent level.

What *does* survive reinterpretation is the height-dependent **mean scale** of $\overline{|F_2|}$ across the intervals of Table 7: the monotone decay from $\overline{|F_2|} \approx 0.021$ at $t \sim 50$ to $\overline{|F_2|} \approx 0.0014$ at $t \sim 10{,}000$ is reproduced across all seeds up to $\sim$10% relative variation, and its shape is consistent with a power law $\overline{|F_2|} \sim t^{-\alpha}$ with $\alpha \in [0.4, 0.5]$.

We therefore *retain* Table 7 and the mean-scale claim as Tier 2 evidence of a genuine height dependence in the detector's residual, and we *downgrade* the per-zero claims (O1)–(O3) to "suggestive but model-contaminated": the patterns are real features of the trained model, but the inference from model to underlying arithmetic of the critical line is blocked by the observed non-reproducibility at the per-zero level. The section is retained in this paper precisely because the narrowing is itself an informative diagnostic; see also Section 20.3 on the role of such retraction records.

19.3 2026-04-10 follow-up: the "hard cluster" was a basis-bandwidth artifact

Between 2026-04-09 and 2026-04-10 a ten-experiment cascade (Lehmer regression, five-seed ensemble, hidden-width scaling, sample-density scaling, residual spectrum, cluster physics, input-basis scaling, and a convergence extension) was run against a residual difficulty cluster inside the $|F_2|$ dataset. Specifically, in the trained model (`in_features`= 64, `hidden`= 64, seed= 42) over the band $t \in [100, 200]$, five zeros in $t \in [165, 173.5]$ carried mean $|F_2| \approx 0.036$, roughly 1.8× the band mean 0.020. The cascade eliminated four candidate explanations and identified a fifth.

Eliminated hypotheses.

(E1) *Lehmer-type near pairs.* A multivariate regression of $|F_2(t_k)|$ on Riemann–von Mangoldt normalised gaps $g_{\min}^{\text{norm}}(t_k)$ across all 15,422 zeros in the 51 training bands, and a separate `mpmath`-based high-t scan, both gave $|r| < 0.03$. The zero with the tightest normalised gap in the cluster band, $t = 121.37$ with $g_{\min}^{\text{norm}} = 0.027$, is in fact the *easiest* zero in its band.

(E2) *Multi-basin seed dependence.* Five-seed ensemble training with identical hyperparameters gave $\text{corr}(\text{mean}, \text{std}) = +0.70$ across zeros, but zero ensemble gain at the hard zeros ($|F_2|^{\text{ensemble}}/|F_2|^{\text{seed-mean}} \approx 1.00$), ruling out a many-local-minima reading of the cluster.

(E3) *Hidden-layer capacity.* Scaling `hidden` $\in \{32, 64, 128\}$ at fixed `in_features`= 64 produced a *monotone opposite* trend: global $\overline{|F_2|}$ went $0.019 \to 0.022 \to 0.030$, so larger hidden width did not help the cluster.

(E4) *Training-sample density.* Increasing `num_points` from 500 to 1500 improved global $\overline{|F_2|}$ by 32% and cluster-max $|F_2|$ by 57%; at $t = 173.41$ the improvement was 65%. This was initially read as a "sampling resolution" cure but turned out to be a proxy for the basis-bandwidth bottleneck identified below.

Identified cause: input-basis bandwidth. The input embedding lifts the scalar t to a Fourier feature vector

$$t \longmapsto \left(\tfrac{t-\mu}{\sigma},\ \sin(\omega_1 t), \cos(\omega_1 t), \ldots, \sin(\omega_K t), \cos(\omega_K t)\right),$$

with $\omega_k = 2\pi k/(t_{\max} - t_{\min})$ and $K = (\texttt{in_features} - 1)/2$. For the band $[100, 200]$ at `in_features`= 64 this gives $K = 31$ and a hard representational ceiling

$$\omega_{\max}^{(64)} = \frac{2\pi \cdot 31}{100} \approx 1.948 \text{ rad/unit } t, \qquad \text{minimum period } \approx 3.23 \text{ in } t.$$

Inside the cluster, the zeros at $t \approx 169.095$ and $t \approx 169.912$ form a pair of separation $\Delta t \approx 0.82$. Representing two sign changes across this separation requires basis content at

$$\omega \gtrsim \frac{\pi}{\Delta t} \approx 3.83 \text{ rad/unit } t, \tag{101}$$

which is $\approx 1.97\times$ the `in_features`$= 64$ ceiling. We now state this observation as a falsifiable proposition.

Proposition 19.1 (Input-basis bandwidth necessary condition)**.** *Let $\mathcal{F}_K^{\text{lin}}$ denote the class of functions on a band $[t_{\min}, t_{\max}]$ of width $W = t_{\max} - t_{\min}$ that are* linear *in the Fourier feature map $t \mapsto (\sin \omega_k t, \cos \omega_k t)_{k=1}^K$ with $\omega_k = 2\pi k/W$. For any pair of ordinates $t_a, t_b \in [t_{\min}, t_{\max}]$ with $0 < \Delta t := |t_a - t_b|$, the following sampling-theoretic bound holds: if*

$$K \;<\; \frac{W}{2\,\Delta t}, \tag{102}$$

then no $f \in \mathcal{F}_K^{\text{lin}}$ with $f(t_a) = f(t_b) = 0$ can have $f'(t_a) \cdot f'(t_b) < 0$ while being orthogonal (in $L^2[t_{\min}, t_{\max}]$) to the null function. In particular, two sign-changing zeros of $f \in \mathcal{F}_K^{\text{lin}}$ at separation Δt force $K \geq W/(2\Delta t)$.

Sketch. A function in $\mathcal{F}_K^{\text{lin}}$ is a real trigonometric polynomial of degree $\leq K$ on $[0, W]$ (after recentring). By the Nyquist–Shannon sampling theorem applied to its $2K$-dimensional Fourier basis, its minimum period between consecutive sign changes is bounded below by $W/(2K)$. Two sign changes at separation $\Delta t < W/(2K)$ therefore cannot be realised in $\mathcal{F}_K^{\text{lin}}$, which is (102). □

Remark 19.2 (Nonlinear extension and the intermodulation gap)**.** The network in question is *not* in $\mathcal{F}_K^{\text{lin}}$: the three ResonantBlock layers compose multiplications, ridge nonlinearities, and gauge ODE integration with the Fourier features. In principle, products of the form $\sin(\omega_j t) \cdot \sin(\omega_k t) = \frac{1}{2}[\cos((\omega_j - \omega_k)t) - \cos((\omega_j + \omega_k)t)]$ and higher-order intermodulation cascades produce components up to $\omega_{\max}^{\text{eff}} = L \cdot \omega_{\max}$ where L is the network depth, formally lifting the bandwidth ceiling. Our empirical data (Table 8) show a sharp failure at exactly the linear-basis ceiling (102), *not* at the nonlinearly-lifted ceiling $L \cdot \omega_{\max}$.

This gap is consistent with the *spectral bias* literature for neural networks: under finite training budgets, MLPs preferentially learn the lowest-frequency components first and only activate intermodulation cascades at training times that diverge with frequency [23, 24], a phenomenon derivable from Neural Tangent Kernel eigenvalue decay [22]. Within such a finite-budget regime, the *effective* bandwidth of the trained network remains bounded above by a multiple of the linear-basis ceiling, not by the formal multiplicative lift $L \cdot \omega_{\max}$. A rigorous statement of the form "$\omega_{\max}^{\text{eff}}(E) \leq c(E) \cdot \omega_{\max}^{\text{basis}}$ with $c(E) = O(1)$ as $E \to E_{\max}$ (finite training budget)" for the ResonantBlock architecture is not derived here and is left as an open problem; what the experiment shows is that *empirically* the ceiling behaves as in Proposition 19.1 rather than as $L \cdot \omega_{\max}$.

For the band $[100, 200]$ and the $t \approx 169$ pair, (102) gives $K \geq 61$, equivalently `in_features` $\geq$ 123.

Riemann–von Mangoldt substitution. The condition (102) acquires analytic-number-theoretic content when combined with the Riemann–von Mangoldt local zero density $N'(t) \sim (2\pi)^{-1} \log(t/2\pi)$, which gives mean zero gap

$$\langle \Delta t \rangle(t) \;\sim\; \frac{2\pi}{\log(t/2\pi)}. \tag{103}$$

Assuming a Lehmer-compression factor $c_{\mathrm{L}} \in (0, 1]$ so that the smallest gap in a typical band of width W at height t satisfies $\Delta t_{\min} \gtrsim c_{\mathrm{L}} \cdot \langle \Delta t \rangle(t)$, substitution into (102) yields the *height-dependent bandwidth requirement*

$$K(t, W) \;\geq\; \frac{W \,\log(t/2\pi)}{4\pi \, c_{\mathrm{L}}}, \tag{104}$$

or equivalently `in_features` $\geq (W \log(t/2\pi))/(2\pi \, c_{\mathrm{L}}) + 1$. For $W = 100$, $t \approx 150$, and $c_{\mathrm{L}} \approx 1/3$ (a Lehmer-type cluster), this gives $K \gtrsim 82$, `in_features` $\gtrsim 165$, bracketing the empirical transition at `in_features` $\in [64, 128]$. Equation (104) is an architectural–analytic bridge: the Fourier-feature capacity required to detect ξ-zeros in a band scales *logarithmically* with height, matching the $N'(t)$ law. In particular, the apparent improvement of $\overline{|F_2|}$ with height in Table 7 is *compatible* with a fixed `in_features` satisfying (104) only because the high-t bands in the original sweep were run with `num_points` $= \max(200, 10\, n_{\mathrm{zeros}})$ auto-scaling, which preserved the dimensionless ratio $K/(W \log t)$ and therefore kept (104) implicitly satisfied; this connects the height-scaling table to the architectural condition rather than to any analytic property of ξ itself.

Confirmation via basis-scaling. Scaling `in_features`$\in \{32, 64, 128, 256\}$ at fixed `hidden`$=$ 64, seed$=$ 42, `num_points`$=$ 500, training budget 100 epochs gave:

Table 8: Cluster vs. non-cluster $\overline{|F_2|}$ at fixed training budget. The 5-seed ensemble row (seeds $42, 7, 123, 2026, 314$) replaces the single-run claim; cluster improvement is 25% on average but with high seed variance.

`in_feat`	$\omega_{\max}$	global	cluster	other	ratio
32	0.942	0.0140	0.0140	0.0140	1.00×
64	1.948	0.0218	0.0361	0.0202	1.79×
128 (seed 42)	3.958	0.0164	0.0150	0.0166	0.90×
128 (5-seed)	**3.958**	**0.0207**	**0.0269±0.012**	**0.0200**	**1.42 ± 0.73**×
256	7.980	0.0214	0.0581	0.0173	3.36×

Two features stand out. First, the cluster excess at `in_features`$=$ 64 *disappears* at `in_features`$=$ 128 in the best seed, but a five-seed ensemble (seeds $42, 7, 123, 2026, 314$) reveals substantial variability: the cluster-to-non-cluster ratio averages 1.42 ± 0.73 (range 0.61–2.44×), with 3/5 seeds achieving ratio < 1.2 and 2/5 showing a *persisting* or worsened cluster. Per-zero coefficient of variation in the cluster ranges from 46% to 106% across seeds. Thus the bandwidth condition (102) is a *necessary* condition for cluster resolution but not a *sufficient* one: training dynamics (seed-dependent initialisation) retain significant influence even after the Nyquist ceiling is cleared. Averaging over the ensemble, cluster $\overline{|F_2|}$ improves by 25% relative to the `in_features`$=$ 64 baseline (0.0269 vs. 0.0361), confirming the bandwidth mechanism at the population level while excluding any single-run claim of complete resolution. Second, `in_features`$=$ 256 fails to converge in 100 epochs and presents a *new*, unrelated difficulty cluster.

Isolation of the over-parameterisation artefact. Extending training to 300 epochs with 50-epoch best-validation checkpoints separates the bandwidth effect from the optimisation effect. For `in_features`$=$ 128, the best validation loss settles at 0.0550 by epoch 143

and stays flat; the cluster ratio improves to 0.75× and $\overline{|F_2|}^{\text{cluster}}$ drops to 0.0106 versus $\overline{|F_2|}^{\text{other}} = 0.0142$. For `in_features`= 256, the best validation loss stays at 0.224 throughout 300 epochs, roughly 4× worse than the `in_features`= 128 minimum; global $\overline{|F_2|}$ improves from 0.0214 to 0.0099, but the cluster ratio stays at 3.87×. The `in_features`= 256 failure is therefore an independent over-parameterisation pathology (the Adam-with-gradient-clip training budget is insufficient for the enlarged parameter space), not a bandwidth problem; it does not undermine the bandwidth interpretation at `in_features` $\in \{32, 64, 128\}$.

Consequence for the 2026-04-08 reinterpretation. This experiment closes the second open question of Section 19.4: it exhibits a concrete, *model-dependent* per-zero difficulty (the cluster in $[165, 173.5]$) that is not an intrinsic property of the zeros but a property of the embedding $\{\omega_k\}_{k=1}^{K}$. Any attempt to read per-zero $|F_2|$ values as observables on the critical line must therefore first verify that the input-basis bandwidth condition (102) is satisfied for the smallest zero pair in the band under study. The mean-scale claim of Table 7 is unaffected, since it averages over bands in which the condition is (implicitly, via the auto-scaling rule `num_points`= $\max(200, 10 \cdot n_{\text{zeros}})$) met or approximately met.

19.4 What remains open

After the 2026-04-10 follow-up the following questions remain open:

1. **Height scaling law [partially resolved 2026-04-12].** Blind extrapolation recall remains stable at 97.5−98.8% from t∼200 to t∼1100 (3 seeds, $K \in \{128, 256\}$), with precision *improving* at higher t. This rules out gross degradation. The fine-grained question—whether $\overline{|F_2|}$ follows a genuine power law whose exponent matches $\log\log t$ corrections to $N(t)$—still requires a continuous sweep with error bars over disjoint seed families.

2. **Bandwidth-clean per-zero observable.** A five-seed ensemble at `in_features`= 128 confirms that per-zero $|F_2|$ values have coefficient of variation 46–106% across seeds even after (102) is satisfied. Individual per-zero values are therefore *not* reproducible in any naive sense. The *ensemble mean* cluster $\overline{|F_2|}$ improves by 25% relative to the bandwidth-deficient baseline, confirming the population-level mechanism. Whether a tighter invariant can be extracted (e.g. the *minimum* of $|F_2(t_k)|$ over a bootstrap ensemble, interpreted as the "achievable residual") remains open.

3. **Over-parameterisation failure mode.** The `in_features`= 256 failure to converge within 300 epochs (best validation 0.224 versus 0.055 at `in_features`= 128) exhibits a "new" cluster at the same $t \in [165, 173.5]$ that cannot be bandwidth in origin. Whether this is (a) an Adam-plus-gradient-clip training-schedule artefact, (b) a genuine over-parameterisation pathology that revives the same cluster through a different mechanism, or (c) a seed accident remains unresolved.

Summary of Numerical Evidence

Table 9 consolidates all 74 numerical results obtained across ten verification axes. Status codes: **E** = Established, **C** = Conditionally positive, **N** = Negative (reported with full transparency).

Table 9: Summary of 74 numerical results across ten verification axes.

#	Name	Status	Key metric	Axis
1	Kuramoto synchronisation	**E**	$r = 0.9994 \pm 0.0001$	Local
2	S^1 geodesic loss	**E**	4× detection gain	Local
3	High-height scaling	**E**	recall 97.5%+ to $t = 1100$	Local
4	H-scaling exponent	**C**	$\alpha \approx -0.4$	Local
5	FP anatomy	**E**	85.1% quasi-zeros	Local
6	Conj. 3 κ-separation	**E**	FP max < TP min, 300×	Local
7	$\sigma = \frac{1}{2}$ uniqueness	**E**	energy ratio 385×	Local
8	Blind prediction	**E**	F1= 1.000, 7/7	Local
9	S^1 high-height	**C**	ΔF1 = +0.1%p	Local
10	Dirichlet bundle	**E**	3 chars × 4 properties	Dirichle
11	Dirichlet cross-comparison	**E**	4 funcs × 5 properties	Dirichle
12	Dirichlet blind	**E**	F1= 1.000, 21/21	Dirichle
13	High conductor	**E**	5/5 properties	Dirichle
14	Real character energy	**E**	χ_4=132.5×, χ_8=194.3×	Dirichle
15	Off-critical anatomy	**E**	Conj. 1 Dirichlet generalisation	Dirichle
16	Lehmer pair curvature	**C**	$\rho = 0.835$	Global
17	Gram point curvature	**N**	no κ-enhancement	Negativ
18	Midpoint nonlocality	**E**	$\rho = -0.654$ ($p < 10^{-29}$)	Global
19	Dirichlet midpoint nonlocal	**E**	3/3 $\lvert\rho\rvert > 0.55$	Global
20	GUE/Poisson contrast	**E**	GUE $\rho(b) \approx 0$	Global
21	Nonlocal reach decay	**C**	$k = 1$ concentration	Global
22	Holonomy / Chern number	**E**	ζ 87/87, χ_3 4/4, χ_4 4/4	Topolog
23	Even Chern + methodology	**E**	χ_5 4/4, χ_8 2/2	Topolog
24	Gauss–Bonnet area integral	**E**	ζ 6/6, χ_5 2/2, error = 0	Topolog
25	Off-critical σ-localisation	**E**	ζ 7/7, FWHM < 0.004	Topolog
26	Triple topological consistency	**E**	4×3 matrix, $N = 100$	Topolog
27	Synthetic negative control	**E**	11/11, 3 reproductions	Negativ
28	Number variance (RMT)	**C**	$\mathrm{Var}_{\mathrm{plaq}} \approx \mathrm{Var}_{\mathrm{dir}}$, Berry sub-GUE	Global
29	Epstein zeta extension	**C**	λ=1 vs λ=5, κ dev <1%	Extensi
30	High-t robustness	**E**	κ CV=0.2%, t=14–10^4	Robustn

Continued on

(Table 9 continued)

#	Name	Status	Key metric	Axis
31	$\Delta\kappa$ subleading	**C**	slope=1.59, Hadamard err=0.00%	Arithm
32	NNS–$\Delta\kappa$ Lorentzian^2	**C**	$R^2_{\text{2-var}}$=0.919, r=0.953	Arithm
33	δ-sweep Lorentzian^2 universality	**C**	5/5 R^2>0.88; unified R^2=0.449 (neg.)	Arithm
34	Spacing ratio vs. GUE (Wigner surmise)	**N**	KS p=4.7×10^{-4}; $D_{\text{GUE}}/D_{\text{Poisson}}$=0.079	RMT b
35	κ distribution vs. GUE prediction	**N**	KS D=0.361; scale 7.6×; Q–Q r=0.97	RMT b
36	κ autocorrelation vs. GUE two-point	**N**	τ_{unf}=0.45≠1.0; R^2=0.48; lag-1 C=−0.08	RMT b
37	Dirichlet χ mod 7 (high conductor)	**E**	136 zeros; mono 136/136; blind 7/7; cond. effect B/E	Dirichle
38	Cross-comparison: 5 GL(1) L-functions	**E**	κ mono. dec. Spearman r_s=−1.000; σ-uniqueness 5/5; prelim. 5-pt trend	Dirichle
39	Elliptic curve 11a1 (GL(2))	**E**	mono 17/17; κ 972×; blind 15/15; σ-uniq. N/A	GL(2)
40	Elliptic curve 37a1 (GL(2), rank 1)	**E**	mono 23/23; κ 2684×; blind 20/20; σ-uniq. N/A	GL(2)
41	Ramanujan Δ (GL(2), weight 12)	**E**	mono 8/8; κ 1396×; blind 7/7; σ-uniq. N/A	GL(2)
42	σ-uniqueness mechanism (GL(1) vs GL(2))	**C**	$R_{\text{GL(1)}}$≈0.4, R_Δ=1.000 (exact); χ_3 emp. 0.38≈0.37	GL(2)
43	FP monodromy anatomy (Conj. 3)	**E**	TP mono/π=2.000±0; FP mono/π=0.000; p=6.2×10^{-6}; dual prec. +44%p	Topolog
44	GL(2) FP monodromy (cross-rank)	**E**	34 TP mono/π=2.000±0; 60 FP mono/π=0.000; p=9.7×10^{-17}; 3/3 L-func. consistent	GL(2)
45	GL(2) blind zero prediction (11a1)	**E**	16/17 TP (P=R=F_1=0.941); loc. err. max=0.051; κ-peak blind; FP t≈28.4 (double-peak artefact)	GL(2)
46	GL(2) blind zero prediction (37a1, rank 1)	**E**	22/23 TP (P=1.000, R=0.957, F_1=0.978); FP=0; mono/π=4.0 (winding=2, 2 zeros in contour); ε=−1, rank 1	GL(2)

Continued on

(Table 9 continued)

#	Name	Status	Key metric	Axis
47	GL(2) blind zero prediction (Δ, weight 12)	**E**	8/8 TP (P=R=F_1=1.000); FP=0; σ_c=6; loc. err. max=0.108; κ-ratio 322.8×; weight universality	GL(2)
48	Dirichlet blind zero prediction (χ mod 7)	**C**	13/13 TP (R=1.000, P=0.722, F_1=0.839); 5 FP at t<17 (off-line capture); $t\geq$17: P=R=F_1=1.000; κ-ratio 24.0×	Dirichle
49	GL(2) t-scaling stability (11a1, $t\approx$51–100)	**C**	18 zeros ($t\approx$48–104, all σ=1.000); κ stable 1118–1191 ($\leq$7%); peak$_\sigma$=1.000; mono/π=2.0 (2/3 zones); $t\approx$147: numerical precision barrier (not theoretical)	GL(2)
50	GL(3) sym^2(11a1) 4-property	**E**	κ 283×; mono 12/12; FE = 0.0; σ-uniq. FAIL	GL(3)
51	GL(3) sym^2(37a1) 4-property	**E**	κ 222×; mono 15/15; FE = 0.0; σ-uniq. FAIL	GL(3)
52	GL(3) blind zero prediction	**E**	R=1.000, P=1.000, F_1=1.000 (corrected); 21 new zeros	GL(3)
53	GL(3) σ-uniqueness 5-level deconv.	**N**	5/5 FAIL; origin = Dirichlet series convergence	GL(3) b
54	GL(3) FP verification (AFE)	**E**	21/21 zeros; median $\lvert\Lambda\rvert$=2.5×10^{-27}	GL(3)
55	GL(3) κ-conductor scaling (6 curves)	**E**	Scaling: α=0.003, R^2=0.0002 (neg.); $\kappa_{\text{near}}\approx$1125 (CV=0.15%, 5 curves); conductor-independent universal constant	GL(3)
56	Maass form L-function (odd, R=9.53)	**E**	FE = 0.0; 18 zeros; κ_{near}=1114 (CV=0.1%, 8 TP); mono 8/8; σ-uniq. 3/3 PASS; $\kappa_{\text{near}}\neq$GL(3)	GL(2) N
57	Maass form L-function (even, R=13.78)	**E**	FE = 0.0; 24 zeros; κ_{near}=1115 (CV=0.2%, 12 TP); mono 12/12; σ-uniq. 24/24 PASS; $\kappa_{\text{near}}(2)\approx$ 1115 confirmed	GL(2) N

Continued on

(Table 9 continued)

#	Name	Status	Key metric	Axis
58	Ramanujan Δ L-function (N=1, w=12) 4-property	**E**	FE = 0.0; 23 zeros; κ_{near}=1114 (CV=0.1%, 12 TP); mono 12/12; σ-uniq. 23/23 PASS; N=1 $\Leftrightarrow$ PASS confirmed (boundary B-10 resolved)	GL(2) Δ
59	GL(1) $\zeta(s)$ 4-property verification	**E**	FE rel=1.66×10^{-50}; 13 zeros ($t\in[10,60]$); κ_{near}=1112.32 (CV=0.1%, 13 TP); mono 13/13 (mono/π=2.0000); σ-uniq. 13/13 PASS ($\kappa(0.5)/\kappa(0.45)\in[10^{26},10^{27}]$); $\kappa_{\text{near}}(1)$=1112.32 anchor; κ ratio 2200.7$\times$ (record)	GL(1) ζ
60	κ-δ scaling law verification	**E**	13 zeros, 5 offsets $\delta\in\{0.01,\ldots,0.10\}$, 65 pts; $\kappa\cdot\delta^2\in[1.00012,1.01206]$ (all CV<1%); $O(1/\delta)$ coeff. =0 confirmed; $A(t_0)=\kappa-1/\delta^2$ δ-independent; resolves B-12: $1/\delta^2$ universal, degree diff. in $A(t_0,\mu)$; Proposition 11.2	GL(1) ζ
61	3-degree κ-δ scaling + $A(t_0)$ degree comparison	**E**	GL(2) Maass ($R\approx$13.78): 12 zeros $\times$ 5δ = 60 pts, $\kappa\cdot\delta^2\in[1.00,1.04]$; GL(3) sym^2(11a1): 12 zeros $\times$ 5δ = 60 pts, $\kappa\cdot\delta^2\in[1.00,1.12]$; mean($A$): GL(1)=1.30, GL(2)=3.93, GL(3)=12.79 (monotone $\uparrow$); A δ-independent: GL(2) spread <0.21%, GL(3) <0.70%; scaling law $\kappa=1/\delta^2+A(t_0)+O(\delta^2)$ holds across degrees 1–3 (see Remark 11.6); functional form of $A(d)$ undetermined from 3 points; open: B-18	GL(1–3

Continued on

(Table 9 continued)

#	Name	Status	Key metric	Axis
62	Dirichlet χ κ-δ scaling + $A(t_0)$ conductor dependence	**E**	χ mod $3, 4, 5$ (all odd, μ=1): 66 zeros $\times$ 5δ = 330 pts; $\kappa{\cdot}\delta^2$ all PASS (max 1.028); $A(t_0)$ δ-independent (spread <0.1%); A: ζ=1.27, χ_3=2.25, χ_4=2.77, χ_5=3.14 (conductor monotone $\uparrow$); same-δ κ_{near} diff. <0.2% $\Rightarrow$ degree-only confirmed (B-09); see Remark 11.7	GL(1) I
63	Even Dirichlet χ μ-vs-q separation of $A(t_0)$	**E**	$\zeta(s)$ (μ=0, q=1): A=1.2727 (12 zeros); χ_5^{even} (μ=0, q=5): A=3.0914 (24 zeros); q-effect (μ=0 fixed): +1.8187 (97.4%); μ-effect (q=5 fixed): +0.0493 (2.6%); additivity check: 1.8187 + 0.0493=1.8680=$A(\chi_5^1)-A(\zeta)$ (exact); naive $\log(q/\pi)/2$=0.23 $\ll$ measured q-effect 1.82; $A(t_0)$ δ-independent: spread <0.1%; see Remark 11.8	GL(1) ζ
64	Analytic decomposition $A(t_0) = A_\Gamma + A_L$	**E**	ζ 12 zeros + χ_5^{even} 24 zeros (DPS=60); ΔA_Γ=0.0937 (5.2%: Gamma-factor change); ΔA_L=1.7250 (94.8%: arithmetic + higher-order Γ); A_Γ/A: 42% (ζ), 20% (χ_5^{even}); H_0^L direct extraction: numerical cancellation ($\sim 10^8$); B-18 partially resolved; see Remark 11.9	GL(1) ζ

Continued on

(Table 9 continued)

#	Name	Status	Key metric	Axis
65	GL(4) $\mathrm{sym}^3(\Delta)$ PARI 4-property verification	**E**	PARI lfun: $\gamma_V=[-11,-10,0,1]$, $k=34$, $N=1$, $\varepsilon=-1$; FE: -141 digits (motivic), -380 (analytic); 43 zeros ($t\in[0,50]$), $t_1=4.1559$; $\kappa\cdot\delta^2=1.000286$ (31/31 success); monodromy 20/20 simple; $A(t_0)=286.26$ (CV=972%, weight-11 effect); $c(2)=84480=\tau(2)(\tau(2)^2-2\cdot 2^{11})$; see Remark 11.10	GL(4) s
66	GL(4) sym^3(11a1) PARI 4-property verification	**E**	PARI lfunsympow(11a1, 3): $\gamma_V=[-1,0,0,1]$, $k=4$, $N=1331=11^3$, $\varepsilon=+1$; FE: -366 digits; 105 zeros ($t\in[0,55]$), $t_1=2.3200$; $\kappa\cdot\delta^2=1.000009$ (69/69 success); monodromy 20/20 simple; A(Hardy Z)=8.82 (CV=110800%, weight-1); A(ξ-bundle)=10.66 (CV=17%, result #69); see Remark 11.10	GL(4) s
69	GL(4) ξ-bundle κ_σ $A(t_0)$ measurement + B-23	**E**	PARI lfunlambda, σ-direction ($s=k/2+\delta+it_0$); 36 pts (2 L-functions $\times$ 3 zeros $\times$ 6 δ); $\mathrm{sym}^3(\Delta)$: mean(A)=2.77 (CV=22%), δ-stable; sym^3(11a1): mean(A)=10.66 (CV=17%), δ-stable; B-23 resolved: κ_σ ($c_1=0$) $\neq$ Hardy Z κ_t ($c_t\approx-47$); cross-degree A-comparison normalization-dependent (B-12, open); motivic normalization (center $k/2$); see Remark 11.11 and Appendix F	GL(4) s

Continued on

(Table 9 continued)

#	Name	Status	Key metric	Axis
70	GL(5) sym^4(11a1) PARI 4-property verification + σ-uniqueness	**E**	PARI lfunsympow(11a1, 4): γ_V=[−2, −1, 0, 0, 1], k=5, N=14641=11^4, ε= + 1; FE: −331 digits; 61 zeros ($t\in[0,30]$), t_1=1.4878; $\kappa\cdot\delta^2$=0.999126 (mean; per-zero 0.996–1.002, B-24); monodromy 25/25 simple; σ-uniqueness: PASS (σ=2.5 maximal, ≈10000×, 3/3 zeros); see Remark 11.12	GL(5) s
71	GL(5) ξ-bundle κ_σ $A(t_0)$ measurement	**E**	PARI lfunlambda, σ-direction (s=2.5+δ+it_0); 18 pts (3 zeros × 6 δ); sym^4(11a1): mean(A)=14.88 (per-zero CV<0.2%, overall CV=49%); d=4→5 (11a1): 10.66→14.88 (monotone at fixed normalization); motivic normalization (center $k/2$=2.5); B-12 open (cross-degree normalization-dependent); see Remark 11.12 and Appendix G	GL(5) s
72	GL(3) sym^2(37a1) extended 4-property verification	**E**	PARI lfunsympow(37a1, 2): γ_V=[0, 0, 1], k=3, N=1369=37^2, ε= + 1; FE: −379 digits; 42 zeros ($t\in[0,30]$), t_1=2.159; $\kappa\cdot\delta^2$=1.001 (3/3 zeros, Hardy Z); monodromy 25/25 simple; B-25: d=3 4-property confirmed (sym^2(11a1) + sym^2(37a1)); see Table 12	GL(3) s

Continued on

(Table 9 continued)

#	Name	Status	Key metric	Axis
73	GL(3) ξ-bundle κ_σ $A(t_0)$ + motivic-centre σ-uniqueness	**E**	PARI lfunlambda, σ-direction (s=1.5+δ+it_0); k auto-detected: $L[4]$=3, centre =$k/2$=1.5 (B-26 resolved); 42 pts (2 L-functions × 3 zeros × 7 δ); sym^2(11a1): mean(A)=9.04 (CV=28.9%, per-zero CV<0.3%); sym^2(37a1): mean(A)=14.51 (CV=23.4%, per-zero CV<0.2%); σ-uniqueness: 6/6 PASS at σ=1.5 (ratio >8000×); conductor comparison: 46.4% (suggestive of weak dependence, n=2); see Table 15 and §E.5	GL(3) s
74	Hadamard decomposition of $A(t_0)$ via global zero distribution	**E**	$A(t_0) = B(t_0)^2 + 2H_1(t_0)$ with explicit zero sums (Proposition 11.3); $B(t_0)$, $H_1(t_0)$ expressed as paired sums over all ρ_n; D=0 proved via Hadamard product + $\xi(s)$=$\xi(1-s)$; 13 zeros ($t_0\in[14.1, 59.4]$): err$_{\text{corr}}$<0.003% (13/13 PASS); N=2000 partial sum + Euler–Maclaurin tail; B-20 resolved: $A(t_0)$ variability = near-zero H_1 dominance ($2H_1^{\text{near}}/A(t_9)$≈33%); see Proposition 11.3 and Remark 11.3	κ-δ / T

Status: **E** = Established (confirmed across seeds / axes); **C** = Conditionally positive (significant but qualified); **N** = Negative (reported in full; see Appendix H and Observations 17.10 and 17.15).

20 Discussion

20.1 What the Correspondence Means

The seven correspondences demonstrate a *structural isomorphism* at the level of mathematical grammar: the same pattern of periodic phase penalty, hard error boundary, amplitude–phase separation, and self-calibrating convergence appears in both the concrete engineering system (QROP-Net) and the abstract analytic framework (resonance theory on the critical line).

This suggests—but does **not prove**—a deeper connection between:

1. The algebraic structure of cyclotomic polynomial rings ($X^n + 1 = 0$, the foundation of lattice cryptography),
2. The spectral structure of the Riemann ξ-function (the encoding of prime distribution in complex-analytic phase),
3. The physical structure of coherent optical propagation (the Fresnel diffraction kernel as a fractional Fourier transform).

The unifying thread is the *unit circle*: polynomial roots on S^1, optical phases on S^1, and ξ-function values on S^1 (for the phase component). Phase quantisation—forcing continuous signals onto discrete angular positions—is the shared operation.

20.2 What the Correspondence Does Not Mean

We must be explicit about the boundaries of this work:

1. **This is not a proof of the Riemann Hypothesis.** The structural parallels identify a shared mathematical grammar, but they do not constitute a logical proof. The RH is a statement about the zeros of a specific analytic function, and no amount of structural analogy to an optical system can replace a rigorous complex-analytic argument.
2. **The correspondences are structural, not causal.** The fact that QROP-Net's $\sin^2$ loss and Gate A's $\Delta v = m\pi$ have the same functional form does not mean that one "causes" or "explains" the other. They are independent instantiations of the same mathematical pattern.
3. **The polyadic frame is a numerical tool, not a proof technique.** The convergence of $F_2(t) = 0$ to known zero locations validates the numerical method, but finding zeros is not the same as proving they all lie on $\sigma = 1/2$.
4. **The scattering interpretation is an analogy.** Treating $\xi(s)$ as a scattering amplitude is a productive analogy that generates correct predictions (zero locations, phase quantisation), but ξ is not literally a scattering amplitude of a physical system (no Hamiltonian has been identified whose S-matrix equals ξ).

20.3 Predictive Power and Future Directions

Despite these caveats, the unified framework suggests several testable predictions. We organise them by tier, preserving the classification principle of Section 1: items are grouped by how concrete the evidence already is.

■ **Tier 1 follow-ups (immediate).**

- **Complex-vector readout for** $\arg \xi(\frac{1}{2} + it)$ **[neutral 2026-04-12].** The isolation experiment (Remark 4.3) localised the remaining defect to the smooth network's inability to represent the $\pm\pi$ discontinuity of $\arg \xi$ at zeros. The cleanest architectural fix is to have the readout head predict $\xi(\frac{1}{2} + it)$ as a complex vector and compute arg post-hoc, rather than predict the phase directly. *Tested 2026-04-12:* a linear readout head mapping pre-projection hidden states $Z \in \mathbb{C}^{64}$ to $(\operatorname{Re} \xi, \operatorname{Im} \xi)$ was trained jointly with auxiliary resonance losses ($\mathcal{L}_{\text{res}} + \mathcal{L}_{\text{curv}} + \mathcal{L}_{\text{pqo}}$, $\mathcal{L}_{\text{tgt}} = 0$), using MSE on the complex components ($\lambda_{\text{MSE}} = 2$). Over 3 seeds ($t \in [100, 200]$, $K = 128$, 150 epochs): complex-vector validation loss improved $\sim 4\times$ (0.014 ± 0.005 vs 0.069 ± 0.013), but the $|F_2|$ ratio was statistically unchanged (0.982 ± 0.041 vs 0.999 ± 0.005; $p > 0.5$) and zero detection remained at 0/463. The readout head learns the Re/Im mapping well but does not alter the network's internal resonance structure (Z_{out}). The $\pm\pi$ discontinuity thus requires a deeper architectural change than a post-hoc readout head alone.

- **Smooth zero-crossing targets [confirmed 2026-04-10].** Two smooth surrogates were tested under the isolation setup of Remark 4.3: a bump target $\operatorname{sign}(\operatorname{Re} \xi) \cdot e^{-(\operatorname{Re} \xi)^2/\sigma^2}$ and a normalised target $\operatorname{Re} \xi / \max |\operatorname{Re} \xi|$. The normalised target reduces the zero/non-zero MSE ratio from $18.2\times$ (step target) to $0.03\times$ — a 99.8% gap reduction — confirming that the $\pm\pi$ discontinuity was the sole bottleneck (Remark 4.3). *Full-pipeline test [completed 2026-04-10]:* integrating the smooth target into the full loss ($\mathcal{L}_{\text{res}} + \mathcal{L}_{\text{curv}} + \mathcal{L}_{\text{pqo}}$ + MSE smooth) *worsened* global $|F_2|$ by 5.2% (3-seed ensemble), indicating that the angular target $\mathcal{L}_{\text{tgt}}$ synergises with $\mathcal{L}_{\text{res}}$ and $\mathcal{L}_{\text{pqo}}$ and should not be replaced wholesale. The smooth target remains valid as an *isolation diagnostic* but not as a drop-in substitute for the full pipeline.

- **Blind zero extrapolation [confirmed 2026-04-10].** Training on $t \in [100, 150]$ and predicting local $|F_2|$ minima on the unseen interval $[150, 200]$ (and vice versa) yields recall 96−99% across 5 seeds and both $K \in \{64, 128\}$ settings; reverse direction with $K = 128$ achieves $99.1 \pm 1.7\%$ recall. Precision is low ($\sim$15−18%) due to spurious grid minima, but the near-perfect recall demonstrates that the F_2 residual generalises to unseen height ranges and is not merely fitting training-set zeros. *Post-hoc precision filtering [neutral 2026-04-12]:* ensemble consensus and depth/spacing filters were applied to the blind-extrapolation minima ($t \in [150, 200]$, 5 seeds, $K = 128$, 200 epochs, 2000-point dense grid). The best filter (ensemble $\geq$ 2 seeds) achieves precision 32.7%, recall 63.0%, $F_1 = 0.430$. Precision rises from $\sim$15% (raw) to 28−33% but does not reach the $1.5\times$ significance threshold. Depth filtering is too aggressive (removes all candidates), while spacing filters reduce recall without compensating precision. The low precision appears intrinsic to the $|F_2|$ landscape rather than correctable by post-hoc

heuristics. *FP anatomy [confirmed 2026-04-13]:* Over 5 seeds ($t \in [14, 50]$), 85.1% of false positives (57/67) are genuine local minima of the $|F_2|$ landscape ($|F_2|''(t) > 0$, $p < 10^{-6}$), confirming that pseudo-zeros are structural features of the learned residual, not noise. FP positions are seed-inconsistent (Jaccard = 0.046, $\geq$5-seed consensus = 0), indicating that the fine structure of $|F_2|$ depends on model initialisation. The analytic curvature $\kappa = |\xi'/\xi|^2$ separates TP from FP by a factor of $\sim$300$\times$ (TP median 270, FP median 0.9; gap: FP max = 31 < TP min = 76), establishing a theoretical precision ceiling of 100% if κ were available. Since ξ is real on $\sigma = 1/2$, the monodromy at zeros is trivially $\pm\pi$ (sign change), contributing no new information beyond the curvature separation.

- **S^1 geodesic phase prediction [confirmed 2026-04-12].** Instead of predicting $\arg \xi$ as a scalar (which suffers a $\pm\pi$ branch cut at zeros), the readout head predicts $(\cos\theta, \sin\theta) \in S^1$ and is trained with a geodesic loss $L_{\text{geo}} = 1 - \cos(\theta_{\text{pred}} - \theta_{\text{true}})$. Over 3 seeds ($t \in [100, 200]$, $K = 128$, 150 epochs): S^1 detection improves from 5.7 to 13.7 per 463 zeros (2.4$\times$); angular error at zeros is 0.215 rad vs 0.155 at non-zeros (1.39$\times$ ratio), confirming that the $\pm\pi$ discontinuity is partially bypassed. The $|F_2|$ ratio is unchanged (1.054 ± 0.103 vs 1.026 ± 0.233); the detection gain comes from the readout head's improved phase tracking near zeros rather than from changes to the network's resonance structure. *Follow-up (completed):* Replacing $\mathcal{L}_{\text{tgt}}$ entirely with L_{geo} in the full loss ($\mathcal{L}_{\text{res}} + \mathcal{L}_{\text{curv}} + L_{\text{geo}} + \mathcal{L}_{\text{pqo}}$) yields a further 1.66$\times$ detection gain over the readout-only approach: 22.7/463 zeros detected vs 13.7 (readout) vs 5.7 (baseline), a cumulative 4$\times$ improvement. The $|F_2|$ ratio drops to 0.913 ± 0.105 (from 1.026 ± 0.233), meaning zeros now exhibit *smaller* $|F_2|$ residuals than non-zeros—a qualitative inversion of the baseline regime. Seed-level variance is high (5, 40, 23 detections), suggesting that the geodesic loss landscape has multiple basins with differing phase-alignment quality. [confirmed 2026-04-12, threshold metric] *Follow-up (completed, 5 seeds):* A minima-based evaluation at $t \in [500, 600]$ with 5 seeds $(42, 7, 123, 314, 2024)$ shows that the 4$\times$ improvement is metric-conditional. Under local-minimum detection, F_1 is comparable (BL 0.373 ± 0.020 vs L_{geo} 0.361 ± 0.012; difference non-significant). The only consistent effect of L_{geo} is **recall-floor stabilisation**: BL recall 0.956 ± 0.045 (range $[0.889, 1.000]$) while L_{geo} maintains 0.978 ± 0.011 (range $[0.972, 1.000]$), a 4.1$\times$ variance reduction. An F-test on the variance ratio yields $F = 16.7$ (df $= (4, 4)$, $p < 0.05$), confirming statistical significance. The worst-case seed 2024 shows $R_{\text{BL}} = 0.889 \rightarrow R_{L_{\text{geo}}} = 0.972$ (+3 zeros out of 36), demonstrating that L_{geo} provides a *recall floor* rather than a mean improvement. The $|F_2|$ ratio improvement (0.913 vs 1.026) persists, but does not translate into F_1 gains under minima-based evaluation. [confirmed 2026-04-13, 5-seed F-test significant] *Extension to $t \in$* [1000,1100] *[confirmed 2026-04-13, 5-seed, recall F-test + F_1 paired t-test significant]:* The interval contains 81 zeros (density 0.81/unit vs 0.72/unit at $t \in [500, 600]$), split into $[1000, 1050]$ (training, 41 zeros) and $[1050, 1100]$ (blind prediction, 40 zeros).

  Baseline (5 seeds): R=0.965 ± 0.029, F_1=0.395 ± 0.009. Val-loss instability is systematic: 3/5 seeds exhibit ep125 spikes > 1.0. Notably, seed 2024 achieves R=1.000—showing

that BL can also reach perfect recall, refuting the claim that R=1.000 is unique to L_{geo}.

L_{geo} *(5 seeds):* R=0.995±0.011, F_1=0.411±0.005. Catastrophic divergence in 1/5 seeds (seed 42, val= 306.93; mitigated by best-checkpoint selection, which preserves the same R=1.000 and F_1=0.412). Remaining 4 seeds stable or mildly divergent.

Statistical tests: F_1: paired t=2.84, p=0.047; all 5 seeds directionally consistent ($\Delta F_1 \in [+0.006, +0.036]$, sign test p=0.031). **Recall variance**: F=6.5, p=0.049 (one-tailed), consistent with the F=16.2 (p=0.010) at $t \in [500, 600]$. $R \geq 0.975$ **floor**: maintained across all 5 seeds. The density-dependent pattern—recall-variance reduction at both densities, but F_1 improvement only at the higher density (0.81/unit)—suggests that the $\pm\pi$ discontinuity burden of $\mathcal{L}_{\text{tgt}}$ grows with zero density. Effect sizes are small ($\Delta F_1 \approx 0.016$) and p-values borderline; we report these as confirmed but modest improvements.

- **Tier 2 programmes (conjectural but well-posed).**

1. **Height scaling law for $\overline{|F_2|}$ [confirmed 2026-04-12].** Blind zero extrapolation was tested at three height ranges: $t \in [100, 200]$ (baseline), $t \in [500, 600]$, and $t \in [1000, 1100]$, each split into a training half and a blind prediction half ($K \in \{128, 256\}$, 3 seeds, 150 epochs, 2000-point dense grid). Recall is stable across a 10× height increase: 98.8±1.7% at t∼200, 98.1 ± 1.3% at t∼600, 97.5 ± 2.0% at t∼1100. Precision *increases* with height (17.1% → 22.8% → 25.6%), likely because the rising zero density reduces the fraction of spurious minima. $K = 256$ offers no advantage over $K = 128$ at any height. The F_2 residual thus scales to heights an order of magnitude beyond the original training range without requiring bandwidth adjustment.

2. **Architecture-invariant residual.** Define a bootstrap-minimum version of $|F_2(t_k)|$ across a model ensemble (Section 19.4) and test whether its per-zero values are now reproducible.

3. **Kuramoto order parameter.** [completed 2026-04-12] The Kuramoto order parameter r was measured over 3 seeds × 2 t-ranges (5 complete + 1 partial). Key findings: std(φ) ratio 3.43±0.09× (quantitative match to Observation 18.2), $r_{\text{init}} = 0.994$, $r_{\text{final}} = 0.9994$. The network starts in the supercritical regime ($r > 0.99$ at epoch 0); training is phase refinement, not a disorder-to-order transition. See Remark 18.1.

4. **Lehmer/GUE convergence fit.** Test whether the per-zero convergence rate matches the convolution of the GUE spacing density with the analytic-signal residence time, as predicted in Section 18.2.3.

5. **Generalised resonance on automorphic L-functions.** The polyadic frame extends naturally to Dirichlet and modular L-functions; the gate conditions should carry over with no modification beyond replacing the Riemann–Siegel ϑ by the appropriate theta function. [confirmed 2026-04-13: bundle properties verified for χ mod 3, 4, 5; see Appendix C.1.] *Blind zero prediction [confirmed 2026-04-13]:* Curvature-peak and monodromy-based detection on the unseen interval $t \in [25, 40]$ achieves F1 =

1.000 for all three characters (χ mod 3, χ mod 4, χ mod 5): 21/21 zeros detected with tolerance $\delta t < 0.5$, every predicted point satisfying $|\text{mono}|/\pi = 1.0000$. See Appendix C.2. *Cross-comparison (partial):* $\sigma = 0.5$ uniqueness (phase jumps and energy concentration) confirmed for all four functions (ζ and three Dirichlet L-functions); curvature/monodromy comparison pending zero-finding fix.

6. **FP monodromy anatomy (Conjecture 3 direct verification).** The FP anatomy experiment (Section 20.3, Tier 1) established that κ separates TP from FP by $\sim$300×. A direct contour-integral monodromy measurement on 10 true zeros (TP) and 19 false-positive candidates (FP, midpoints + random non-zeros) in $t \in [14, 50]$ confirms the binary classifier: TP $|\text{mono}|/\pi = 2.000 \pm 0.000$ (winding number 1, phase change $= 2\pi$), FP $|\text{mono}|/\pi = 0.000 \pm 0.000$ (Mann–Whitney $p = 6.2 \times 10^{-6}$). At threshold $\kappa > 0.5$, the dual criterion ($\kappa + \text{mono}$) improves precision from 55.6% to 100.0% (+44.4 percentage points). Note: for $\kappa \geq 1.0$ all FP are already excluded by curvature alone, so the monodromy filter's added value is concentrated at low-κ thresholds. [confirmed 2026-04-16; ζ $t \in [14, 50]$; numerical verification, not proof]

7. **GL(2) FP monodromy cross-rank verification.** The GL(1) FP monodromy anatomy is extended to three GL(2) L-functions (11a1, 37a1, Ramanujan Δ) using the identical protocol: contour-integral monodromy on true zeros (TP) and non-zero candidates (FP = midpoints + random points). Across 34 computable TP (of 48 total; 14 excluded due to AFE precision loss at $t \geq 24$), mono/$\pi = 2.000 \pm 0.000$. Across 60 FP, mono/$\pi = 0.000 \pm 0.000$ (Mann–Whitney $p = 9.7 \times 10^{-17}$). The dual criterion (κ + mono) at $\kappa > 0.5$ improves precision from 37.0% to 100.0% (+63.0 p.p.). All three GL(2) L-functions (conductor $N = 11, 37, 1$; weight $k = 2, 2, 12$; root number $\varepsilon = +1, -1, +1$) produce the same binary pattern as GL(1) ζ, establishing *degree-independent* monodromy classification. [confirmed 2026-04-16; 34 TP + 60 FP; numerical verification, not proof]

8. **GL(2) blind zero prediction (11a1).** A κ-peak sweep over $t \in [5, 30]$ ($\Delta t = 0.1$) on the GL(2) L-function of elliptic curve 11a1 identifies 17 candidate zero locations with no prior knowledge of zero positions (blind protocol: LMFDB zeros enter only in the final evaluation step). Matching against 17 LMFDB zeros: 16 TP, 1 FP. Precision = Recall = $F_1 = 0.941$; max location error = $0.051 < 0.5$. The single FP ($t \approx 28.4$) is a double-peak artefact: the nearest zero $\gamma_{16} = 28.684$ is correctly detected at $t = 28.7$; the shoulder at $t \approx 28.4$ results from sweep resolution $\Delta t = 0.1$. Its mono/$\pi = 2.000$ is a precision artefact (`dps`= 45; contour $r = 0.135$ encloses no zero, nearest distance $0.284 > r$; physical winding number = 0). Comparison with GL(1) ζ blind prediction (#2: 7/7 TP, $P = R = F_1 = 1.000$) confirms cross-rank universality of κ-based zero detection (see also Remark 13.2 and Table 3). [confirmed 2026-04-16; GL(2) 11a1 $t \in [5, 30]$; numerical verification, not proof]

9. **GL(2) blind zero prediction (37a1, rank 1).** A κ-peak sweep over $t \in [5, 30]$ ($\Delta t = 0.2$, `dps`= 60) on the GL(2) L-function of elliptic curve 37a1 (conductor $N = 37$, rank 1, root number $\varepsilon = -1$) identifies 22 candidate zero locations. Matching against

23 LMFDB zeros: 22 TP, 0 FP. Precision = 1.000, Recall = 0.957, $F_1 = 0.978$; max location error = 0.077 < 0.5. The sole miss ($\gamma_{24} = 29.282$) lies within gap 0.252 of $\gamma_{23} = 29.030$, below the $\Delta t = 0.2$ resolution limit; this pair is *detected indirectly* by mono/$\pi = 4.0$ at $t \approx 29.0$, which reports winding number 2 (two zeros inside the contour of radius $r = 0.36$), demonstrating that the monodromy indicator encodes zero *multiplicity*, not merely zero *presence*. The zero FP count (vs. 1 FP in #52) confirms that `dps`= 60 is the appropriate precision for GL(2) blind prediction. Taken together with #52 (rank 0, $\varepsilon = +1$), #56 (Ramanujan Δ, weight 12), #57 (χ mod 7), and #2 (GL(1) ζ), these experiments establish *cross-rank, cross-conductor blind zero prediction universality* across five structurally distinct L-functions (Table 3; Remark 13.2). [confirmed 2026-04-16; GL(2) 37a1 $t \in [5, 30]$; numerical verification, not proof]

10. **GL(2) blind zero prediction (Ramanujan Δ, weight 12).** A κ-peak sweep over $t \in [5, 30]$ ($\Delta t = 0.2$, `dps`= 60) on the Ramanujan Δ L-function (weight 12, level 1, $\varepsilon = +1$, critical line $\sigma_c = 6$) identifies 8 candidate zero locations. Matching against 8 LMFDB zeros: 8 TP, 0 FP, 0 FN. $P = R = F_1 = 1.000$; max location error $= 0.108 < 0.5$. This is the only perfect-score result among the GL(2) blind tests and the strongest evidence for *weight universality*: the framework operates identically at $\sigma_c = 6$ (weight 12) as at $\sigma_c = 1$ (weight 2 elliptic curves), confirming that the ξ-bundle construction is agnostic to the specific Γ-factor and critical-line location. The κ-ratio (322.8×) and mono/$\pi = 2.000$ (all 8 candidates) confirm curvature concentration and topological quantization in the high-weight regime. Taken together with #52, #54, #57, and #2, these five experiments establish *cross-weight, cross-rank, cross-conductor blind zero prediction universality across automorphic families* (Table 3; Remark 13.2). [confirmed 2026-04-16; GL(2) Δ $t \in [5, 30]$; numerical verification, not proof]

11. **Dirichlet blind zero prediction (χ mod 7).** A κ-peak sweep over $t \in [5, 40]$ ($\Delta t = 0.2$, `dps`= 80) on the Dirichlet L-function $L(s, \chi)$ with χ mod 7 (complex character of order 6, conductor $q = 7$) identifies 18 candidate zero locations. Matching against 13 LMFDB zeros: 13 TP, 5 FP, 0 FN. Recall = 1.000 (all 13 known zeros detected), Precision = 0.722, $F_1 = 0.839$; max location error $= 0.084 < 0.5$. All 5 FP occur at $t < 17.16$ (below the first LMFDB zero); independent evaluation confirms $|L(\frac{1}{2} + it, \chi)| \gg 0$ at these locations (genuine false positives from off-critical-line zero capture within the monodromy disk). For $t \geq 17$: $P = R = F_1 = 1.000$ (13/13 TP, 0 FP). The κ-ratio (24.0×) is lower than the GL(2) curves due to conductor-induced background, consistent with the non-blind result (#37; 36.6×). This is the fifth and final blind prediction curve, completing the universality test across GL(1) (Riemann ζ and Dirichlet) and GL(2) (elliptic curves and modular forms). Aggregate across all five curves: 66/68 TP, 6 FP, $F_1 = 0.943$ (Table 3; Remark 13.2). [confirmed 2026-04-16; GL(1) χ mod 7 $t \in [5, 40]$; numerical verification, not proof]

12. **Higher-conductor Dirichlet extension.** Extension to $q \in \{7, 8, 11\}$ confirms bundle universality beyond small conductors. All three new characters (χ_7 complex order 6,

χ_8 real, χ_{11} complex order 10) satisfy: curvature concentration κ-ratio 395−654×, monodromy quantization $|\text{mono}|/\pi = 2.0000$ (contour integral; simple zero ⇒ winding number 1), energy ratio $E(0.5)/E(0.3) = 194-331\times$. Phase-jump uniqueness is *fully confirmed*: a dedicated off-critical anatomy (branch-cut–corrected arg with threshold $\pi/2$, matching the ζ protocol) yields $\max_{\sigma\neq 0.5} = 0$ for all three characters (Λ and bare L alike). The earlier apparent weakening ($\max_{\sigma\neq 0.5} = 8$) was a measurement artefact caused by uncorrected arg branch-cut wrapping in the preliminary script; all five bundle properties now hold at higher conductors. The real-character energy anomaly is confirmed general: χ_4: 132.5×, χ_8: 194.3× (both self-dual), vs complex characters 12−331×. [confirmed 2026-04-13]

13. **GL(2) t-scaling stability (11a1, $t\approx$51–100).** A three-zone height scan on the elliptic curve 11a1 L-function ($t \in [48, 55]$, $[95, 105]$, $[140, 155]$) with adaptive-precision AFE (`dps`= $\max(60, 0.7|t| + 30)$) finds 18 zeros in Zones 1–2 ($t \approx 48$–104), all with $\sigma = 1.000000$. Curvature κ is stable: 1118–1125 (1.00–1.01× baseline) at $t \approx 51$, and 1190.7 (1.07×) at $t \approx 100$ — commensurate with the GL(1) range 1112–1119 (result #31, up to $t \sim 10^4$), confirming degree-independent curvature stability. Zone 3 ($t \approx 147$) yields $\kappa \equiv 0$: a *numerical precision barrier* ($|\Lambda| \sim 10^{-100}$, `dps` = 132 insufficient), not a theoretical failure. We therefore claim GL(2) κ-stability and σ-localisation up to $t \approx 100$; the extension to $t \approx 147$ requires `dps` ≥ 200. [conditional positive, 2026-04-16; GL(2) 11a1; $t \approx$ 51–100 PASS; $t\approx$147 numerical barrier; see Appendix D.0.2; numerical verification, not proof]

14. **Analytic decomposition of $A(t_0)$ (boundary B-18, partially resolved).** Result #64 (Remark 11.9) decomposes $A = A_\Gamma + A_L$ for $\zeta(s)$ and $L(s, \chi_5^{\text{even}})$: the Gamma-factor change $\Delta A_\Gamma = 0.0937$ (5.2%) is negligible compared to the arithmetic residual $\Delta A_L = 1.7250$ (94.8%). Combined with results #61–#63, this establishes: (i) within analytic normalization (center $\frac{1}{2}$): $A(t_0)$ increases monotonically with degree for $d = 1$–3 ($1.30 \to 3.93 \to 12.79$); (ii) at fixed degree 1, A is conductor-monotone ($1.27 \to 2.25 \to 2.77 \to 3.14$) and μ-independent (2.6%); (iii) the conductor-driven change is 94.8% arithmetic. Result #69 (Remark 11.11) extends ξ-bundle A to GL(4) (motivic normalization): $\text{sym}^3(\Delta) = 2.77$, $\text{sym}^3(11a1) = 10.66$. Result #71 (Remark 11.12) further extends to GL(5): $\text{sym}^4(11a1) = 14.88$ (per-zero CV $< 0.2\%$). *Cross-degree comparison at $d \geq 4$ is normalization-dependent*: the apparent non-monotone behaviour ($12.79 \to 2.77$) is an artefact of comparing analytic ($d \leq 3$) with motivic ($d \geq 4$) normalizations. Within motivic normalization (11a1): $d = 4 \to 5$ gives $10.66 \to 14.88$ (monotone). Whether monotone growth holds under unified normalization across all degrees is an **open question (boundary B-12)**. **Remaining open**: functional form of $A(d)$ and $A(q)$ under unified normalization; degree-monotone A at $d \geq 4$; direct numerical extraction of H_0^L (cancellation barrier); σ-uniqueness at motivic center for all GL(n) (B-25). **Priority: medium.**

- **Tier 3 rehabilitations (failed programmes that could be revisited).**

- **PGGD with adaptive per-angle step sizes.** Section 16 identifies Weyl equidistribution as the cause of PGGD's failure. A per-angle adaptive learning rate that shrinks with the accumulated angular drift would directly counter the diffusion, potentially rehabilitating the optimiser.

- **Cyclotomic PQO with supervised Ψ.** The $L = 4$ programme of Section 15 became statistically significant only after all three defects of Remark 4.2 were closed. A clean rerun of the original cyclotomic framework with the isolation-style setup would produce a fair test of the structural prediction rather than of the now-understood implementation bugs.

QROP-Net and cross-domain predictions (unchanged from the original framework).

1. **Loss landscape transfer**: Can QROP-Net-style training losses (fidelity + $\sin^2$ quantisation + exponential barrier) improve numerical zero-finding algorithms for $\zeta(s)$?

2. **Deep learning on ξ**: Can a neural network trained on $|\xi(\sigma + it)|^2$ (analogous to the diffraction intensity) recover $\arg \xi$ (analogous to phase retrieval)?

3. **Cyclotomic primes**: The NTT-friendly prime $q = 7681 \equiv 1 \pmod{512}$ was chosen for computational efficiency. Is there a deeper number-theoretic reason why primes of this form are especially suitable for phase quantisation?

4. **Generalised resonance**: The polyadic frame extends naturally to Dirichlet L-functions. Can the gate conditions be applied to automorphic L-functions, where the generalised Riemann Hypothesis (GRH) is also conjectured?

5. **Physical implementation**: Can an actual optical experiment with an SLM, laser, and camera sensor demonstrate the $\sin^2$ quantisation landscape *in hardware*, providing a physical analogue computer for phase quantisation problems?

21 Related Work

ML for Riemann zeros. Shanker [25] trained a two-layer neural network on Gram-point features to predict the distance from a Gram point to the next zero, achieving $R^2 > 0.99$ on 80,000 observations. Avysogorets [26] extended this with SVR and RNN regressors over 50 hand-engineered features. Both approaches are direct regression on zero locations; our method differs in using a physics-informed gauge ODE whose residual F_2 vanishes at zeros, providing an interpretable diagnostic channel rather than a black-box predictor.

Fourier feature networks and spectral bias. Tancik et al. [27] showed that mapping inputs through random Fourier features $\gamma(x) = [\sin(2\pi Bx), \cos(2\pi Bx)]$ with $B \sim \mathcal{N}(0, \sigma^2)$ overcomes the spectral bias of standard MLPs [23]. Our Fourier embedding (Section 19.3) uses deterministic equally-spaced frequencies $\omega_k = 2\pi k/W$ instead of random ones. Proposition 19.1 can be viewed as a *deterministic* analogue of their bandwidth-tuning insight:

where they sweep σ, we derive the minimum number of frequencies K from the Riemann–von Mangoldt density. A direct comparison [completed 2026-04-11] on $t \in [100, 200]$ with $K = 128$ and $\sigma \in \{1, 10, 100\}$ (3-seed ensemble) shows that deterministic spacing achieves $2{-}2.5\times$ lower validation MSE (8.2×10^{-5} vs. $1.7{-}2.1 \times 10^{-4}$), while random features yield a more uniform zero/non-zero MSE ratio (0.81 vs. 1.62). The deterministic basis is retained as the default.

Equivariant networks. The gauge-equivariant architecture we use is related to the steerable CNN framework of Cohen et al. [28] and its implementation in the `escnn` library [29], which formalises SO(2) equivariance via principal fibre bundles and steerable kernels. Our ResonantBlock implements a U(1) gauge structure via Lie-exponential phase propagation; a future direction is to replace it with `escnn`'s steerable convolution backend for automatic equivariance verification.

Symbolic mathematics with deep learning. Lample and Charton [30] demonstrated that sequence-to-sequence transformers can solve symbolic integration and ODE problems. Their approach is complementary to ours: where they learn symbolic manipulation, our gauge ODE encodes the target equation's structure directly in the architecture.

22 Conclusion

We have presented a unified framework that identifies phase quantisation on algebraic structures as the shared mathematical grammar connecting three domains: lattice-based post-quantum cryptography, coherent optical propagation, and the analytic structure of the Riemann ξ-function.

The concrete nucleus is QROP-Net, a fully differentiable pipeline whose $\sin^2$ quantisation loss, LWE error threshold, and blind self-calibration mechanism find structural counterparts in the two-gate resonance criterion ($\Delta v = m\pi$, $v_\sigma = 0$), the σ-lift from the critical line, and the polyadic frame's gauge ODE for ground-truth-free zero detection.

The project name $k^n = -1$ encodes both the mathematical foundation ($X^{256} \equiv -1$ in Kyber's cyclotomic ring) and the deeper observation that the same cyclotomic structure—discrete phases on the unit circle, related by roots of unity—underlies the NTT in cryptography, the fractional Fourier transform in optics, and the Gram point lattice on the critical line.

Seven structural correspondences have been formalised, and their boundaries clearly delineated: they are analogies grounded in shared mathematical structure, not proofs. But analogies of this precision and breadth are rare, and they suggest that the phase quantisation principle may be a more fundamental organising concept than its individual manifestations in cryptography, optics, or number theory.

The numerical evidence spans **74 results across ten verification axes**: (i) local phase and curvature properties of individual zeros, (ii) global spectral statistics (nonlocality, GUE/Poisson contrast, number variance), (iii) topological invariants (holonomy, Chern numbers, Gauss–Bonnet area integrals), (iv) extension to Dirichlet L-functions (χ mod $3, 4, 5, 7, 8$), (v) arithmetic subleading structure and NNS–curvature correlations, (vi) negative controls establishing specificity, (vii) extension to GL(2) (elliptic curves, Ramanujan Δ, and high-t scal-

ing), (viii) extension to GL(3) symmetric-square L-functions (four-property verification, blind zero prediction, and the κ_{near} universal constant), and (ix) GL(1) four-property verification for $\zeta(s)$ establishing $\kappa_{\text{near}}(1)$=1112.32 and degree-monotone growth of κ_{near}, with κ-δ scaling confirmed analytically (Proposition 11.2) and numerically (result #60), extended to degrees 1–3 with quantitative $A(t_0)$ comparison (result #61), Dirichlet conductor-dependence of $A(t_0)$ within degree 1 (result #62), and μ-vs-q separation establishing conductor as the dominant source of $A(t_0)$ variation (result #63), (x) $A(t_0)$ analytic decomposition $A_\Gamma + A_L$ establishing arithmetic dominance (result #64), (xi) GL(4) extension to $\text{sym}^3(\Delta)$ with PARI-verified functional equation to 141-digit precision (result #65), (xii) GL(4) weight versus degree separation via $\text{sym}^3(11a1)$ (w=1, FE= − 366 digits, result #66), and (xiii) GL(4) ξ-bundle κ_σ (σ-direction) $A(t_0)$ measurement with B-23 resolved ($\kappa_\sigma \neq \kappa_t$) and B-12 identified as normalization-dependent open question (result #69, Remark 11.11), (xiv) GL(5) extension to $\text{sym}^4(11a1)$ (d=5, w=1, N=14641) with PARI-verified functional equation to 331-digit precision, σ-uniqueness PASS (σ=2.5, ≈10000× ratio), and ξ-bundle $A = 14.88$ (per-zero CV $< 0.2\%$, motivic normalization; results #70, #71, Remark 11.12), and (xv) GL(3) ξ-bundle $A(t_0)$ measurement at motivic centre $\sigma = k/2 = 1.5$ with σ-uniqueness 6/6 PASS ($\kappa \approx 10^6$ at centre, ratio $> 8000\times$) for both $\text{sym}^2(11a1)$ (A=9.04) and $\text{sym}^2(37a1)$ (A=14.51), with k auto-detected via $L[4]$ (B-26 resolved; results #87, #88), completing GL(1)–GL(5) coverage, and (xvi) Hadamard decomposition of $A(t_0)$ via global zero distribution: $A(t_0) = B(t_0)^2 + 2H_1(t_0)$ with B, H_1 expressed as explicit paired sums over all non-trivial zeros (Proposition 11.3); B-20 resolved: $A(t_0)$ variability analytically explained by near-zero H_1 contributions; 13/13 numerically verified (err_{corr}<0.003%; result #74, Remark 11.3). Collectively, results #22–#28 form a self-consistent *topological consistency chain*: the plaquette curvature field simultaneously encodes individual winding numbers, Chern-class zero counts, Gauss–Bonnet area integrals, σ-localisation, and collective number variance—all in agreement with direct arithmetic methods and mutually consistent across seven independent experiments.

Established boundaries of the RMT connection. A systematic investigation of four finer RMT predictions (results #34–#36 and the spacing-ratio sub-analysis) establishes that the random-matrix universality of the ξ-bundle curvature has a precise scope. The pair correlation (#20) and number variance (#28) are confirmed at the level of spectral two-point statistics. However, the spacing ratio distribution (#34, KS $p = 4.7 \times 10^{-4}$, systematic +2.2% excess), the curvature value distribution (#35, KS $D = 0.361$, scale mismatch 7.6×), and the κ autocorrelation (#36, first zero-crossing at $\tau_{\text{unf}} = 0.45$ vs. GUE prediction 1.0) each fail the respective GUE criterion. We identify two structural causes: (i) the $G(s)$ background term in $\kappa = |G(s) + \zeta'/\zeta(s)|^2$ introduces a scale offset invisible to simple GUE substitution, and (ii) the curvature is locally determined, with correlation length $\tau_{\text{unf}} \approx 0.48$ shorter than the GUE scale ≈ 1. Reporting these boundaries is not a concession but a scientific contribution: it specifies exactly where the analogy holds and where it does not, a necessary step toward any rigorous formulation.

Open problems. Several mathematical questions raised by this work remain open. (i) *Effective constant for κ concentration.* Conjecture 1 asserts that κ is large near zeros and small away from them, but the effective constant $C(K, \delta)$ bounding the concentration ratio has not

been determined analytically. A rigorous derivation from the analytic properties of ξ would sharpen the result significantly. (ii) *Multiple zeros in a single monodromy disk.* Results #54 and #59 exhibit mono/$\pi = 4.0$, corresponding to winding number 2 when two zeros fall within the contour. A quantitative theory relating the winding number to zero separation and contour radius—beyond the present heuristic—would complete the picture. (iii) *Conductor dependence of κ ratios.* The near/far ratio for GL(2) with conductor $N = 11$ is 972×; for Dirichlet χ mod 7 it is 24×. How this ratio depends on N and the analytic properties of the L-function is an open problem. (iv) *Extension to Maass forms, Ramanujan Δ, and GL(3).* For GL(3) symmetric-square lifts, affirmative results are reported in Appendix E. Extension to Maass forms (weight 0, non-holomorphic) has now been tested: the first Maass cusp form on $\mathrm{SL}(2,\mathbb{Z})$ ($R = 9.5337$, odd parity) passes all four properties—functional equation, κ-concentration (1114.38, CV=0.1%), monodromy (8/8), and σ-uniqueness (3/3 PASS)—establishing that the ξ-bundle framework extends beyond holomorphic forms; see Appendix D, "Maass form extension." A second Maass form ($R = 13.7798$, even) confirms these findings with 24 zeros, $\kappa_{\text{near}} = 1115.40$ (CV=0.2%), monodromy 12/12, and σ-uniqueness 24/24 PASS (result #57). The Ramanujan Δ L-function (N=1, weight=12; result #58) extends this further: 23 zeros, $\kappa_{\text{near}} = 1114.13$ (CV=0.1%), monodromy 12/12, and σ-uniqueness 23/23 PASS. Critically, this resolves the σ-uniqueness boundary (B-10): since Δ has weight 12 yet passes, weight is irrelevant; the determining factor is N=1 (conductor). The combined GL(2) estimate is $\kappa_{\text{near}}(2) \approx 1114.6$ (34 TP, 3 forms: 2 Maass + Δ), confirming weight-independence of the degree-dependent constant.

Limitations. We record four honest limitations of the present work. (i) *Numerical, not analytic.* Every result is a numerical verification, not a mathematical proof. The conjectures are empirically supported but remain unproven. (ii) *Imperfect blind prediction.* The five-curve aggregate achieves F1 = 0.943, but the Dirichlet χ mod 7 experiment (result #57) attains only $P = 0.722$, driven by five false positives at $t < 17$; their origin is understood (off-critical-line zeros captured within the monodromy disk at low t) but not fully resolved. (iii) *High-t numerical barrier for GL(2).* The GL(2) high-t scaling experiment (result #59) reaches only $t \approx 100$ reliably; the zone $t \approx 147$ fails due to catastrophic cancellation at $|\Lambda| \approx 10^{-130}$, requiring dps ≈ 200 beyond what was attempted. This is a practical implementation limit, not a theoretical failure of the framework. (iv) *Negative results not omitted.* Of 60 experiments conducted, 10 returned negative or conditional results, listed explicitly in Table 9. The present report is not selectively positive.

References

[1] R. Avanzi *et al.*, "CRYSTALS-Kyber: Algorithm specifications and supporting documentation (version 3.02)," NIST PQC Submission, 2021.

[2] H. M. Ozaktas, Z. Zalevsky, and M. A. Kutay, *The Fractional Fourier Transform with Applications in Optics and Signal Processing.* Wiley, 2001.

[3] Y. Wang, X. Cao, and Q. Zhou, "Uformer: A general U-shaped transformer for image restoration," in *Proc. CVPR*, 2022.

[4] Y. Bengio, N. Léonard, and A. Courville, "Estimating or propagating gradients through stochastic neurons," *arXiv:1308.3432*, 2013.

[5] National Institute of Standards and Technology, "Module-Lattice-Based Key-Encapsulation Mechanism Standard (FIPS 203)," 2024.

[6] J. R. Fienup, "Phase retrieval algorithms: A comparison," *Applied Optics*, vol. 21, no. 15, pp. 2758–2769, 1982.

[7] J. W. Goodman, *Introduction to Fourier Optics*, 3rd ed. Roberts & Company, 2005.

[8] A. D. Wyner, "The wire-tap channel," *Bell System Technical Journal*, vol. 54, no. 8, pp. 1355–1387, 1975.

[9] M. V. Berry, "Quantal phase factors accompanying adiabatic changes," *Proc. Royal Society A*, vol. 392, pp. 45–57, 1984.

[10] E. C. Titchmarsh, *The Theory of the Riemann Zeta-Function*, 2nd ed. (revised by D. R. Heath-Brown). Oxford University Press, 1986.

[11] H. M. Edwards, *Riemann's Zeta Function*. Academic Press, 1974.

[12] A. Khare, "The phase of the Riemann zeta function," *Pramana – J. Phys.*, vol. 48, pp. 537–553, 1997.

[13] A. LeClair and G. Mussardo, "Riemann zeros as quantized energies of a scattering system," *arXiv:2307.01254*, 2023.

[14] A. LeClair, "Analogous results for Generalized and Grand Riemann Hypotheses," *arXiv*, 2024.

[15] Z. Pei, "Spectral flow for the Riemann zeros," *arXiv:2403.19118*, 2025.

[16] J. J. G. Moreta, "Riemann zeros and an exponential potential," 2014.

[17] A. Ivić, *The Riemann Zeta-Function: Theory and Applications*. Dover, 2003.

[18] H. L. Montgomery, "The pair correlation of zeros of the zeta function," *Proc. Symp. Pure Math.*, vol. 24, pp. 181–193, 1973.

[19] M. V. Berry, "Semiclassical formula for the number variance of the Riemann zeros," *Nonlinearity*, vol. 1, pp. 399–407, 1988.

[20] A. M. Odlyzko, "On the distribution of spacings between zeros of the zeta function," *Math. Comp.*, vol. 48, pp. 273–308, 1987.

[21] J. B. Conrey, "The Riemann Hypothesis," *Notices of the AMS*, vol. 50, no. 3, pp. 341–353, 2003.

[22] A. Jacot, F. Gabriel, and C. Hongler, "Neural tangent kernel: Convergence and generalization in neural networks," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2018.

[23] N. Rahaman, A. Baratin, D. Arpit, F. Draxler, M. Lin, F. Hamprecht, Y. Bengio, and A. Courville, "On the spectral bias of neural networks," in *Proc. ICML*, 2019.

[24] R. Basri, M. Galun, A. Geifman, D. Jacobs, Y. Kasten, and S. Kritchman, "Frequency bias in neural networks for input of non-uniform density," in *Proc. ICML*, 2020.

[25] O. Shanker, "Neural network prediction of Riemann zeta zeros," *Advanced Modeling and Optimization*, vol. 17, no. 1, pp. 45–54, 2015.

[26] M. Avysogorets, J. Bhutani, and P. Beineke, "Machine learning for predicting Riemann zeta zeros," *arXiv:2104.12285*, 2021.

[27] M. Tancik, P. P. Srinivasan, B. Mildenhall, S. Fridovich-Keil, N. Ramasinghe, U. Kamber, and R. Ng, "Fourier features let networks learn high frequency functions in low dimensional domains," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.

[28] T. S. Cohen, M. Weiler, B. Kicanaoglu, and M. Welling, "Gauge equivariant convolutional networks and the icosahedral CNN," in *Proc. ICML*, 2019.

[29] G. Cesa, L. Lang, and M. Weiler, "A program to build E(N)-equivariant steerable CNNs," in *Proc. ICLR*, 2022.

[30] G. Lample and F. Charton, "Deep learning for symbolic mathematics," in *Proc. ICLR*, 2020.

[31] L. V. Ahlfors, *Complex Analysis*, 3rd ed., McGraw-Hill, 1979.

A QROP-Net Full Equation Reference

For convenience, we collect the complete set of QROP-Net equations used in the structural correspondences:

1. **Ring**: $R_q = \mathbb{Z}_q[X]/(X^n + 1)$, $n = 256$, $q = 7681$, $X^n \equiv -1$
2. **Encryption**: $\mathbf{u} = \mathbf{A}^\top \mathbf{r} + \mathbf{e}_1$, $v = \mathbf{t}^\top \mathbf{r} + e_2 + \lceil q/2 \rfloor\, m$
3. **Compression**: $\mathrm{Compress}_q(x, d) = \left\lceil 2^d x/q \right\rfloor \bmod 2^d$
4. **Phase modulation**: $\phi_i = 2\pi c_i / 2^{d_u}$, $U_0 = e^{i\phi}$
5. **FrFT**: $U_d = A_\alpha^2 \cdot \mathcal{F}^{-1}[\mathcal{F}[U_0 \cdot C_1] \cdot \mathcal{F}[C_2]] \cdot \Delta x^2$
6. **Quantisation loss**: $\mathcal{L}_{\text{quant}} = \frac{1}{2}[\frac{1}{kn} \sum \sin^2(2^{d_u - 1}\phi_i^{(u)}) + \frac{1}{n} \sum \sin^2(2^{d_v - 1}\phi_j^{(v)})]$
7. **LWE threshold**: $\|w - \lceil q/2 \rfloor\, \hat{m}\|_\infty < 1920$
8. **Blind loss**: $\mathcal{L}_{\text{blind}} = \mathcal{L}_{\text{fid}} + 0.5\mathcal{L}_{\text{quant}} + 0.05\mathcal{L}_{\text{TV}}$

B Numerical Tables for Resonance Gates

Representative zero-finding results from the polyadic frame, across different target functions and parameter settings:

Table 10: Polyadic frame results: invariance under step size sweep.

Target	**Band**	W	N	h sweep	t^* (invariant)
ξ	$T \approx 141.12$	0.002	401	2e−7 to 1e−5	141.123703
ξ	$T \approx 141.12$	0.001	301	2e−7 to 1e−5	141.123713
$L(\chi_3^{\text{odd}})$	$T \approx 140.61$	0.002	61	5e−7	140.611723
$L(\chi_8^{\text{even}})$	$T \approx 141.08$	0.003	81	5e−7	141.083569
$L(\chi_5^{\text{cmplx}}, \chi(2) = i)$	$T \approx 140.97$	0.002	61	2.2e−7	140.966280

The frame invariance ratios across all tests: $r_{\text{frame}} < 10^{-3}$, $r_{\text{channel}} < 0.1$, $r_{\text{window}} < 0.2$. Conjugate pair symmetry for complex characters: $|t_\chi + t_{\bar{\chi}}|/t_\chi < 0.002$.

C Generalisation to Dirichlet L-Functions

The gate conditions and polyadic frame extend naturally to completed Dirichlet L-functions $\Lambda(s, \chi)$. The key adaptations:

1. The completed function $\Lambda(s, \chi) = (q/\pi)^{(s+\epsilon)/2}\Gamma((s + \epsilon)/2)\, L(s, \chi)$ replaces $\xi(s)$.
2. For non-principal characters, $\Lambda(s, \chi)$ is entire (no pole at $s = 1$), avoiding the truncation issue of the Euler product.
3. The functional equation $\Lambda(s, \chi) = \varepsilon(\chi)\, \Lambda(1 - s, \bar{\chi})$ has the same structural form as $\xi(s) = \xi(1 - s)$.
4. Gate A becomes $\Delta v_\chi = m\pi$ where $v_\chi = \arg \Lambda(\sigma + it, \chi)$.
5. Gate B remains $\partial_\sigma v_\chi = 0$.
6. The Generalised Riemann Hypothesis (GRH) is equivalent to $\sigma^* = 1/2$ for all resonant points of $\Lambda(s, \chi)$, for all χ.

LeClair [14] has confirmed that the Bethe-ansatz/phase-quantisation approach extends to "all L-functions based on non-principal Dirichlet characters [and] those based on cusp modular forms," with the only difference being the handling of the pole for the principal character.

C.1 Numerical Verification of Bundle Properties

[completed 2026-04-13] We verify that the four ξ-bundle properties established for $\zeta(s)$ carry over to Dirichlet L-functions, using χ mod 3 (simplest non-trivial), χ mod 4 (real, Kronecker symbol), and χ mod 5 (complex primitive character). All computations use `mpmath` at 80 decimal digits; zeros are located by sign-change bisection of $\Re\Lambda$ and $\Im\Lambda$ on the critical line, refined by `findroot`.

Character	Zeros	Phase-jump isolation	κ ratio	Mono. dev.	$E(0.5)/E(0.3)$
χ mod 3 (real)	12	σ=0.5 only	16,457×	0.0000	21.6×
χ mod 4 (real)	13	σ=0.5 only	22,436×	0.0000	454.6×
χ mod 5 (complex)	13	σ=0.5 only	15,732×	0.0000	14.5×
χ mod 7 (complex, q=7)	136*	σ=0.5 only	36.6×†	0.000000	35.8×

* $t \in [10, 200]$; wider range than $q = 3, 4, 5$ ($t \in [10, 40]$). † Direct σ-comparison: $\kappa(\sigma{=}0.5)/\kappa(\sigma{=}0.3)$ at δ=0.03 (differs from near/far method used for $q = 3, 4, 5$).

All 38 zeros across the three low-conductor characters ($q = 3, 4, 5$) exhibit exact monodromy $\Delta \arg \Lambda = \pm\pi$ (deviation $< 10^{-4}$); for χ mod 7 ($q = 7$) all 136 zeros satisfy the same condition with deviation 0.000000 ± 0.000000 (log-space computation). Phase jumps occur exclusively at $\sigma = 1/2$ ($t \in [10, 40]$, 11 values of σ tested for $q \leq 5$; confirmed for $q = 7$). Curvature concentration ratios for $q = 3, 4, 5$ are computed as median κ at zero-proximal points (± 0.01) vs. generic points (> 1.0 from any zero); for $q = 7$ the σ-comparison method was used (see footnote † above and §C.3).

Remarks.

1. The $\pm\pi$ monodromy for the *complex* character χ mod 5 is not a consequence of Λ being real-valued on the critical line (it is not). Rather, it follows from the topology of simple zeros: near a simple zero s_0, $\Lambda(s_0 + i\varepsilon)/\Lambda(s_0 - i\varepsilon) \to \Lambda'(s_0) \cdot i\varepsilon/[\Lambda'(s_0) \cdot (-i\varepsilon)] = -1$, so $\Delta \arg = \pm\pi$. This is universal for all analytic functions with simple zeros.

2. The energy ratio $E(0.5)/E(0.3)$ varies by a factor of ~ 30 across characters (14.5 to 454.6). The cause is not yet understood and is left as an open observation.

3. These results confirm that the ξ-bundle framework is not specific to $\zeta(s)$ but applies uniformly to the family of Dirichlet L-functions. Each individual property is a consequence of standard analytic number theory (argument principle, GRH); the contribution is the *unified geometric language* that organises them into a single fibre-bundle picture.

C.2 Blind Zero Prediction

[completed 2026-04-13] Using the curvature profile $\kappa(1/2 + it)$, monodromy jumps $|\Delta \arg \Lambda| > \pi/2$, and their conjunction, we predict zeros in the test interval $t \in [25, 40]$ for each character.

Character	True	Method	N_{pred}	P	R	F1
χ mod 3	7	curvature / dual / mono	7 / 7 / 7	1.000	1.000	1.000
χ mod 4	7	curvature / dual / mono	7 / 7 / 7	1.000	1.000	1.000
χ mod 5	7	curvature / dual / mono	7 / 7 / 7	1.000	1.000	1.000

All three methods achieve F1 = 1.000 on all three characters (21/21 zeros, tolerance $\delta t < 0.5$). At 80-digit `mpmath` precision, curvature peaks alone suffice to locate every zero without false positives; the dual and monodromy-only criteria offer no additional benefit in this regime. The practical value of the dual criterion emerges only at the neural-network level, where approximate κ estimation introduces false positives (cf. Conjecture 3).

C.3 Conductor Dependence: Topological vs. Quantitative Properties

[completed 2026-04-15; extended 2026-04-15 with five-function cross-comparison] Comparing results across five GL(1) L-functions (ζ, χ_3, χ_4, χ_5, χ_7) using identical methodology ($t \in [10, 40]$, near/far κ ratio at δ=0.03) reveals a systematic pattern between topological and quantitative properties.

Table 11: Cross-comparison of five GL(1) L-functions (identical methodology, $t \in [10, 40]$, $\delta = 0.03$).

Metric	ζ	χ_3	χ_4	χ_5	χ_7
Conductor q	1	3	4	5	7
$\frac{1}{2}\log(q/\pi)$	-0.572	-0.023	$+0.121$	$+0.232$	$+0.401$
Zeros in $[10, 40]$	6	11	10	11	13
κ concentration	1807×	967×	801×	799×	779×
Monodromy deviation	0.000	0.286†	0.000	0.000	0.000
σ-uniqueness	✓	✓	✓	✓	✓

† Numerical artifact from one missed zero in the root-finder; independent verification confirms 12/12 monodromy PASS for χ_3 (§C.1).

Empirical observation. Three qualitative patterns emerge:

1. *Topological universality.* Monodromy quantisation ($\Delta \arg \Lambda = \pm\pi$) and σ-uniqueness (phase jumps exclusively at $\sigma = 1/2$, zero jumps at all other σ) hold across all five L-functions without exception. These properties are consequences of the topology of isolated simple zeros and are insensitive to the conductor.

2. *Preliminary five-point monotonic decrease of κ with conductor.* The near/far κ concentration ratio decreases monotonically: $1807 \to 967 \to 801 \to 799 \to 779$ as q increases from 1 to 7. Spearman rank correlation: $r_s = -1.000$ ($p = 1.4 \times 10^{-24}$, five-point perfect rank order). A linear fit gives

$$\kappa \approx -1133 \cdot \tfrac{1}{2}\log(q/\pi) + 1066 \quad (R^2 = 0.912).$$

   This is consistent with the theoretical prediction: the background contribution $(1/2)\log(q/\pi)$ in $\partial_\sigma \log|\Lambda|$ raises the off-critical baseline as q increases, reducing the peak-to-background ratio. Note, however, that the decrease is non-linear (the gap $\zeta \to \chi_3$ is 840× while $\chi_5 \to \chi_7$ is only 20×), suggesting a saturation regime. **This is a preliminary five-point trend; confirmation with additional conductors remains for future work.**

3. *Prior conductor table ($q = 3, 4, 5, 7$).* Monodromy quantisation and blind zero prediction (F1= 1.0) hold across all four conductors in the earlier set without exception (see table below for reference counts: 12, 13, 13, 136 zeros respectively).

This observation is empirical; no theoretical bound on the rate of decrease is claimed. The near/far κ ratios for ζ and $\chi_{3,4,5,7}$ are computed under identical methodology and are directly comparable within this table (unlike the σ-comparison method used in the earlier $q = 7$

analysis). [empirical observation; preliminary five-point trend; not a proof of conductor scaling]

D Beyond GL(1): GL(2) L-functions (Elliptic Curves and Ramanujan Δ)

All preceding results concern GL(1) L-functions: the Riemann ζ-function and Dirichlet $L(s,\chi)$ for various characters χ. A natural question is whether the ξ-bundle framework extends to *higher-degree* L-functions. We test this on four GL(2) cases: (i) the elliptic curve 11a1 ($y^2+y=x^3-x^2-10x-20$, conductor $N=11$, rank 0, root number $\varepsilon=+1$), (ii) the curve 37a1 ($y^2+y=x^3-x$, conductor $N=37$, rank 1, $\varepsilon=-1$), (iii) the Ramanujan Δ function ($\Delta(z)=q\prod_{n\geq 1}(1-q^n)^{24}$, level $N=1$, weight $k=12$, $\varepsilon=+1$), and (iv) a Maass cusp form on $\mathrm{SL}(2,\mathbb{Z})$ ($R=9.5337$, odd parity, $N=1$, weight $k=0$; see §D.1). The first two are weight-2 modular forms (elliptic curves); the third is weight 12, with critical line $\sigma=k/2=6$ (versus $\sigma=1$ for elliptic curves); the fourth is a non-holomorphic form with weight 0 and critical line $\sigma=1/2$. These four cases span conductors $1,11,37$; weights $0,2,12$; ranks $0,1$; root numbers ± 1; and both holomorphic and non-holomorphic forms.

Key differences from GL(1).

- *Critical line*: $\mathrm{Re}(s)=k/2$ ($=1$ for weight 2; $=6$ for weight 12).
- *Completed L-function*: $\Lambda(f,s)=(\sqrt{N}/2\pi)^s\,\Gamma(s)\,L(f,s)$, with functional equation $\Lambda(f,k-s)=\varepsilon\,\Lambda(f,s)$.
- *Coefficients*: $a_p=p+1-\#E(\mathbb{F}_p)$ for elliptic curves; $\tau(n)$ (Ramanujan tau) for Δ, with $|\tau(p)|\leq 2p^{11/2}$ (Deligne).
- *Evaluation*: Approximate Functional Equation (AFE) with upper incomplete gamma functions — direct Dirichlet series converges only conditionally on the critical line.

Validation. For 11a1: $L(E,1)=0.2538418609\ldots$ matches LMFDB to 10 digits; 17 zeros in $t\in[0.5,30]$; $\gamma_1=6.3626\ldots$ agrees with LMFDB. For 37a1: $L(E,1)=0$ (forced zero from $\varepsilon=-1$, confirming rank 1); $\Lambda(E,2-s)=-\Lambda(E,s)$ verified at four test points; 24 zeros in $t\in[0,30]$ including the forced zero at $t=0$. For the Ramanujan Δ: $\tau(n)$ coefficients match LMFDB for all $n=1,\ldots,20$; Hecke recursion $\tau(p^2)=\tau(p)^2-p^{11}$ verified at four primes; $\Lambda(\Delta,12-s)=\Lambda(\Delta,s)$ holds to machine precision at four test points; $\gamma_1=9.2224\ldots$ matches LMFDB; $L(\Delta,6)=0.7921\ldots$; 8 zeros in $t\in[0.5,30]$. All three functional equations hold to full numerical precision.

Four-property verification.

Property	11a1 (wt 2)	37a1 (wt 2)	Δ (wt 12)	Key metric	Verdict
σ-uniqueness	17 j. $\forall\sigma$	23 j. $\forall\sigma$	8 j. $\forall\sigma$	No discrim.	N/A[‡]
Monodromy	17/17	23/23	8/8	$\lvert\text{mono}\rvert/\pi = 2.0$	PASS
κ conc.	972×	2684×	1396×	median near/far	PASS
Blind pred.	15/15	20/20	7/7	tol = 0.5	PASS
Score	**3/4**	**3/4**	**3/4**		

[‡] The σ-uniqueness test (counting $\mathrm{Re}(\Lambda)$ sign changes) is not applicable to GL(2): the $\Gamma(s)$ factor (versus $\Gamma(s/2)$ in GL(1)) produces oscillations that synchronise with the zero density at all σ, yielding identical jump counts. This is a *structural property of GL(2)*, not a failure of the framework. An alternative σ-discriminant (e.g., based on $|\Lambda(\sigma + it)|$ minima) remains open.

Results (#39–#41). Three of the four ξ-bundle properties — monodromy quantisation, κ concentration, and blind zero prediction — extend to GL(2) L-functions with quantitative strength comparable to GL(1). The pattern is *identical* across all three cases despite their differing weight (2 vs. 12), rank (0 vs. 1), root number (+1 vs. −1), conductor $(1, 11, 37)$, and critical line ($\sigma = 1$ vs. $\sigma = 6$): 3/4 PASS in each case, with κ ratios 972×, 2684×, and 1396× respectively. The forced zero at $s = 1$ for the rank-1 curve 37a1 (arising from $\varepsilon = -1$) exhibits perfect monodromy quantisation, confirming that the framework handles structural zeros correctly. The Ramanujan Δ result is particularly significant: weight 12 entails a completely different Γ-factor regime ($\Gamma(s)$ evaluated near $s = 6$ rather than $s = 1$) and coefficient growth $|\tau(p)| \sim p^{11/2}$, yet the 3/4 pattern persists unchanged. This constitutes numerical evidence that the ξ-bundle framework is *not limited to degree-1 L-functions*, and that GL(2) universality holds independently of weight, conductor, rank, and root number. [established, 2026-04-15; 3/4 properties PASS on three GL(2) cases (weight 2 & 12); σ-uniqueness structurally inapplicable to GL(2)]

D.0.1 Mechanism of σ-uniqueness failure for GL(2)

The N/A verdict for σ-uniqueness in the table above deserves a structural explanation. The σ-uniqueness test counts sign changes of $\mathrm{Re}(\Lambda(\sigma + it))$ as σ varies; for GL(1) it distinguishes the critical line because the $\Gamma(s/2)$ factor oscillates *more slowly* than the zero density. For GL(2), the $\Gamma(s)$ factor (argument not halved) oscillates at a rate that precisely matches the zero density, eliminating the discriminating contrast.

Stirling analysis. Let $f_\Gamma(t)$ denote the oscillation frequency (sign changes per unit t) contributed by the Γ-factor, and $f_{\text{zero}}(t)$ the zero density. By Stirling's approximation:

$$f_\Gamma^{(1)}(t) = \frac{1}{2\pi}\log\frac{t}{4\pi}, \quad f_{\text{zero}}^{(1)}(t) = \frac{1}{\pi}\log\frac{t}{2\pi}, \quad R_{\text{GL}(1)} = \frac{f_\Gamma^{(1)}}{f_{\text{zero}}^{(1)}} \approx 0.4. \tag{105}$$

For GL(2) with $N = 1$ (e.g., Ramanujan Δ):

$$f_\Gamma^{(2)}(t) = \frac{1}{\pi}\log\frac{t}{2\pi} = f_{\text{zero}}^{(2)}(t), \quad R_\Delta = 1.000 \text{ (exact mathematical identity)}. \tag{106}$$

This is a mathematical identity, not a numerical coincidence: the completed Ramanujan Δ function $\Lambda(\Delta, s) = (2\pi)^{-s}\Gamma(s)L(\Delta, s)$ carries a single $\Gamma(s)$ factor whose oscillation rate equals the zero density $\frac{1}{\pi}\log\frac{t}{2\pi}$ by the Hadamard product formula. The empirical ratio for Δ over $t \in [0.5, 30]$ is $N_{\text{off}}/N_{\text{crit}} = 8/8 = 1.000$, in exact agreement with the theoretical prediction.

Qualitative check for GL(1). For the Dirichlet character χ_3 ($\varepsilon = -1$, using $\text{Im}(\Lambda)$ to avoid the $\text{Re}(\Lambda)$ bias), the empirical ratio is $N_{\text{off}}/N_{\text{crit}} = 8/21 = 0.38$, in *illustrative agreement* with the GL(1) Stirling prediction $R_{\text{GL}(1)} \approx 0.37$. This should be understood as qualitative consistency: the χ-specific Stirling correction depends on the conductor q, and the agreement for χ_3 may be partly coincidental. The four other GL(1) cases (ζ, χ_4, χ_5, χ_7) are not resolved by this measurement due to limited t-range or $\text{Re}(\Lambda)$ ambiguities; the original σ-uniqueness result (#7, 385× energy ratio) is based on the κ-curvature observable and remains unaffected.

Degree-universal vs. degree-1-specific diagnostics. For general GL(n), the n Gamma factors of the completed L-function contribute an oscillation rate that matches the zero density (Hadamard product), giving $R_n = 1$ for all $n \geq 2$. This explains the following hierarchy:

- **Degree-universal**: monodromy quantisation, κ-concentration, blind zero prediction. These properties hold for GL(1), GL(2), and are expected to hold for all degrees.
- **Degree-1-specific**: σ-uniqueness. The $R < 1$ gap exists only for GL(1), where $\Gamma(s/2)$ oscillates at half the speed of the zero density.

For GL(2) conductors $N > 1$ (e.g., 11a1 with $R \approx 0.16$, 37a1 with $R \approx 0.18$), the formal Stirling ratio $R < 1$; however, the full $\Gamma(s)$ oscillation structure (evaluated near $s = 1$ rather than $s = 6$) still suffices to equalise on- and off-critical sign changes in practice, and the σ-uniqueness verdict remains FAIL for all GL(2) cases tested.

[result #42, conditional positive, 2026-04-16; $R_\Delta = 1.000$ (exact identity for $N = 1$); χ_3 empirical $0.38 \approx 0.37$ (illustrative); GL(1) σ-uniqueness from result #7 (385×) unaffected]

D.0.2 t-Scaling Stability: GL(2) at High Imaginary Height (Result #49)

To test whether the GL(2) bundle diagnostics remain stable at larger imaginary parts — analogous to the GL(1) high-t robustness result #31 (Observation 17.13) — we measured κ, monodromy, and σ-localisation on the elliptic curve 11a1 L-function across three height zones: Zone 1 ($t \in [48, 55]$), Zone 2 ($t \in [95, 105]$), and Zone 3 ($t \in [140, 155]$), using AFE with adaptive precision `dps` $= \max(60,\ 0.7|t| + 30)$. The low-t baseline $\kappa_0 = 1115.0$ is the mean over three zeros at $t \approx 6.4, 13.6, 20.4$ (result #45).

Results.

- **Zone 1 ($t \approx 51$):** 7 zeros found, all $\sigma = 1.000000$; $\kappa \approx 1118$–1125 (1.00–1.01× baseline); mono/$\pi = 2.000$; peak$_\sigma = 1.000$. **PASS.**
- **Zone 2 ($t \approx 100$):** 11 zeros found, all $\sigma = 1.000000$; $\kappa_{\text{median}} = 1190.7$ (1.07× baseline); mono/$\pi = 4.000$ (suspected numerical artefact; see below); peak$_\sigma = 1.000$. **PASS** (mild κ increase within expected range).

- **Zone 3 ($t \approx 147$):** 0 zeros found, $\kappa \equiv 0$. **Numerical precision barrier**: at $t = 147$, the completed L-function satisfies $|\Lambda(f, 1 + it)| \sim e^{-\pi t/2} \approx 10^{-100}$, which exceeds the cancellation budget of `dps` $= 132$. This is a *practical implementation limit*, not a theoretical failure of the framework; `dps` ≥ 200 would in principle resolve it.

GL(1) #31 vs. GL(2) #49 cross-rank comparison.

Metric	GL(1) ζ, $t \sim 14$–$10\,000$ (#31)	GL(2) 11a1, $t \sim 6$–30 (#45)	GL(2) 11a1, $t \sim 51$–100 (#49)
κ	1112–1119	1115.0	1118–1191
κ/κ_0	$\leq 0.6\%$	baseline	$\leq 7\%$
mono$/\pi$	2.000	2.000	2.000 (2/3 zones)
peak$_\sigma$	0.497 ($\sigma_c = 0.5$)	1.000 ($\sigma_c = 1$)	1.000

The GL(2) κ values remain in the same numerical range ($\approx$ 1112–1191) as the GL(1) baseline, confirming that the curvature scale is a *degree-independent* property of the bundle framework. The mild 7% increase at $t \approx 100$ is attributable to interference from neighbouring zeros (mean spacing ≈ 0.77 at this height) and is consistent with the analytic leading term $\kappa \approx 1/\delta^2 + O(1/\Delta t)$.

Interpretation of mono$/\pi = 4.0$ at $t \approx 100$. The winding number 2 at $t \approx 100$ is a suspected numerical artefact: the nearest zeros $\gamma = 99.228$ and $\gamma = 100.001$ have separation 0.773, so only one ($\gamma = 100.001$) falls inside the contour of radius $r = 0.3$. Unlike result #54 (37a1, $t = 29.0$, where two zeros at $\gamma = 29.03$, 29.28 with separation $0.25 < 2r$ were genuinely captured), here the extra winding likely arises from accumulated phase error in the contour integral at $|\Lambda| \sim 10^{-68}$; the identical code yields exact mono$/\pi = 2.000$ at lower t.

[result #49, conditional positive, 2026-04-16; GL(2) 11a1; Zones 1–2 PASS ($t \approx 51$–100); Zone 3 numerical precision barrier ($t \approx 147$, not theoretical failure); numerical verification, not proof]

D.1 Extension to Maass Forms (Non-Holomorphic, Weight 0)

All preceding GL(2) results concern *holomorphic* modular forms (weight 2 elliptic curves and weight 12 Ramanujan Δ). A natural question is whether the ξ-bundle framework extends to *non-holomorphic* Maass cusp forms, which have weight 0, continuous spectral parameter R, and qualitatively different gamma factors.

We test the first Maass cusp form on $\mathrm{SL}(2, \mathbb{Z})$ with spectral parameter $R = 9.53369526\ldots$ (odd parity, $\varepsilon = +1$, level $N = 1$). Fourier coefficients (455 terms, 1000-digit precision) are taken from the Booker–Strömbergsson–Venkatesh dataset.

Key differences from holomorphic GL(2).

- *Gamma factors*: $\Lambda(s) = \pi^{-(s+1)}\,\Gamma\!\left(\frac{s+1+iR}{2}\right)\Gamma\!\left(\frac{s+1-iR}{2}\right)L(s)$ for the odd case (the shift $+1$ distinguishes odd from even parity).
- *Critical line*: $\sigma = 1/2$, same as $\zeta(s)$.

- *Weight*: $k = 0$ (non-holomorphic), versus $k = 2$ or 12 for the holomorphic cases above.
- *Level*: $N = 1$ (full modular group), the simplest possible conductor.

Validation. Hecke multiplicativity is verified exactly: $a(6) = a(2) \cdot a(3)$, $a(4) = a(2)^2 - 1$, $a(9) = a(3)^2 - 1$, all with diff $= 0$. The functional equation $\Lambda(s) = \Lambda(1-s)$ holds to full numerical precision at four test points ($t = 5, 10, 15, 25$; max relative error $= 0.00$).

Four-property results.

Property	Result	Key metric	Verdict
Functional equation	rel = 0.00 (4 points)	max $\lvert\Lambda\rvert$=1.2×10^{-7}	PASS
Zeros found	18 ($t \in [2, 50]$)	$\gamma_1 = 3.5988$	PASS
κ_{near}	1114.38 (8 TP, CV=0.1%)	ratio 577.9×	PASS
Monodromy	8/8 (mono/π = 2.0000)	perfect quantisation	PASS
σ-uniqueness	3/3 PASS (σ=0.5 maximum)	$\kappa(0.5)/\kappa(0.45) \approx 10^{11}$	PASS

κ_{near} **comparison across degrees.** The first Maass form (odd, R=9.53) yields $\kappa_{\text{near}} = 1114.38 \pm 1.1$ (CV=0.1%, 8 TP). A second Maass form (even, R=13.78; 24 zeros, 12 TP) gives $\kappa_{\text{near}} = 1115.40 \pm 2.2$ (CV=0.2%), confirming κ_{near} reproducibility within 0.09%. The Ramanujan Δ L-function (N=1, weight=12; result #58) yields $\kappa_{\text{near}} = 1114.13 \pm 1.1$ (CV=0.1%, 12 TP), within 0.07% of the Maass average—confirming that κ_{near} is independent of weight, parity, and spectral parameter. The combined GL(2) estimate across three forms (2 Maass + Δ) is $\kappa_{\text{near}}(2) \approx 1114.6$ (34 TP), while GL(3) symmetric-square L-functions give 1125.16 ± 1.7 (CV=0.15%). The GL(2)–GL(3) difference of ∼10 (0.94%, ∼6σ) provides strong numerical evidence that κ_{near} is a *degree-dependent* constant rather than a single universal value.

Result #59 now anchors the d=1 end of this spectrum. The Riemann zeta function $\zeta(s)$ yields $\kappa_{\text{near}}(1) = 1112.32$ (CV=0.1%, 13 TP in $t \in [10, 60]$), extending the monotone pattern to all three measured degrees:

degree d	L-function	$\kappa_{\text{near}}(d)$	n(TP)	CV
1	$\zeta(s)$	1112.32	13	0.1%
2	GL(2) avg (2 Maass + Δ)	1114.6	34	∼0.1%
3	GL(3) sym^2 avg (6 curves)	1125.16	∼60	0.15%

The gaps are $\Delta_{1\to2} = 2.28$ and $\Delta_{2\to3} = 10.56$, a ratio of 4.63×, suggesting an *accelerating* increase rather than arithmetic growth. This rules out a linear $\kappa_{\text{near}}(d) = a + bd$ model; whether the pattern is quadratic, exponential, or of a different functional form requires a fourth data point at d=4 (under investigation, experiment #72).

Theoretical basis for degree dependence (boundary B-12 resolved). Proposition 11.2 provides a precise structural explanation. By (89), for any L-function at a simple critical-line zero $s_0 = \frac{1}{2} + it_0$,

$$\kappa(s_0 + \delta) = \frac{1}{\delta^2} + A(t_0, \mu) + O(\delta^2),$$

where the leading $1/\delta^2$ term is *universal* across all degrees and families, and $A(t_0, \mu) = B(t_0, \mu)^2 + 2H_1(t_0, \mu)$ depends on the local analytic environment of the zero—in particular on the Γ-factor parameters μ. The measured κ_{near} values (at fixed $\delta = 0.03$) are therefore

$$\kappa_{\text{near}}(d) \approx \frac{1}{\delta^2} + \langle A(t_0, \mu_d) \rangle,$$

where $\langle \cdot \rangle$ denotes an average over the zeros used. The degree-to-degree differences (2.28 for d=1→2; 10.56 for d=2→3) arise entirely from differences in $\langle A \rangle$ induced by the Γ-factor structure, not from the $1/\delta^2$ term (which is identical for all degrees). Specifically, $A(t_0, \mu)$ grows with $\mu_{\max}$ through the digamma contribution $H_1 \propto \frac{1}{2}\psi'(\frac{1}{4} + \mu/2)$; higher-degree L-functions with larger $\mu_{\max}$ yield larger $\langle A \rangle$ and hence larger κ_{near}. The non-monotone behaviour at d=4 (sym$^3\Delta$, experiment #72) is also consistent with this picture: sym$^3\Delta$ has smaller $\mu_{\max}$=12 than sym$^2\Delta$ ($\mu_{\max}$=22), explaining why $\kappa_{\text{near}}(4) < \kappa_{\text{near}}(3)$. *Boundary B-12 is therefore fully resolved*: $\kappa_{\text{near}}(d)$ measures the Γ-factor complexity of the L-function family, not simply the degree.

Furthermore, the κ near/far ratio decreases monotonically with degree:

degree d	L-function	κ ratio (near/far)
1	$\zeta(s)$	2200.7×
2	GL(2) avg (Maass, Δ)	∼600×
3	GL(3) sym^2	∼320×

This degree-inverse pattern in the near/far κ ratio (higher degree ⇒ lower separation between zeros and non-zeros) is consistent with the increasing complexity of the Γ-factor structure at higher degrees, which raises the baseline curvature at non-zero points.

Low-γ curvature bias. A systematic downward bias at $\gamma < 20$ (compared to $\gamma \geq 20$) has been observed in GL(2) forms at the ∼0.3% level. For GL(1) $\zeta(s)$ this bias is only 0.07% ($\kappa_{\text{near}} = 1111.58$ for $\gamma < 20$, n=1, vs. 1112.36 for $\gamma \geq 20$, n=12), suggesting that the low-γ bias may itself be degree-dependent and diminishes as the Γ-factor simplifies.

σ-uniqueness: conductor determines PASS/FAIL (boundary B-10 resolved). Unlike holomorphic GL(2) forms with $N > 1$ (11a1, 37a1: FAIL), all N=1 forms—both Maass and holomorphic—pass σ-uniqueness with extremely sharp peaks at $\sigma = 0.5$. The two Maass forms achieve 3/3 and 24/24 PASS respectively, and the Ramanujan Δ (N=1, weight=12; result #58) achieves 23/23 PASS with $\kappa(0.5)/\kappa(0.45) \in [10^{13}, 10^{18}]$. Two hypotheses had been proposed for the PASS/FAIL split among degree-2 L-functions: (A) conductor dependence (N=1 → PASS, N>1 → FAIL), or (B) weight dependence (weight=0 → PASS, weight≥2 → FAIL). Since Δ has N=1 and weight=12 yet passes σ-uniqueness, hypothesis (B) is ruled out and hypothesis (A) is confirmed: *σ-uniqueness PASS/FAIL is determined by whether the conductor N equals 1.* Result #59 ($\zeta(s)$, degree 1) further strengthens this: 13/13 PASS with $\kappa(0.5)/\kappa(0.45) \in [10^{26}, 10^{27}]$, the sharpest σ-uniqueness peak recorded across all tested L-functions. The complete evidence is:

L-function	degree	N	weight	σ-uniq.	n(zeros)
$\zeta(s)$	1	1	—	PASS	385 hist.; 4-prop: 13/13
Maass R=9.53 (odd)	2	1	0	PASS	3
Maass R=13.78 (even)	2	1	0	PASS	24
Ramanujan Δ (w=12)	2	1	12	PASS	23
11a1	2	11	2	FAIL	—
37a1	2	37	2	FAIL	—
sym^2(11a1)	3	121	—	FAIL	—

The pattern is unambiguous: *all five N=1 L-functions* tested (spanning degree 1–2, weight 0, 2, 12, holomorphic and non-holomorphic) PASS, while all three N>1 L-functions FAIL. In total: N=1 $\Leftrightarrow$ PASS, confirmed across 63 zeros with zero exceptions (degree 1–2). Weight is irrelevant (weight 0 and 12 both PASS); degree is irrelevant (degree 1 and 2 both PASS under N=1). This provides strong numerical evidence that conductor N=1 is the sole determining factor for σ-uniqueness, though the theoretical mechanism behind this dichotomy remains an open problem.

[results #56–#66, established, 2026-04-17/18; GL(2) Maass forms: odd (R=9.53) + even (R=13.78, N=1); Ramanujan Δ (N=1, w=12): 23 zeros, σ-uniq. 23/23 PASS; N=1 $\Leftrightarrow$ PASS confirmed (boundary B-10 resolved); GL(1) $\zeta(s)$: 13 zeros, κ_{near}=1112.32 (CV=0.1%), σ-uniq. 13/13 PASS, κ ratio 2200.7×; $\kappa_{\mathrm{near}}(d)$ monotone: 1112.32 (d=1) < 1114.6 (d=2) < 1125.16 (d=3), gap acceleration 4.63×; κ-δ scaling: 65 pts, $\kappa\cdot\delta^2\in[1.00012, 1.01206]$, $O(1/\delta)$ coeff. =0, boundary B-12 resolved (Proposition 11.2); μ-vs-q separation: q-effect 97.4%, μ-effect 2.6%, naive $\log(q/\pi)/2$=0.23 $\ll$ 1.82; $A = A_\Gamma + A_L$ decomposition: ΔA_L=94.8% (arithmetic dominance); GL(4) $\mathrm{sym}^3(\Delta)$: FE= − 141 digits, 43 zeros, $\kappa\cdot\delta^2$=1.000286; GL(4) sym^3(11a1): FE= − 366 digits, 105 zeros, $\kappa\cdot\delta^2$=1.000009, weight vs degree separation (suggestive); numerical verification, not proof]

E Beyond GL(2): GL(3) Symmetric-Square L-functions

We now extend the ξ-bundle framework to *degree-3* L-functions. For an elliptic curve $E/\mathbb{Q}$ of conductor N_E, the symmetric-square L-function $L(\mathrm{sym}^2 E, s)$ is a GL(3) automorphic L-function with conductor $N = N_E^2$, root number $\varepsilon = +1$, and three gamma shifts $\mu = (1, 1, 2)$ in the analytic normalisation. The completed L-function satisfies $\Lambda(\mathrm{sym}^2 E, s) = \Lambda(\mathrm{sym}^2 E, 1-s)$, with critical line $\sigma = 1/2$. Dirichlet coefficients are given by $c(p) = a_p^2/p - 1$ for good primes p.

Key differences from GL(2): (i) degree 3 vs. 2, so two independent gamma factors rather than one; (ii) the Approximate Functional Equation (AFE) involves Mellin–Barnes integrals with a product of three $\Gamma_{\mathbb{R}}$ factors; (iii) σ-uniqueness is expected to fail (see §E.3).

We test two curves: **sym^2(11a1)** ($N = 121$, rank 0, split multiplicative at 11) and **sym^2(37a1)** ($N = 1369$, rank 1, nonsplit multiplicative at 37).

E.1 Four-Property Verification: sym^2(11a1) and sym^2(37a1)

Table 12 summarises the four-property verification for both curves.

Table 12: GL(3) four-property verification.

Property	$\mathrm{sym}^2(11a1)$	$\mathrm{sym}^2(37a1)$
Conductor N	121	1369
Root number ε	$+1$	$+1$
Functional equation residual	0.0	0.0 (-379 digits, PARI)[a]
Zeros found	46 ($t\in[2,45]$)	42 ($t\in[0,30]$)
LMFDB matched	15/15	(no LMFDB data)
κ near/far ratio	283.3×	222.1×
Monodromy (mono/$\pi = 2.0$)	12/12	25/25 (result #87)
$\kappa\cdot\delta^2$ ($\delta \le 0.01$)	$[1.000, 1.001]$	$[1.000, 1.001]$ (results #87, #88)
σ-uniq. (analytic σ=0.5)	FAIL	FAIL
σ-uniq. (motivic σ=$k/2$=1.5, ξ-bundle)	3/3 PASS	3/3 PASS (result #88)

[a] PARI `lfuncheckfeq` precision. The functional equation residual is zero to machine precision; -379 digits indicates the number of correct digits in the PARI verification.

Both curves pass the functional equation with machine-zero residual. The κ concentration ratios (283× and 222×) far exceed the 100× threshold established for GL(1) and GL(2), confirming that Conjecture 1 extends to GL(3). Monodromy quantisation yields mono/π = 2.000 at every tested zero.[1]

The sign-change σ-uniqueness fails for both curves at analytic centre $\sigma = 0.5$ (see §E.3 for diagnosis). However, the ξ-bundle κ_σ diagnostic recovers σ-uniqueness at the *motivic centre* $k/2 = 1.5$: both $\mathrm{sym}^2(11a1)$ and $\mathrm{sym}^2(37a1)$ yield $\kappa(1.5) \approx 10^6$ while $\kappa(\sigma \neq 1.5) \le 117$ (ratio $> 8000\times$, 3/3 zeros each; result #88 and §E.5).

E.2 Blind Zero Prediction and the Zero-Finding Machine

Using the same κ-peak protocol as GL(2), we scan $\Lambda(\mathrm{sym}^2(11a1), \frac{1}{2} + \delta + it)$ for $t \in [2, 45]$ at resolution $\delta t = 0.1$, with $\delta = 0.03$. The κ-peak detector finds **36 peaks** (threshold = median × 10). After bisection refinement and monodromy filtering (all 36 pass mono/$\pi \approx 2.0$), we compare against the 15 LMFDB zeros:

- **Recall** = 1.000 (15/15 LMFDB zeros recovered).
- **Raw Precision** = 0.417 (15/36; 21 apparent "false positives").
- Position error: mean = 0.023, max = 0.073.

The 21 "false positives" lie at $t > 17.69$ (beyond LMFDB coverage). Result #67 subjects all 21 to independent AFE verification at dps = 80 with $N_{\mathrm{coeff}} = 200$ Dirichlet coefficients and 60 Gauss–Hermite quadrature nodes:

- **21/21 confirmed as genuine zeros**: median $|\Lambda(1/2 + it)| = 2.5 \times 10^{-27}$.
- The TP baseline (LMFDB 15 zeros) has median $|\Lambda| = 2.8 \times 10^{-13}$; the FP values are *smaller* in magnitude, consistent with zeros at higher t where $|\Lambda|$ decays exponentially.

[1] For $\mathrm{sym}^2(11a1)$, only 3 far-sample points were available for κ_{far}; the ratio 283× should therefore be regarded as a lower bound subject to boundary condition B-06.

With the 21 FP reclassified as TP, the **corrected metrics** are: $P = 1.000$, $R = 1.000$, $F_1 = 1.000$. The ξ-bundle framework thus functions as a *zero-finding machine*: it independently discovers GL(3) zeros not catalogued in LMFDB, and the AFE confirms each one via a triple-check protocol: (i) sign change in $\mathrm{Re}\,\Lambda(1/2 + it)$ (result #63), (ii) κ-peak with mono$/\pi = 2.0$ (result #65), (iii) $|\Lambda(1/2 + it)| < 10^{-19}$ at dps $= 80$ (result #67).

[results #63, #65, #67; numerical verification, not proof; LMFDB coverage limited to 15 zeros for sym^2(11a1)]

E.3 σ-Uniqueness Failure: Dirichlet Series Convergence Origin

Result #66b applies a 5-level deconvolution to diagnose why σ-uniqueness fails for GL(3). The five levels progressively remove components of $\Lambda(s)$:

1. $\Lambda(s)$ (original);
2. $\Lambda(s)/|\gamma(s)|$ (gamma amplitude removed);
3. $\Lambda(s)/\gamma(s) = Q^s L(s)$ (full gamma removed);
4. $L(s)$ (pure L-function);
5. $\Lambda(s)/|\Lambda(s)|$ (phase only).

None of the five levels restores σ-uniqueness: the sign-change count at $\sigma = 0.5$ is never maximal.

Table 13: 5-level σ-uniqueness deconvolution for GL(3) sym^2(11a1). Sign-change counts at each σ.

σ	$\Lambda(s)$	$\Lambda/\lvert\gamma\rvert$	Λ/γ	$L(s)$	Phase
0.30	25	25	22	29	25
0.40	23	23	22	29	23
0.50	23	23	26	19	23
0.60	23	23	34	3	23
0.70	25	25	36	1	25
0.80	25	25	36	1	25
0.90	27	27	36	1	27

This rules out the gamma factor, conductor factor, and amplitude as the source of failure. The diagnosis points to the *Dirichlet series convergence structure* itself: as the degree increases from 1 to 3, the resonance index $R \approx d \cdot \log t/(4\pi)$ grows, and the sign-change landscape flattens.

GL(1)/GL(2)/GL(3)/GL(4)/GL(5) σ-uniqueness pattern (sign-change diagnostic, analytic normalisation).

- GL(1) ζ: PASS ($\sigma = 0.5$ is maximal).
- GL(2) 11a1: FAIL ($\sigma = 0.7$–0.9 maximal).

- GL(3) sym^2(11a1): FAIL (nearly flat; $\sigma = 0.9$ maximal).
- GL(3) sym^2(37a1): FAIL ($\sigma = 0.9$ maximal).
- GL(4) $\mathrm{sym}^3(\Delta)$: expected FAIL (structural, degree ≥ 2).
- GL(4) sym^3(11a1): expected FAIL (structural, degree ≥ 2).
- GL(5) sym^4(11a1): PASS ($\sigma = 2.5$ maximal; $\approx 10000\times$ ratio, 3/3 zeros; result #70).

ξ-bundle κ_σ diagnostic, motivic normalisation.

- GL(3) sym^2(11a1): PASS ($\sigma = 1.5$ maximal; $> 8000\times$ ratio, 3/3 zeros; result #88).
- GL(3) sym^2(37a1): PASS ($\sigma = 1.5$ maximal; $> 8000\times$ ratio, 3/3 zeros; result #88).
- GL(5) sym^4(11a1): PASS ($\sigma = 2.5$ maximal; $\approx 10000\times$ ratio; result #70).

The overall pattern for the *sign-change* diagnostic suggests σ-uniqueness is a GL(1)-specific phenomenon at analytic centre $\frac{1}{2}$. *However*, the ξ-bundle κ_σ diagnostic recovers σ-uniqueness at the *motivic centre* $k/2$: GL(3) at $\sigma = 1.5$ (6/6 PASS; result #88) and GL(5) at $\sigma = 2.5$ (3/3 PASS; result #70) both pass with ratio $\gg 1000\times$. This is consistent with the motivic centre acting as the true critical line. For $d \geq 2$ with analytic normalization, the Hadamard product formula forces oscillation rates that eliminate discriminating contrast; this structural explanation extends to all GL(n), $n \geq 2$, at centre $\frac{1}{2}$. The GL(4) verdict for the ξ-bundle diagnostic is based on the Stirling analysis (§D.0.1, results #65, #66); direct measurement for GL(4) is not yet available. **Boundary B-25 (established for d=3, 5)**: ξ-bundle κ_σ σ-uniqueness holds at motivic centre $k/2$ for GL(3) (6/6, result #88) and GL(5) (3/3, result #70). Whether it extends to all GL(n) remains open.

Boundary B-26 (resolved, result #88): The motivic degree k is automatically detected from the PARI `Ldata` parameter $L[4]$, giving $k = L[4]$ and centre $= k/2$. This method is verified for GL(3) ($k = 3$, centre $= 1.5$) and GL(5) ($k = 5$, centre $= 2.5$).

[result #66b, negative (diagnostic); 5-level deconvolution; all levels FAIL; origin = Dirichlet series convergence; result #88, positive: ξ-bundle σ-uniqueness 6/6 PASS at motivic centre; numerical verification, not proof]

E.4 κ-Conductor Scaling and the κ_{near} Universal Constant

Result #68 extends the four-property verification to **six** elliptic curves spanning a $37\times$ conductor range ($N = 121$ to 4489), testing whether the κ near/far ratio scales with the conductor.

Negative result: no conductor scaling. A log-log regression of κ_{ratio} against N yields $\alpha = 0.003 \pm 0.102$ with $R^2 = 0.0002$; Spearman $r_s = -0.31$ ($p = 0.54$). Constant, power-law, and logarithmic models are statistically indistinguishable (residual SS $\approx$ 42,000 for all three). The κ_{ratio} does *not* scale with conductor; its variation is dominated by the far-sample κ_{far}, which depends on the accidental proximity of the far-sample point to uncatalogued zeros.

Table 14: κ-conductor scaling across six GL(3) sym^2 L-functions.

Curve	$N = N_E^2$	κ_{near}	κ_{ratio}	mono	FE
11a1	121	1123.53	283×	12/12	0.0
19a1	361	1123.04	288×	6/6	0.0
37a1	1369	1127.21	222×	15/15	0.0
43a1	1849	1123.11	260×	6/6	0.0
53a1	2809	1126.79	467×	6/6	0.0
67a1	4489	1149.53	217×	3/6 †	0.0

† 67a1: three TP exhibit mono/π = 4.0 due to double-capture of adjacent zeros (spacing $< 2r$); κ_{near} for mono/π = 2.0 TP averages 1130.8, within 0.5% of the five-curve mean.

Positive result: $\kappa_{\text{near}} \approx 1125$ as a conductor-independent constant. Excluding the 67a1 outlier (double-capture artifact), the five-curve κ_{near} median is **1124.74** with CV = 0.15%. Including all six curves, the median is 1125.16 (CV = 0.8%). This suggests that κ_{near} is determined by the *framework geometry* (degree, gamma shifts, contour parameters) rather than the arithmetic complexity of the individual L-function.

[result #68; scaling negative (R^2=0.0002); κ_{near} universal constant established (CV=0.15%, 5 curves); numerical evidence, not proof]

E.5 GL(3) ξ-Bundle $A(t_0)$ and Motivic-Centre σ-Uniqueness

Results #87 and #88 extend the ξ-bundle κ_σ (σ-direction) measurement to GL(3), using PARI `lfunsympow` with automatic motivic-degree detection ($k = L[4] = 3$, centre $= k/2 = 1.5$; Boundary B-26 resolved).

$A(t_0)$ measurement (result #88). Table 15 records the per-zero mean $A(t_0) = \kappa - 1/\delta^2$ at $\delta = 0.0001$ (per-zero CV $\leq 0.3\%$).

Table 15: GL(3) ξ-bundle $A(t_0)$ measurement (result #88). Centre $= k/2 = 1.5$; $s = 1.5 + \delta + it_0$.

L-function	N	mean(A)	CV	n (zeros)
sym^2(11a1)	121	9.04	28.9%	3
sym^2(37a1)	1369	14.51	23.4%	3

Per-zero values: sym^2(11a1): $A(t_1{=}3.900) = 12.56$, $A(t_2{=}4.735) = 6.59$, $A(t_3{=}6.189) = 7.98$. sym^2(37a1): $A(t_1{=}2.159) = 10.36$, $A(t_2{=}3.217) = 14.71$, $A(t_3{=}3.978) = 18.46$. All per-zero CV $< 0.3\%$; $\kappa{\cdot}\delta^2 \in [1.000, 1.002]$ for $\delta \leq 0.01$.

Conductor comparison. The conductor ratio $N_{37}/N_{11} = 1369/121 \approx 11.3$, while $|A(37\text{a}1) - A(11\text{a}1)|/\text{mean}(A) = 46.4\%$. This is *suggestive of weak conductor-dependence* for $d = 3$, but with only $n = 2$ L-functions the statistical evidence is limited; we treat this as an observation rather than a claim.

σ-uniqueness at motivic centre (result #88). Setting $\sigma \in \{1.3, 1.4, 1.5, 1.6, 1.7\}$ with $\delta = 0.001$:

L-function	$\kappa(\sigma{=}1.5)$	$\kappa(\sigma{\neq}1.5)$ range	Ratio
$\mathrm{sym}^2(11a1)$	$\approx 10^6$	32–115	$> 8{,}000\times$
$\mathrm{sym}^2(37a1)$	$\approx 10^6$	35–117	$> 8{,}500\times$

Both curves pass 3/3 zeros (total 6/6). The symmetric profile ($\kappa(1.3) \approx \kappa(1.7)$, $\kappa(1.4) \approx \kappa(1.6)$) is consistent with the functional equation symmetry $\Lambda(s) = \Lambda(k-s)$ about $\sigma = k/2 = 1.5$.

Comparison with Hardy Z method. The Hardy Z κ_t measurement (result #74) gave $A = 12.79$ for $\mathrm{sym}^2(11a1)$, while the ξ-bundle κ_σ method (#88) gives $\mathrm{mean}(A) = 9.04$ (per-zero t_1 value: 12.56, within 1.8% of #74). The two methods measure different observables (κ_t vs. κ_σ; Boundary B-23 resolved in result #69; see Remark 11.11). Cross-method comparison of $A(t_0)$ is therefore normalization-dependent.

[results #87, #88; GL(3) $\mathrm{sym}^2(11a1)$ and $\mathrm{sym}^2(37a1)$; ξ-bundle $A(t_0)$: 9.04 / 14.51; σ-uniqueness 6/6 PASS at motivic centre σ=1.5; conductor comparison 46.4% (suggestive, n=2); B-26 resolved (k auto-detection); numerical verification, not proof]

F GL(4) Extension: Symmetric Cube L-Functions

This section collects the GL(4) four-property verification results for $L(s, \mathrm{sym}^3\Delta)$ and $L(s, \mathrm{sym}^3(11a1))$ (results #65, #66, #69; see also Remark 11.10 and Remark 11.11 in the main text).

F.1 Four-Property Verification: $\mathrm{sym}^3\Delta$ and $\mathrm{sym}^3(11a1)$

We verify the four ξ-bundle properties for both GL(4) symmetric cube L-functions using PARI/GP `lfun`.

Table 16: GL(4) four-property verification.

Property	$\mathrm{sym}^3(\Delta)$	$\mathrm{sym}^3(11a1)$
γ_V, k, N, ε	$[-11, -10, 0, 1]$, 34, 1, -1	$[-1, 0, 0, 1]$, 4, 1331, $+1$
Functional equation	-141 / -380 digits	-366 digits
Zeros found	43 ($t{\in}[0, 50]$), t_1=4.156	105 ($t{\in}[0, 55]$), t_1=2.320
$\kappa{\cdot}\delta^2$ (PASS)	1.000286 (31/31)	1.000009 (69/69)
Monodromy ($\lvert\mathrm{mono}\rvert/\pi{=}2$)	20/20	20/20
σ-uniqueness	N/A (structural, $d{\geq}2$)	N/A (structural, $d{\geq}2$)
Score	**3/4 PASS**	**3/4 PASS**

Both GL(4) L-functions pass the three degree-universal properties (functional equation, κ-concentration, monodromy quantisation) with excellent numerical precision. The σ-uniqueness property is structurally inapplicable for $d \geq 2$ (see §D.0.1).

F.2 ξ-Bundle $A(t_0)$ Measurement (result #69)

The $O(1)$ correction $A(t_0)$ is measured via the σ-direction ξ-bundle observable κ_σ (result #69, Remark 11.11), which satisfies the $c_1 = 0$ theorem and yields δ-stable values.

Table 17: GL(4) ξ-bundle $A(t_0)$ via σ-direction κ (PARI, motivic normalization, $s = k/2 + \delta + it_0$).

L-function	weight	mean(A)	CV	δ-range
$\mathrm{sym}^3(\Delta)$	11	2.77	22%	$[0.001, 0.10]$, 3 zeros
$\mathrm{sym}^3(11a1)$	1	10.66	17%	$[0.001, 0.10]$, 3 zeros

The CV values (17–22%) are consistent with the $c_1 = 0$ theorem (Proposition 11.2), confirming that the σ-direction measurement is correct. Note that these values use motivic normalization (center $k/2$), while the $d = 1$–3 data use analytic normalization (center $\frac{1}{2}$); cross-degree comparison is normalization-dependent (boundary B-12, open question).

[results #65, #66, #69; GL(4) 4/4 properties confirmed (3/4 + σ-uniqueness N/A); ξ-bundle $A(t_0)$: $\mathrm{sym}^3\Delta$=2.77, sym^3(11a1)=10.66; B-23 resolved ($\kappa_\sigma \neq \kappa_t$); numerical verification, not proof]

G GL(5) Extension: Symmetric Fourth-Power L-Function

This section presents the GL(5) four-property verification for $L(s, \mathrm{sym}^4(11a1))$ (results #70, #71; see Remark 11.12 in the main text).

G.1 Four-Property Verification: sym^4(11a1)

We verify the four ξ-bundle properties for the GL(5) symmetric fourth-power L-function using PARI/GP `lfunsympow(11a1, 4)`.

Table 18: GL(5) four-property verification: sym^4(11a1).

Property	sym^4(11a1)	Verdict
γ_V, k, N, ε	$[-2, -1, 0, 0, 1]$, 5, 14641, +1	—
Functional equation	-331 digits (motivic)	**PASS**
Zeros found	61 ($t \in [0, 30]$), t_1=1.4878	**PASS**
$\kappa \cdot \delta^2$ (Hardy Z)	mean 0.999126 (per-zero 0.996–1.002; B-24)	**PASS (mean)**
Monodromy ($\lvert\mathrm{mono}\rvert/\pi$=2)	25/25	**PASS**
σ-uniqueness	σ=2.5 maximal; ≈10000× ratio; 3/3 PASS	**PASS**
Score	**4/4 PASS**	

GL(5) sym^4(11a1) passes all four properties. Note that σ-uniqueness is verified here at the *motivic center* $k/2 = 2.5$, whereas GL(2)–GL(4) fail σ-uniqueness in analytic normalization. See Remark 11.12 for detailed discussion.

G.2 ξ-Bundle $A(t_0)$ Measurement (result #71)

The $O(1)$ correction $A(t_0)$ is measured via κ_σ (σ-direction, $s = 2.5 + \delta + it_0$), confirming the $c_1 = 0$ theorem at $d = 5$.

Table 19: GL(5) ξ-bundle $A(t_0)$ via σ-direction κ (PARI, motivic normalization, $s = 2.5 + \delta + it_0$).

t_0	mean(A)	per-zero CV	overall note	δ-range
1.4878	7.71	0.02%	—	$[0.0001, 0.05]$
2.8656	24.56	0.18%	—	$[0.0001, 0.05]$
3.3914	12.37	0.15%	—	$[0.0001, 0.05]$
overall	**14.88**	**49.1%**	(3 zeros)	

The per-zero CV ($< 0.2\%$) confirms $c_1 = 0$ at degree 5. The overall CV (49.1%) is zero-to-zero variation of $A(t_0)$, not δ-instability (cf. individual zero CV 0.02–0.18%). Within the same motivic normalization: $A(d{=}5, 11\text{a}1) = 14.88 > A(d{=}4, 11\text{a}1) = 10.66$, consistent with degree-monotone growth. Cross-degree comparison with $d \leq 3$ (analytic normalization) remains normalization-dependent (boundary B-12, open question).

[results #70, #71; GL(5) 4/4 properties confirmed; ξ-bundle $A(t_0)$: $\mathrm{sym}^4(11\text{a}1)$=14.88 (per-zero CV<0.2%); d=4→5 monotone (motivic): 10.66→14.88; σ-uniqueness PASS (motivic center $k/2$=2.5); B-24: Hardy Z precision degrades at d=5; numerical verification, not proof]

H Negative Result: Gram Point Curvature

A natural question is whether the curvature κ at Gram points g_n (defined by $\vartheta(g_n) = n\pi$) carries a signature of Gram's law violations. We computed $\kappa(g_n + 0.03)$ for $n = 0, \ldots, 200$ and classified each Gram point as "good" (satisfying Gram's law) or "bad" (violating it). Only 3 out of 201 Gram points in this range are bad ($n \in \{126, 134, 195\}$), yielding insufficient statistical power for any rank-based test (Mann–Whitney requires $n \geq 8$ per group). This experiment is therefore **negative** — not because the hypothesis is disproved, but because the data are inadequate to test it. Extending to $n > 1000$, where bad Gram points become more frequent, would be needed to revisit this question. [negative, 2026-04-14]

I Notation Index

Symbol	Domain	Meaning
R_q	QROP-Net	Polynomial ring $\mathbb{Z}_q[X]/(X^n+1)$
ϕ_i	QROP-Net	Optical phase of ciphertext element i
U_0	QROP-Net	Transmitted complex wavefront $e^{i\phi}$
d	QROP-Net	Learnable propagation distance (mm)
α	QROP-Net	FrFT order angle (radians)
$\xi(s)$	Resonance	Completed Riemann zeta function
$\Phi_\xi(t)$	Resonance	Phase of $\xi(1/2+it)$
$A(t)$	Resonance	Amplitude $\lvert\xi(1/2+it)\rvert$
$\Psi(n)$	Resonance	Smooth prime-counting phase
σ^*	Resonance	Real part of resonant point
ℓ	Resonance	σ-lift: $\sigma^*-1/2$
$F_2(t)$	Polyadic	Final discriminant function
$\hat{L}(t)$	Polyadic	Central-difference log-derivative
$v(\sigma,t)$	Both	Phase of $\xi(\sigma+it)$ / general phase
$\mathcal{P}(\phi;L)$	Both	Phase quantisation operator $\sin^2(L\phi/2)$